

# Foundations for Surreal and Surcomplex Mathematics

Sets, Classes, Universes, Type Theories,  
and a Pinned Lean Audit

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## Abstract

The phrase “formalize the surreal numbers” hides at least four distinct projects: construct canonical numbers and prove the ordered-field laws; characterize the resulting simplicity hierarchy; develop algebra over sufficiently large set-sized surreal subfields; and formalize functions, supports and analysis over the full class. They share mathematics but not size requirements and not proof-assistant dependencies.

Surreal numbers are individually set-sized and collectively a proper class. Their complexification  $\mathbf{SC} = \mathbf{No}[i]$  inherits that size without introducing a new algebraic paradox. The genuine foundational issues are the permitted sizes of cuts, the coding of quotients, the strength of recursion, the scope of support and function quantifiers, and the interpretation of topology. This report compares definable classes in ZF and ZFC, explicit class theories (GB, GBC, specified NBG), Kelley–Morse, Grothendieck universes and Tarski–Grothendieck, Feferman reflection, birthday fragments, set-sized Hahn workspaces, constructive set theory, classical dependent type theory, higher inductive-inductive and univalent approaches, and an internal set-theoretic model inside Lean. It restates, size-correctly and with proofs, the workspace, summability, topology and phase principles of five supplied surcomplex research manuscripts; it gives an elementary finite-algebra descent theorem, a support-local nonlinear recursion proposition, two near-mirror examples over  $\mathbb{C}((t^{\mathbb{Q}}))$  with opposite exponent signs and opposite conclusions, and a family of topological counterexamples that pin down exactly where the degeneracy theorems live. It then reports one dated, pinned, read-only inspection of a contemporary Lean surreal repository, separating what the inspected source establishes from what remains an unverified bridge, and proposes a layered implementation architecture.

The central lesson is the discipline of three meanings of “all”: all members of a specified set; all values at a specified universe level subject to smaller-index rules; and the entire definable class. The unifying rule behind that discipline is that *an individual object can have small data without the totality of all such objects being small*. Smallness, algebraic closure, summability and ambient topology must be separate stated hypotheses rather than connotations of the word “surreal.”

**Status.** This report merges three independently written audits. Their three status declarations are reproduced here as a union rather than rewritten into one.

*From the first audit.* “This is a researched mathematical exposition and implementation blueprint. The supplied manuscripts are treated as motivating research texts, not as machine-certified libraries. The Lean examples accompanying this article have not been compiled in this environment. No unconditional consistency proof, completed formalization of the manuscripts, or exhaustive survey of every repository is claimed.”

*From the second audit.* “All consistency assurances are relative to the chosen foundational theory and its metatheory. This article supplies constructions, explicit size restrictions, and proof obligations; it does not claim an absolute proof of the consistency of set theory or Lean.” Its illustrative Lean files and source-audit record “are not represented as an executed formal verification.”

*From the third audit.* “This is a researched mathematical exposition and implementation design, not a machine-checked formalization of the attached research manuscripts. ‘Paradox-free’ means avoiding the identified invalid constructions within explicitly stated foundations; no absolute consistency proof of those foundations is claimed. Lean source observations are pinned where possible and dated. Illustrative Lean snippets have not been compiled in this environment.”

*For the original merged document.* Three of the five shipped Lean files were compiled and axiom-audited, under a toolchain different from the one the sources pin. Those historical checks are recorded in [section 2.6](#). Subsequent repository formalization is a separate body of evidence, summarized in [section 2.1](#) and tracked theorem by theorem in `docs/FORMALIZATION.md`. The present exposition is not a claim that all its results have been formalized.

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# 1 Scope, the five source manuscripts, and the recommended separation

## 1.1 Four distinct projects, three layers, four meanings of “rigorous”

There are at least four different projects hidden in the phrase “formalize the surreal numbers.” One can construct canonical numbers and prove their ordered-field laws; characterize the resulting simplicity hierarchy; develop algebra over sufficiently large set-sized surreal subfields; or formalize functions, supports and analysis over the full class. These projects share mathematics, but they do not have the same size requirements or the same proof-assistant dependencies.

Cutting the same material a different way, the problem has three layers that should not be merged. The first is a *foundational construction*: what objects are surreal numbers, what does it mean to quantify over them, and why are recursive definitions legitimate? The second is a *mathematical interface*: which properties of ordered fields, cuts, simplicity, normal forms and exponentiation are available? The third is *formal verification*: how can a proof assistant represent these objects and check arguments without silently changing their size or their topology?

Correspondingly, “rigorous” names four different achievements. A precise mathematical construction specifies its objects and permissible operations. An axiomatic construction additionally names the assumptions from which existence and theorems follow. A proof-assistant development supplies proof terms accepted by a specified kernel with an audited axiom set. A relative-consistency analysis compares the assumptions with another theory. A Lean declaration asserting an ordered field together with arbitrary-cut filling can be syntactically accepted even when the asserted axioms imply a contradiction; conversely a class construction in ZFC can be rigorous with no proof-assistant implementation at all.

The surcomplex numbers of the supplied work are

$$\mathbf{SC} = \mathbf{No}[i] = \{x + iy : x, y \in \mathbf{No}, i^2 = -1\},$$

where the braces denote a *class*, not a set asserted to contain all such pairs. The algebraic input is that  $\mathbf{No}$  is real closed, and consequently  $\mathbf{SC}$  is algebraically closed in the finite-polynomial sense. These are established results of surreal theory, not consequences of a notational definition [8, 9, 10]. Neither statement supplies a complex-analytic topology, a general summation operation, or a set of all class functions.

## 1.2 What “paradox-free” is allowed to mean here

A rigorous treatment answers four separate questions. *Formation*: are the purported objects legal sets, classes or types in the chosen foundation? *Construction*: do the definitions actually produce the stated ordered field and its complexification? *Metatheory*: what assumptions justify confidence in the underlying formal system? *Verification*: has a proof assistant checked the particular definitions and proofs, with a recorded axiom dependency? A positive answer to the first does not imply a positive answer to the second. An axiomatic structure with impossible fields can be syntactically well formed and uninhabited. Conversely, an inconvenient universe error is usually a size mismatch, not evidence that surreal numbers are inconsistent.

In this report, *paradox-free formalization* means a construction with explicit formation rules, legitimate recursion and comprehension, and no hidden assumptions that contradict its own order or universe laws. Consistency remains relative to the foundation used. More sharply still, “paradox-free” denotes only the explicit exclusion of the specific invalid constructions analysed below — the

self-cut, the proper-class-member quotient, the unrestricted support sum, and incompatible equality principles — and nothing more. For a consistent, effectively axiomatized theory strong enough for arithmetic, the second incompleteness theorem prevents the theory from proving its own standard arithmetized consistency statement under that theorem’s hypotheses. This obstructs neither the construction of surreal numbers nor any ordinary mathematical existence theorem. It tells us not to advertise a construction in ZFC or Lean as an absolute proof that its foundation cannot derive a contradiction [20, §5].

The preferred architecture for the written mathematics is

$$\begin{aligned}
 & \boxed{\text{set-sized algebra and analysis}} \\
 + & \boxed{\text{explicit size-aware comparison maps}} \\
 + & \boxed{\text{a class or universe interpretation}}.
 \end{aligned} \tag{1.1}$$

The first component carries most proofs. The second prevents accidental changes of meaning. The third supplies the genuinely unbounded interpretation of **No** and **SC**.

### 1.3 The five supplied manuscripts, and who read which

This report merges three independently prepared audits, referred to throughout as **M1**, **M2** and **M3**. They were given *different* collections of source manuscripts. Five distinct surcomplex research manuscripts are involved, and *only one of them was read by all three audits*. The full ledger — titles, filenames, recorded hashes, and which audit read which text — is [section A](#). The essential facts are these.

**A, the Analysis manuscript.** *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory* [1]. Read by all three audits. M1 and M3 cite it under the filename `surcomplex_analysis(3).tex`; M2 cites it under the filename `surcomplex_analysis(2).tex`. The SHA-256 values recorded independently by M2 and by M3 are identical (`e6e2ef4a...17748f4`), so this is one file carrying two labels. We do not silently pick a filename and we assert nothing about which label is correct.

**AG, the Analytic Geometry manuscript.** *Infinitesimal Analytic Geometry over the Surcomplex Numbers* [2]. Read by **M1 only**. No hash was recorded for it by any audit.

**T, the Trigonometry manuscript.** *Trigonometry on the Surcomplex Plane* [3]. Read by **M2 only**; hash recorded by M2.

**P, the Polynomial Algebra manuscript.** *Surcomplex Polynomial Algebra: Order Geometry, Newton Profiles, and Scale-Resolved Stability* [4]. Read by **M3 only**; hash recorded by M3.

**R, the Hartogs manuscript.** *Hartogs Coherence, Finite Maps, and Residue Duality over the Surcomplex Numbers* [5]. Read by **M3 only**; hash recorded by M3.

**Tagging convention.** Because the audits read different texts, every claim in this report that is derived from AG, T, P or R carries a visible tag naming the single audit that read that source. No sentence in this report says “the supplied manuscripts” when it means a text that only one member saw. Claims derived from A may be attributed to the three audits jointly, and are so marked. This convention is not decorative: it is the only thing preventing a merged document from reading as an audit of one coherent corpus when it is in fact an audit of five texts with a single common element.

Two further non-claims about the sources travel with this ledger. First, P refers to a separate *global-divisors* manuscript that was never supplied for any of the three tasks; no assessment of that fourth text is offered here (M3). Second, other archives named in the bibliographies of the supplied manuscripts — in particular in the trigonometry bibliography — were not examined (M2). All five manuscripts are user-supplied research expositions, not independently certified libraries or refereed publications, and their original files are not duplicated in any of the three archives.

## 1.4 What each manuscript assumes

*A* (all three audits) explicitly chooses NBG with global choice and stipulates that individual supports, index families, coefficient sequences and the data of its constructions are sets. It distinguishes fine-local, Hahn-coherent and all-scale theories. Its statements about arbitrary small rescaling and about all-scale rigidity use the full surreal class, not merely one fixed field. Its contour functional is coefficientwise; it is not integration along arbitrary fine-continuous real-parameter paths. Its fine topology is described by balls with arbitrary positive surreal radii, not by a purported set of all open subclasses.

*AG* (M1 only) works with a nonzero divisible set-sized ordered subgroup  $\Gamma \subset \mathbf{No}$  and writes

$$t^\gamma = \omega^{-\gamma}, \quad K_\Gamma = \mathbb{C}((t^\Gamma)), \quad R_n = \mathbb{C}\{z_1, \dots, z_n\}, \quad \mathcal{A}_{n,\Gamma} = R_n((t^\Gamma)). \quad (1.2)$$

Its coefficients are convergent complex germs *without* a shared ordinary radius. It claims several-variable division, a Nullstellensatz, finite complete-intersection deformation, residue duality and finite-zero workspace invariance, with explicit mathematical proofs and without machine verification.

*T* (M2 only) retains the same support convention. Its circular functions are initially defined on finite surreal arguments, its geometric lengths range over all surreals, and its extensions to infinite circular arguments require extra phase data. These are substantive domain restrictions, not presentational conveniences.

*P* (M3 only) confines polynomials, supports, matrices and the relevant recursions to sets; its degrees are ordinary natural numbers; it introduces divisible set-sized exponent groups  $\Gamma$  and algebraically closed fields  $K_\Gamma$ .

*R* (M3 only) uses the same workspaces but also considers an arbitrary class function on a full infinitesimal thickening, and invokes class replacement after restricting that function to an ordinary set of complex points.

The three manuscripts M3 read state that they have not been independently proof-assistant verified; the same holds of the two M1 read and the two M2 read. Their theorems are therefore *targets and supplied arguments*, not imported formal certificates. Size-correctness of  $\mathcal{A}_{n,\Gamma}$  does not certify its Noetherianity, and formalization of an abstract field does not certify a proposed analytic residue formula (M1).

## 1.5 Three levels of assertion, maintained throughout

The following discipline, taken from M1, is used in every section.

1. Classical construction facts are attributed to established literature.
2. Statements about *A*, *AG*, *T*, *P* and *R* are explicitly marked source-derived, with the tag naming the audit that read the source.

3. Propositions proved in this report are elementary foundational or algebraic deductions, included to make the proposed architecture precise; no novelty claim, no priority claim and no resolution of a published open conjecture is attached to any of them.
4. Statements about software describe *inspected declarations or documentation*, not a successful independent rebuild. Proposed module names and future APIs are design suggestions.

## 1.6 A practical answer before the details

For a conventional mathematical exposition, a carefully specified version of GBC is a convenient language: individual numbers and supports are sets; **No**, its operations and particular global analytic functions are classes. The basic theory can instead be expressed in ZFC through definable classes, with no new large-cardinal axiom merely for surreal arithmetic.

For Lean, the most effective architecture is usually different. Work with a universe-indexed surreal type, prove theorems about *small* families at that level, and implement the analytic core over ordinary set-sized Hahn fields. Reuse the existing game-based surreal field rather than starting its algebra from zero. Prove explicit comparison theorems between the game, sign and Hahn presentations. The full normal-form theorem and the analytic enrichments should be tracked as mathematical dependencies, not inferred from the existence of suggestively named modules.

This *split* — one recommendation for the mathematics, a different and compatible one for the implementation — is the spine of the recommendation in [section 21](#). It is the only framing among the three audits that names the two questions separately, and we adopt it for that reason.

## 2 The verification boundary of this report

The three September 21 source audits were source inspections: none compiled a Lean file, rebuilt a repository, or ran `#print axioms`. The original merge additionally compiled three illustrative files, as recorded in [section 2.6](#). The repository has since developed a larger Lean library. These are three different verification records, with different revisions and scopes; the later work does not retroactively turn the historical source inspections into executed audits.

### 2.1 The current implementation and the historical proposal

At repository checkpoint `ebda0e0`, the checked-development ledger `docs/FORMALIZATION.md` records the constructed sign-sequence ordered field, its game-quotient comparison, small-cut and fine-topology results, the concrete surcomplex pair field, and many generic Hahn-field results. It also records canonical evaluation of small formal normal forms as an order isomorphism with inverse extraction. That last result does not yet supply the arithmetic and strong-sum preservation required by the full bridge in [section 19.9](#). Real closedness of the concrete sign carrier and the resulting surcomplex algebraic-closure specialization remain separate obligations at this checkpoint.

The implementation uses Lean 4.32.0 and the dependency revisions in `lake-manifest.json`, rather than the source audits' toolchain. Its `SurrealAudit.lean` target imports the library and checks transitive axiom dependencies of every imported declaration in the `Surreal` and `SurrealHahnSeries` namespaces, allowing only `propext`, `Quot.sound`, and `Classical.choice`. This is broader than the three illustrative-file checks, but it still requires a statement-level mapping to each manuscript claim. The ledger distinguishes complete mappings from prerequisites, conditional

results and pending work; neither a successful build nor an order isomorphism alone verifies the whole article.

The layers and suggested module paths in [section 19](#) retain their role as a plan. For implementation status, follow the ledger and actual imports in `Surreal.lean`; do not infer completion from a proposed module name. The files in this report’s `code/` and `data/` directories remain the historical originals. The source manuscripts and their READMEs are retained in Git history at checkpoint 251bd32, under the report’s former `sources/` directory; they are no longer duplicated in the working tree.

## 2.2 One inspection, three readers

All three audits inspected the *same* public Lean repository at the *same* immutable revision, on the same day, by reading source through the public web interface. All three record the same commit hash, the same toolchain, the same field instance, the same small-support Hahn type, and the same absent bridges. Printed three times, or printed once in confident merged prose with three citations, that agreement would be indistinguishable from independent confirmation. It is not.

Three agreeing readings of one source corroborate each other only about what that source says. They do not corroborate that the source compiles, that its theorems are what their names suggest, or that its axiom closure is clean. Accordingly the findings of that inspection are printed *once*, in [section 18](#), framed as a single act of source reading with three concurring readers, and nowhere in this report are they presented as three pieces of evidence.

## 2.3 What was executed, and by whom

The only step any of the three preparation environments actually executed was a  $\text{\LaTeX}$  build of its own article.

| Audit | Executed                                                                                                                                                                                                                                                       | Not executed                                                                                                                                           |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1    | pdf $\text{\LaTeX}$ build; PDF compiled and rendered for layout inspection.                                                                                                                                                                                    | Every Lean file. No Lean or Lake executable was available. No build log, no exit status, no <code>#print axioms</code> output anywhere in the archive. |
| M2    | pdf $\text{\LaTeX}$ 3.141592653-2.6-1.40.26 (TeX Live 2025/dev/Debian); 32 pages, 17 main sections, 2 appendices, 28 references; zero unresolved reference or citation warnings; zero overfull or underfull box warnings; all 32 pages rendered and inspected. | Lean compilation recorded in machine-readable form as “NOT EXECUTED; Lean/Lake were not installed in the preparation environment.”                     |

| Audit | Executed                                                                                                                                                                                                                                                                                                                       | Not executed                                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M3    | <code>latexmk/pdflatex</code> completed successfully; 40 pages, 94 PDF bookmarks, 36 references; cross-references stabilized with no unresolved references; 0 overfull boxes; 0 missing glyphs; all pages rendered and visually inspected. SHA-256 recorded for both its own <code>.tex</code> and its own <code>.pdf</code> . | Lean compilation not run; external Lean project rebuild not run; transitive <code>#print axioms</code> audit not run; Mizar execution not run; supplied-manuscript formal verification not performed. |

In M3’s own words, its build report “certifies document-production checks only.” M2’s build report states its scope as “PDF production checks and source inspection, not formal mathematical verification.”

## 2.4 Three archives, three different code inventories, none compiled

The three archives do not ship the same things, and the difference matters for what a future reader could actually check. The “State” column below records each file *as its archive delivered it*; “Not compiled” is a statement about that archive, not about the present report, which compiled three of the five (section 2.6).

| Audit | Lean files shipped                                 | Import footprint                            | State as delivered                                                                                                                                                                                                                                                 |
|-------|----------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1    | <code>FoundationsSketch.lean</code><br>(38 lines)  | <code>Init only</code>                      | Not compiled. Contains <code>noUniversalStrictBound</code> , <code>RawGame</code> , <code>SmallCutData</code> , and one <code>#print axioms</code> command whose output is recorded nowhere.                                                                       |
| M1    | <code>ComplexifySketch.lean</code><br>(35 lines)   | <code>Mathlib.Algebra.Ring.Basic</code>     | Not compiled. Proves exactly one theorem, $i^2 = -1$ . No zero, no addition, no subtraction, no inverse, no <code>Ring</code> and no <code>Field</code> instance.                                                                                                  |
| M1    | <code>InspectCurrentLibrary.lean</code> (24 lines) | Pinned repository plus <code>Mathlib</code> | Not run. Seven <code>#check/#synth</code> commands whose output is recorded nowhere.                                                                                                                                                                               |
| M2    | <code>LogicalGuards.lean</code>                    | <code>import Mathlib</code> (all of it)     | Not compiled. Contains <code>noUnrestrictedUpperBounds</code> , <code>RawGame</code> , <code>QuadraticPair</code> with <code>normSq_mul</code> and <code>gramIdentity</code> proved by <code>ring</code> , plus three unexecuted <code>#print axioms</code> lines. |
| M2    | <code>SourceAudit.lean</code>                      | Pinned repository                           | Not run.                                                                                                                                                                                                                                                           |
| M3    | <i>none</i>                                        | —                                           | M3 ships no Lean files at all. Its five code blocks exist only as <code>lstlisting</code> snippets typeset inside its PDF, each marked in the document itself as an illustrative design.                                                                           |

Note the asymmetry recorded in the third column. M1’s `FoundationsSketch` imports `Init` only and is therefore checkable without any `Mathlib` build; M2’s `LogicalGuards` imports the whole of `Mathlib`. So even the one theorem both archives contain — the machine-level form of the unrestricted-cut obstruction — has two entirely different dependency footprints. Neither was run.

M2’s README lists a `SHA256SUMS` file among the package contents, and the delivered archive did contain one. It is not present in this repository: the collection carries no checksum manifests, so the file was removed when the package was unpacked. Nothing was dropped in transit, and the README did not overstate its archive’s contents. M2’s README also gives `clone/checkout/build/run` commands for both POSIX `sh` and PowerShell and explicitly labels them “a reproduction recipe, NOT commands executed while preparing this package,” with the instruction: *build the upstream project with its original pinned dependency manifest, and do not run `lake update` and then describe that as a check of this exact revision*. That instruction is reproduced here because it is the single most likely way for a future reader to produce a result that looks like a check of the pinned revision and is not one.

## 2.5 What the merged document is, and what its length does not buy

This report is longer than any of its three sources. It cites five source manuscripts, proves more than thirty propositions, and carries more than fifty references. None of that is execution. The union of three read-only audits is a larger blueprint, not a certificate. Specifically:

- zero of the three audits compiled any Lean file;
- zero rebuilt the inspected repository;
- zero recorded any `#print axioms` output;
- zero performed a transitive axiom audit;
- zero ran Mizar;
- zero machine-verified any of the five supplied manuscripts.

Every one of those six remains true of the three source audits. The original merged report changed the first and the third only for the three files named in [section 2.6](#). Its own checks did not rebuild the external repository, audit all its declarations, run Mizar, or verify a complete supplied manuscript. Subsequent work in this repository is recorded separately in [section 2.1](#).

## 2.6 What was compiled for this report

The three audits each stated that no Lean or Lake executable was available where they were prepared. Both were available where this merge was prepared, so the shipped files were put through them. This subsection is the record.

*The toolchain was not the pinned one.* The three audits pin `leanprover/lean4:v4.35.0-rc2` with `Mathlib dec5b2b780537b6eaf7f5e5f000c12f7387fb24d`. That toolchain is not installed on the machine used here. The compilations below ran under Lean 4.32.0 against a built `Mathlib` at the matching release. This is a *nearby but different* toolchain, and nothing below licenses any claim about the pinned revision or about code that depends on it.

| File                     | Imports                | As shipped              | Axioms reported                                                                        |
|--------------------------|------------------------|-------------------------|----------------------------------------------------------------------------------------|
| M1 FoundationsSketch     | Init only              | compiles, exit 0        | noUniversalStrictBound:<br>none at all                                                 |
| M1 ComplexifySketch      | Mathlib                | compiles, exit 0        | three of five none; neg and<br>imaginaryUnit_sq:<br>propext                            |
| M2 LogicalGuards         | Mathlib                | <b>does not compile</b> | after repair: noUnrestrict<br>edUpperBounds none; two<br>others propext,<br>Quot.sound |
| M1 InspectCurrentLibrary | CombinatorialG<br>ames | not attempted           | repository absent                                                                      |
| M2 SourceAudit           | CombinatorialG<br>ames | not attempted           | repository absent                                                                      |

Neither `sorryAx` nor `Classical.choice` occurs in the reported dependencies of these illustrative files. Every axiom reported is `propext` or `Quot.sound`, both ordinary kernel axioms of Lean 4 rather than holes. In particular the two size-obstruction theorems that carry the argument of [section 3](#) — M1’s `noUniversalStrictBound` and M2’s `noUnrestrictedUpperBounds` — depend on no axioms whatever, so both are fully constructive.

*M2’s LogicalGuards.lean does not compile as shipped.* The kernel reports `invalid 'import' command`, it must be used in the beginning of the file. The file opens with a `/-! ... -/` module docstring and only then writes `import Mathlib`. A module docstring is a command, and no command may precede an import. M1’s files open with a plain `/- ... -/` comment, which is legal because a comment is not a command; the entire difference is the exclamation mark. Moving the import above the docstring makes the file compile, and the three axiom probes it already contains then produce the results tabulated above. The file ships here exactly as delivered, defect included.

*What this does and does not establish.* It establishes that three small illustrative files are syntactically and logically well formed under a current toolchain, and that the two impossibility theorems are constructive. It establishes nothing about the pinned repository, about the surreal field instance, about the absent normal-form bridge or real-closedness instance, or about any theorem in the five supplied manuscripts. Compiling a thirty-eight-line elementary impossibility theorem verifies that theorem and nothing beyond it.

Source inspection and compilation are different verification levels. A successful execution of the small illustrative examples in the M1 and M2 archives — which is now on record for three of them — does not constitute verification of the full surreal theory or of any manuscript. The deliverable here is a mathematical exposition, a foundational audit and an implementation blueprint. [Section E](#) collects the complete union of the three limitation lists.

### 3 The size obstructions every foundation must respect

#### 3.1 Set-sized numbers, a proper-class totality

One canonical presentation identifies a surreal with a sign sequence

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \mathbf{Ord},$$

with birthday  $b(s) = \alpha$ . Each such function is a set. At every fixed ordinal bound  $\beta$ , the sequences of length less than  $\beta$  form a set: the relevant function sets exist, and their union is indexed by the

set  $\beta$ . The total collection over all ordinals is a definable class. The empty sequence is zero; the constantly positive sequence of length  $\alpha$  represents the ordinal  $\alpha$  [9, 8].

All three audits prove the following by the same Replacement argument on the birthday map. We print it once, in M3's slightly stronger form, which carries the bounding clause as part of the statement.

**Proposition 3.1** (There is no set of all surreal numbers). *In the sign-sequence construction the class  $\mathbf{No}$  is not a set. More strongly, every set of surreal numbers has its birthdays bounded by an ordinal.*

*Proof.* For a set  $B$  of sign sequences, Replacement gives the set  $\{\text{dom}(s) : s \in B\}$  of their domains. The ordinal

$$\beta = \sup\{b(x) + 1 : x \in B\}$$

bounds their birthdays. If  $B$  contained every surreal it would contain the constantly positive sequence of length  $\beta$ , whose birthday is  $\beta$  and so is not less than  $\beta$ : a contradiction. The empty case is harmless.  $\square$

The same argument is the size mechanism behind the Burali-Forti obstruction: a collection containing a distinct canonical object at every ordinal stage cannot itself be one of the bounded sets in that construction. It is not a defect in surreal arithmetic. The ordered-pair construction  $\mathbf{SC} = \mathbf{No} \times \mathbf{No}$  with the intended complex arithmetic is likewise a class: an individual pair is a set, but the embedding  $x \mapsto (x, 0)$  precludes a set containing all pairs.

### 3.2 A size ledger

The following seven-row ledger is M1's, and is a pedagogic artifact rather than a theorem. It pairs each notation with the reading that is safe and the reading that is not.

| Expression                        | Safe reading                                                        | Dangerous reading                                                       |
|-----------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------|
| $x \in \mathbf{No}$               | A set code satisfying a definable predicate                         | An element of a universal set containing all surreal codes              |
| $\{L \mid R\}$                    | $L, R$ are sets, with every left option below every right option    | A cut of two arbitrary proper classes                                   |
| $\sum r_\alpha \omega^{a_\alpha}$ | A set-length normal form with suitable support                      | An unspecified sum over all ordinals                                    |
| $\mathbf{SC}[[X]]$                | A class of set-coded countable coefficient sequences                | A set-sized ring without a size restriction                             |
| $\mathcal{A}_{n,\Gamma}$          | A set ring for a fixed set $\Gamma$                                 | All possible exponent groups silently combined into one set             |
| A class function                  | A class of ordered pairs of sets, satisfying functionality          | A set-valued element of an unrestricted space of all class functions    |
| A class-valued sheaf              | Coded section and restriction relations, or a larger-universe sheaf | A small-category-valued functor with unacknowledged proper-class values |

### 3.3 The unrestricted-cut obstruction: the interface regression test

A Conway presentation  $\{L \mid R\}$  uses *sets* of previously available numbers with  $l < r$  for every  $l \in L$  and  $r \in R$ , and denotes the simplest number strictly between the two sides. It does not assert that every pair of subclasses with that separation property admits a separator. All three audits prove the following, by substituting the whole carrier on the left against the empty right side, and all three identify it as the single most useful regression test in the subject.

**Proposition 3.2** (The unrestricted cut axiom is inconsistent). *Let  $X$  carry an irreflexive relation  $<$ . There is no operation assigning to every subcollection  $A \subseteq X$  an element  $b(A) \in X$  with  $a < b(A)$  for every  $a \in A$ . Equivalently: for a nonempty strict order  $S$ , the assertion that every pair of subclasses  $L, R \subseteq S$  with  $L < R$  has an element strictly between them is false once the permitted pairs include  $(S, \emptyset)$ .*

*Proof.* Take  $A = X$  and then  $a = b(X)$ . The required inequality is  $b(X) < b(X)$ , contradicting irreflexivity. In the two-sided form, separation of  $S$  from the empty right side is vacuous, so a separator  $a \in S$  would satisfy  $x < a$  for every  $x \in S$ ; take  $x = a$ .  $\square$

Thus the surreal cut  $\{\mathbf{No} \mid \emptyset\}$  is not a new, extremely large surreal. It is an *illegal input*. The valid theorem bounds every *set* of surreal numbers by another surreal; in a universe-indexed implementation it bounds every sufficiently *small* family, not every subset of the entire larger carrier. Moving from sets to classes does not validate the unrestricted axiom; it makes the excluded input easier to describe.

This is stronger guidance than “beware of Russell’s paradox.” It supplies a one-line regression test for an axiom interface: *any interface granting strict upper bounds for all subsets of its own nonempty carrier is already inconsistent, regardless of its name*. The same reasoning works internally in type theory whenever a rule accepts the entire carrier as an indexing family (M2), and it recurs in [section 16](#) as the reason the universe shift in the raw-game constructor is the construction rather than an inconvenience.

**Lemma 3.3** (Bounds for sets of surreals; the small positive lower bound). *Every set  $B \subseteq \mathbf{No}$  is bounded above and below in  $\mathbf{No}$ . For every set  $A \subseteq \mathbf{No}_{>0}$  there is  $\delta > 0$  with  $\delta < a$  for all  $a \in A$ .*

*Proof.* The legal cut  $\{B \mid \emptyset\}$  gives an upper bound and its negation a lower bound. The legal cut  $\{0 \mid A\}$  gives the final assertion; for  $A = \emptyset$  take  $\delta = 1$ .  $\square$

Here each input is a set, so [theorem 3.2](#) does not apply. There is no greatest surreal, yet every set of surreals has an upper bound; the collection of all surreals is precisely not one of those sets. All three audits give this lemma, in the second form under the names “small positive lower-bound principle” (M2) and “a lower bound for a set of positive tolerances” (M1). Two further remarks travel with it. The hypothesis that  $A$  is a set is essential. And  $\delta$  need not lie in any previously specified subfield containing  $A$  (M2) — a point that becomes the whole content of [theorem 12.3](#).

### 3.4 Four different assertions involving “all”

The following four statements have different meanings (M2):

1. Every ordinal of the ambient set universe embeds in the surreal class.
2. Every ordinal small relative to a specified universe embeds in its associated surreal type.

3. Every set of surreal coefficients fits in a sufficiently large set-sized workspace.
4. There is one set-sized workspace containing every surreal number.

The first three have rigorous formulations. The fourth contradicts the first. A theorem producing a workspace *after* a set of inputs has been given does not produce a universal workspace. This is the same distinction as the three meanings of “all” identified by M1 — all members of a specified set; all values at a specified universe level subject to smaller-index rules; the entire definable class — which that audit identifies as the actual source of apparent paradox.

### 3.5 Bounds are not suprema

All three audits prove that the surreal ordered field is not Dedekind complete, with three different witnesses for one fact: M1 halves an upper bound of  $\mathbb{N}$ , while M2 and M3 subtract one.

**Proposition 3.4.** *The surreal ordered field is not Dedekind complete, even for set-sized bounded subsets.*

*Proof.* The set  $\mathbb{N}$  is bounded above, for instance by  $\omega$ . If  $h$  is an upper bound then  $h \geq 2n + 1$  for every  $n \in \mathbb{N}$ , so  $h/2 > n$  for every  $n$  and  $h/2$  is a strictly smaller upper bound. Equivalently,  $s \geq n + 1$  for all  $n$  makes  $s - 1$  another upper bound. Either way there is no least upper bound.  $\square$

More generally (M3), if a nonempty set  $B$  has no greatest member and  $b$  is an upper bound, then every member of  $B$  is strictly below  $b$ , and the legal cut  $\{B \mid \{b\}\}$  produces a smaller upper bound. The cut construction selects the *simplest* separator; it does not select a least upper bound in the numerical order. Consequently “complete” must always be qualified: real closedness, saturation for small cuts, valuation completeness and Dedekind completeness are not interchangeable. A Lean declaration asserting ordered-completeness for all surreals must therefore be rejected, and the phrase “complete ordered field”, which occurs in introductory software prose, must not be read as a formal Dedekind-completeness theorem. All three audits make this last point; [section 18](#) records where the phrase actually occurs.

### 3.6 Russell’s paradox, universal function spaces, and universal sets

In ZF, Separation forms  $\{x \in a : \varphi(x)\}$  from a pre-existing set  $a$ ; it does not form a universal set. In an explicit class theory the Russell class is allowed as a class but cannot be a set member. Similarly, the graph of a class function  $\mathbf{No} \rightarrow \mathbf{No}$  may be a class, but the collection of *all* such class functions is not automatically another class whose elements are those graphs. Classes in NBG are not eligible to be members merely because they have been named (M3). The same warning applies to “all open subclasses,” “the category of all classes,” and “all equivalence classes of a class-sized relation.” Each is useful informal language; a formalization needs a level of coding, a universe bound, a scheme of assertions, or a stronger framework.

Theories with a universal set, such as NFU and other stratified theories, are not a shortcut (M1, M2). They restrict comprehension, type assignment, or the class of admissible constructions. A theory with a universal carrier and a strict order cannot also prove that every subset of that carrier has a strict upper bound inside it: [theorem 3.2](#) is untouched. One cannot combine the existence of a universal set from one foundation with unrestricted ZFC-style subset and cut rules from another. A full surreal development in such a theory would require its own translation of ordinal, support and smallness conditions. *No claim of such a completed translation is made here* (M1).

## 4 Constructions of the carrier

### 4.1 Raw games and the numeric quotient

A raw game is a well-founded, set-branching tree with designated left and right options at each node. Working with raw games first avoids requiring a numeric comparison predicate in the constructor itself. Define comparison recursively on pairs of trees, then define which games are numeric. For numeric raw games the comparison law is

$$x \leq y \iff (\forall x^L \in L_x, \neg(y \leq x^L)) \wedge (\forall y^R \in R_y, \neg(y^R \leq x)). \quad (4.1)$$

A raw game is numeric when its options are numeric and every left option is strictly less than every right option. Classical comparison and numeric induction establish that numerical equivalence  $x \sim y \iff x \leq y \wedge y \leq x$  is an equivalence relation and that the quotient order is linear. Equal values can have different presentations: both  $\{\emptyset \mid \emptyset\}$  and  $\{-1 \mid 1\}$  represent zero. The distinction between the preorder on presentations and equality of numbers must remain visible until quotienting [8].

The recursive operations begin with

$$-x = \{-R_x \mid -L_x\}, \quad (4.2)$$

$$x + y = \{L_x + y, x + L_y \mid R_x + y, x + R_y\}, \quad (4.3)$$

$$xy = \{l_x y + x l_y - l_x l_y, r_x y + x r_y - r_x r_y \mid l_x y + x r_y - l_x r_y, r_x y + x l_y - r_x l_y\}, \quad (4.4)$$

where each displayed family ranges over the appropriate option sets. Numericity, compatibility with  $\sim$ , associativity, distributivity and the inverse construction are separate proof obligations. Merely printing (4.4) does not provide a field structure, and inversion is not justified by displaying the multiplication cut.

A rigorous development records three independent claims at each recursive definition (M2): the recursive calls are well founded; the option families are sets; and the constructed left options lie below the right options. Well-foundedness alone does not imply that a game is a number.

### 4.2 Recursion measures: what is and is not a valid substitute

There are two distinct orders. Numerical  $<$  is not well founded — it has the descending chain  $0, -1, -2, \dots$  — while simplicity, that is, taking a strict prefix of a sign sequence or passing to an option, is well founded. Recursive comparison and arithmetic must decrease the latter kind of complexity.

For a pair  $(x, y)$ , the Hessenberg natural-sum measure

$$b(x) \# b(y) \quad (4.5)$$

is strictly increasing in each argument, and replacing either argument by an option strictly decreases it. Two *different* wrong candidates are excluded by the two audits that considered the question, and both exclusions survive here because they eliminate different errors:

- the ordinary **maximum** of the two birthdays is not a valid general substitute, since decreasing the smaller argument can leave the maximum unchanged (M1);

- ordinary **ordinal addition** is not a valid substitute, since the natural sum is strictly increasing in each coordinate while ordinary ordinal addition is not strictly increasing in its left coordinate (M3).

M2 records only the weaker statement that some well-founded relation on pairs of birthdays or option trees justifies the recursion. Formal termination arguments must name such a relation; “one of the inputs became simpler” is not a checked termination certificate.

There is also a route that avoids a global pair-recursion theorem entirely (M1). Fix the two inputs and collect all their subpositions; their descendant closures are sets. Recursion then takes place on a set-sized well-founded relation, and uniqueness proves agreement as the input-dependent domains enlarge. For implementations with indexed option presentations this dependency-local argument is preferable to assuming that all syntactic forms of a given numerical birthday form a small collection. The general principle behind it is [theorem 6.4](#).

### 4.3 Why a literal proper-class quotient is unsafe

The equivalence class of a numerical value among all raw presentations can be a proper class. All three audits use the same motivating failure: the cuts  $\{-\alpha \mid \alpha\}$  for positive ordinals  $\alpha$  all have numerical value zero, while their presentations have unbounded rank. So “the class of equivalence classes” cannot mean a class whose elements are those proper classes; in GB or NBG, proper classes are not elements.

M1 names three safe alternatives: canonical sign sequences; restricting the prequotient to a set or a fixed type universe before quotienting; and Scott’s trick. M2 and M3 both promote the third to a numbered proposition with proof. We print that proof once.

**Proposition 4.1** (Set codes for a definable class quotient). *Let  $E$  be a definable equivalence relation on a definable class  $D$  of sets. For  $x \in D$  let  $\rho_x$  be the least rank of a set  $E$ -equivalent to  $x$ , and put*

$$C_x = \{y \in D \cap V_{\rho_x+1} : E(x, y) \text{ and } \text{rank}(y) = \rho_x\}.$$

*Then  $C_x$  is a nonempty set, and  $C_x = C_z$  if and only if  $E(x, z)$ .*

*Proof.* The ranks under consideration form a nonempty definable class of ordinals containing  $\text{rank}(x)$ ; a least one is obtained by searching the set  $\text{rank}(x) + 1$ . All candidates of that rank lie in  $V_{\rho_x+1}$ , and Separation forms  $C_x$ . If  $E(x, z)$  then the two classes of equivalent representatives agree, hence so do their least ranks and their codes. Conversely a member of their common nonempty code is equivalent to both  $x$  and  $z$ , so transitivity gives  $E(x, z)$ .  $\square$

The honest caveat, stated by both M2 and M3, is that the resulting collection of codes can still be a proper class: the repair is that every *code* is a set, not that the quotient has become small. Scott coding preserves an extensional quotient with no global choice of representatives, but transporting recursive functions and simplification through these codes adds overhead. That is an engineering reason to prefer canonical signs or a native type-theoretic quotient, not a foundational defect.

### 4.4 Sign expansions as the canonical carrier

Sign sequences avoid the quotient at the level of the carrier. For two distinct sequences, compare at their first disagreement, treating termination as an intermediate symbol:

$$- < \text{termination} < +.$$

A set of disagreement positions has a least member, so this is a classical linear order. Each sign sequence has canonical left and right options obtained by truncating immediately before a plus or a minus respectively; these families are indexed by subsets of its ordinal length and are therefore sets, and they reconstruct the sequence as the simplest separator. This is a particularly clean foundation for birthdays, initial embeddings and no-size-collapse theorems [9, 13].

*Remark 4.2* (Simplicity is not birthday precedence). Write  $x \prec_s y$  for “ $x$  is a proper initial segment of  $y$ ” and  $x \prec_b y$  for “ $b(x) < b(y)$ .” The second does not imply the first: the sequence  $+$  is shorter than  $--$  but is not its initial segment (M3). Initial segment extension is the simplicity relation, *not* the numerical order (M2). This distinction returns in section 15.5, where the birthday-precedence relation of a recent research announcement must be kept apart from sign-prefix simplicity.

*Remark 4.3* (The ordinal embedding is not an arithmetic homomorphism). The ordinal embedding  $\alpha \mapsto$  (all-plus sequence of length  $\alpha$ ) preserves the canonical order but does not identify ordinal arithmetic with field arithmetic: in **No**,  $1 + \omega = \omega + 1$  by commutativity, while in ordinary ordinal arithmetic  $1 + \omega = \omega$  (M2, M3). The positive fact is that on embedded ordinals, surreal addition and multiplication agree with the *natural* (Hessenberg) operations, not generally with the usual noncommutative ordinal operations (M3) [8]. A formal library must use separate operations with explicit comparison results rather than treating an ordinal coercion as a homomorphism.

The tradeoff is that addition, multiplication and the normal-form theorem are less immediate on strings. A productive formalization uses both descriptions and proves an order-preserving equivalence between numeric-game quotients and sign sequences. The Mizar development discussed in section 18 is an established example of using more than one presentation rather than requiring one representation to serve every proof [37]; its existence is also evidence that a proof assistant need not have a primitive surreal inductive-inductive constructor to formalize the classical theory (M2).

## 4.5 Hahn fields as workspaces, not a circular definition

A Hahn field  $\mathbb{R}((t^\Gamma))$  can be constructed from an independently given ordered abelian group  $\Gamma$ . Its coefficients, support and convolution have direct set-sized definitions, which makes Hahn fields excellent algebraic and analytic workspaces.

The slogan **No**  $\cong \mathbb{R}((\omega^{\mathbf{No}}))$  has a different status. Its exponent class is the object being described, the supports are reverse well ordered and set-sized, and the evaluation map uses the Conway monomial construction. It summarizes a *normal-form theorem*; it is not a nonrecursive set definition that solves its own size problem, and it does not define **No** as an unrestricted set of functions from **No** to  $\mathbb{R}$ . A normal form is a set-coded expression

$$x = \sum_{\xi < \lambda} r_\xi \omega^{a_\xi}, \quad \lambda \in \mathbf{Ord}, \quad r_\xi \in \mathbb{R} \setminus \{0\}, \quad a_\xi > a_\eta \text{ for } \xi < \eta, \quad (4.6)$$

whose coding data — the ordinal  $\lambda$ , the coefficient function and the exponent function — are sets, even though exponent *values* range through the class **No**. It is not an arbitrary proper-class function **No**  $\rightarrow \mathbb{R}$ . Existence and uniqueness of (4.6) constitute the Conway normal-form theorem, not a consequence of size bookkeeping [8, 9]. A formal development may use this theorem as a proved bridge; it must not postulate the bridge merely because a Hahn-series type already exists.

In a class theory an individual class-Hahn series should be stored by a set support and a set coefficient function on that support; its zero extension to all surreal exponents is a class function, not a set function with a proper-class domain. In a larger type universe that zero extension may be an ordinary function type, but the small-support condition still has to be imposed (M2). These are different encodings of one size discipline.

## 5 ZF, ZFC, and definable classes

### 5.1 Definable classes require no new objects

In first-order ZFC, “ $x$  is a surreal sign sequence” is a formula and class notation is an abbreviation. A definable class addition is represented by a formula  $\text{Add}(x, y, z)$  together with the theorem that for all surreal  $x, y$  there is exactly one surreal  $z$  satisfying it. A class function  $f : \mathbf{No} \rightarrow \mathbf{No}$  is represented by a formula  $\varphi_f(x, y)$  for which one proves

$$\forall x (\text{Surreal}(x) \Rightarrow \exists! y [\text{Surreal}(y) \wedge \varphi_f(x, y)]).$$

There is no set of all inputs or of all graph pairs, yet each restriction to a set of inputs has a set of outputs by Replacement. Statements about finitely many such operations are ordinary first-order assertions about sets. This suffices for the classical field construction [8, 9].

For an ordinal  $\alpha$  the birthday-bounded collection

$$\mathbf{No}_{<\alpha} = \bigcup_{\beta < \alpha} \{-, +\}^\beta$$

is a set, by Power Set, Separation, Replacement and Union, and every set of sign sequences sits in some such stage. The full  $\mathbf{No}$  is the definable class union of these stages, not a set obtained by applying Union to a nonexistent set of all stages. No inaccessible cardinal is needed for this class construction.

### 5.2 Which axioms do what

The relevant axioms are concrete (M3). Power Set and Separation produce bounded collections of sign sequences and supports. Replacement and Union collect birthdays, coefficient families and recursively generated sets. Foundation supports rank and well-founded induction. Infinity provides the ordinary natural numbers. These are sufficient resources for the usual set-theoretic construction.

Replacement is not decorative (M2): it bounds the birthdays and collects the supports of a set family. Power Set is visible in unrestricted function sets and in the set of all Hahn coefficient functions on a fixed exponent set (M1). Weakening the set theory requires rechecking these steps; even removing Power Set has subtleties, since Replacement and Collection cease to have some of their familiar equivalences [27]. Zermelo set theory without Replacement, Kripke–Platek set theory, intuitionistic ZF and constructive ZF are not interchangeable alternatives: bounded finite-day constructions may require very little, while collection of arbitrary ordinal-indexed supports and the full normal-form machinery needs more.

### 5.3 Schema versus class quantification

ZFC has no variables ranging over classes. A theorem for a *specified definable*  $F$  can use its defining formula in Replacement. A theorem quantified over all class functions is instead interpreted as a schema over definitions, or stated in an explicit class theory. These formulations are related, but their quantifiers are not literally identical, and confusing a schema about definable functions with quantification over all classes can strengthen or weaken a statement unintentionally (M2).

For example, R (M3) takes a class function  $F$  on an infinitesimal thickening, restricts it to an ordinary set  $U$ , and collects  $F(c)$  for  $c \in U$ . In ZFC this is legitimate when the graph of  $F$  is

given by a formula for which the required functionality has been proved. An unformalized phrase “arbitrary external assignment” does not supply that formula. [Section 13.3](#) returns to this step, which is the decisive size step in R’s automatic Hartogs support theorem.

## 5.4 Choice, representatives and normal forms: a choice audit

ZF is classical logic without choice; it is not constructive set theory. Canonical sign sequences, ordinal rank bounds and unique normalized constructions should not be assigned a global-choice dependency merely because they are discussed with classes. The deterministic sign-sequence representation avoids choosing a raw representative from each class. Adding choice permits the usual classical field theory, ordinary complex analysis, and the selection arguments used in the supplied support proofs without a separate weak-choice analysis, which is why ZFC is a practical baseline for the set-sized layers.

The three audits agree on the following itemized audit for this subject.

- The simplest element of a separated surreal cut is unique, so its selection is canonical (M2).
- Unions of supports need no representative choice once normal forms exist (M2); the union of a specified set of supports, its rational span, and coefficientwise extension of a fixed linear map are all canonical (M1).
- Selecting roots of a *set* of polynomials uses ordinary choice (M2). AG’s choice of a complex-linear splitting is a choice concerning a set-sized vector-space surjection (M1).
- Choosing representatives or bases across a proper class of unrelated structures can require global choice (M2). A global family of splittings for *all* workspaces is a different request and should not be added unless actually needed (M1).
- A chosen section of a quotient in Lean often uses classical choice, as the inspected `Surreal.out` does. That is a convenience of the implementation, not a proof that every mathematical account of **No** needs global choice (M1).

None of this entails that every proof using global choice essentially needs it. Conversely, a *particular* proof using arbitrary representatives, a well-ordering of a coefficient field, or extraction of a decreasing sequence from a non-well-ordered set may require a choice principle not analysed here (M3). This report does not claim a sharp reverse-mathematical calibration of every step in normal forms, real closedness, Hahn-field algebraic closedness or analytic duality (M1), and identifying the weakest sufficient subsystem would be a separate investigation (M2).

## 6 Class theories and recursion strength

### 6.1 Fix the convention before invoking a theory name

Names for class theories vary, and all three audits insist that the convention be fixed before the letters are used. Here *GB* means a two-sorted Gödel–Bernays-style theory with set and class variables, elementary class comprehension (class parameters allowed, quantified variables restricted to sets), and the usual class-replacement principle; choice assumptions are stated separately. *GBC* denotes the version with Global Choice and the usual ZFC set part. NBG is a closely related presentation whose naming conventions differ over whether choice or global choice is built in; when translating

A’s “NBG with global choice,” GBC is the intended working convention. None of the comparisons below depends on treating the letters NBG as a guarantee of an unstated choice convention.

A class consists of sets, and only sets can be members. The elements of a proper class are still sets, not other proper classes. Consequently **No**, **SC**, the graph of surreal addition, the graph of a definable global function, the finite elements  $\mathcal{O} \subset \mathbf{No}$ , the infinitesimals  $\mathbf{m} \subset \mathbf{No}$ , the surcomplex unit circle, and the purely infinite part  $\Pi$  are legitimate classes; their individual values remain sets. Elementary class comprehension allows set quantifiers and existing class parameters but not arbitrary bound class quantifiers in the defining formula; class replacement ensures that the image of a set under a class function is a set. These principles are exactly suited to recording a surreal carrier, its operations, a class function  $F$ , and the set of its values on an ordinary domain.

*Remark 6.1* (Second-sort quantification is not a class of classes). The assertion “for every class homomorphism  $\chi : \Pi \rightarrow \mathbf{T}(\mathbf{No})$ ” is a legitimate second-sort quantification. It is *not* the assertion that there exists a class whose elements are all such homomorphisms; the latter would require classes to be elements, hence a higher-order framework or a coding restriction. This is the proper reading of the global phase classification in  $\mathbf{T}(\mathbf{M2})$ , and it is the same discipline that [section 13.2](#) applies to a class-valued sheaf.

*Remark 6.2* (Conservativity, stated exactly). The standard predicative class theories are conservative over the corresponding set theory for sentences about sets; in particular GBC is conservative over ZFC for first-order set sentences. This is a relative proof-theoretic fact. It does *not* make the theories identical: GBC can quantify over classes, and models can have different collections of classes over the same underlying set universe. It must not be strengthened to the assertion that every particular model of ZFC has an expansion to GBC with exactly the same sets, nor to the assertion that every model of ZFC already has a definable global well-order; expansion questions are subtler, especially for global choice [\[21, 22, 20\]](#).

*Related (batch 89)*. Without Replacement, global choice is not conservative: Glazer proved that a global selector, with Separation extended to it, is not conservative over Zermelo set theory with Choice (arXiv:2312.11902). Part XV of the report `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders` (its source 31) refines this to a selector required to choose only from nonempty countable sets (Theorem XV.5.1, `swo:gcz:thm:main`), through an interface to Glazer’s published countermodels that has not been reviewed in this repository, and shows that without expanded Separation such a selector is conservative (Proposition XV.5.5, `swo:gcz:prop:nosep`). In the full-class rank models  $(V_\lambda, \mathcal{P}(V_\lambda))$ , global choice holds at every limit height  $\lambda > \omega$ , while full class Replacement holds exactly at strongly inaccessible heights (Theorem XV.9.3, `swo:gcz:thm:replacement`). None of this changes the remark above, which concerns GBC over ZFC.

## 6.2 The recursion design principle

The words “transfinite” and “class-length” alone do not determine proof-theoretic strength.

**Design principle 6.3** (Audit the output size as well as the index size). A recursion with a proper-class domain can still be a local, set-valued recursion. Conversely, a recursion of ordinal length can ask for a new *class* at each stage and require a substantially stronger principle. The length of the index alone does not determine the strength needed.

For the first kind, ordinary well-founded recursion, rank bounds and class comprehension often suffice: each stage  $\mathbf{No}_{<\alpha}$  is a set, each sign sequence is a set, and the predecessors involved in

constructing the value of a fixed arithmetic expression can be gathered into a set. A set-like well-founded relation — one with only set-many predecessors per element — is the standard favourable case [24]. Operations obtained by agreement of such local recursions do not require an axiom of unrestricted class recursion. M3 makes this precise as the only numbered theorem in the group on recursion strength.

**Proposition 6.4** (Local recursion and coherent assembly). *Suppose every input  $x$  has a set  $D_x$  of dependencies, closed under the predecessors required by a given recursive rule; suppose the predecessor relation on  $D_x$  is well founded, the rule has a unique set-valued output once predecessor outputs are known, and the recursions on different  $D_x$  agree wherever both are defined. Then the value at each input is obtained by set recursion. If the data and rule are definable, the assembled operation is a definable class function; in GB the same argument permits fixed class parameters.*

*Proof.* Perform well-founded recursion on the set  $D_x$ ; its values and graph are sets by Replacement. For two dependency sets, uniqueness of recursively computed values proves agreement by induction on the common required predecessors. Define the global graph by saying that  $y$  is the value at  $x$  in such a local recursion; existence and compatibility make this graph functional. No stage of this argument constructs a class-valued approximation as an element of a sequence of classes.  $\square$

For a raw game, the transitive closure of its coded option tree is set-sized. For the positive-support recursions in P and R (M3), the dependency domain is the explicit set  $E^*$  of section 10.2. These are the relevant localization certificates. Theorem 6.4 is explicitly *not* a blanket certification of every informal class recursion: more elaborate surreal operations still require their actual recursive dependency proofs.

*Related (batch 80).* The report *Lexicographic Orders of Well-Orderings of the Surreal Numbers*, held in this repository at `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/` (labels `swo:`), uses only recursions of the kind covered by theorem 6.4. Each of its four sources says so in its foundational section (its Subsections 5.3, 5.4.4, 5.5.3 and 5.6.2): every transfinite recursion in its proofs runs along the ordinals with a set value at each stage, determined by the set of earlier values and fixed class parameters, and the local solutions are assembled by uniqueness and elementary comprehension; none invokes ETR. An example is the recursion building well-orders of the sets  $V_\alpha$  in the proof of its Theorem 13.1 (`swo:cf:choice`), over GB without set choice. It leaves recursions with class-valued stages to its open questions, such as Research question 113.3 (`swo:n16.3`; numbered 27.3 when this note was written, renumbered by later parts of that report) on iterated residual decompositions, and warns that its set-valued justification does not carry over unchecked to non-set-like class well-orders.

M1 supplies the corresponding audit checklist: a formalization audit should record (i) the domain of recursion, (ii) the codomain of each stage, (iii) the decreasing relation, and (iv) the size of the information inspected at a step.

### 6.3 ETR, class forcing, and two invalid moves

*Elementary transfinite recursion* (ETR) concerns elementary definitions that may recursively produce class-sized sections along class well-orders. Its general availability is genuine extra strength over GBC in its usual forms. Iterated satisfaction and truth constructions are the standard examples: at a stage one may need a relation on all set assignments rather than one set. Even an **Ord**-indexed construction can exhibit this difficulty when the stages are classes, and results on class forcing locate important recursions at  $\text{ETR}_{\text{Ord}}$  [26, 25, 23].

Two moves are therefore invalid, and all three audits reject both:

- “NBG permits transfinite recursion, therefore every recursion along all ordinals is allowed” — the conclusion is false;
- “we must use Kelley–Morse because **No** is a proper class” — the strength is not justified by the size of the carrier.

One must identify the actual recursion and prove that its dependency domains and outputs have the advertised sizes. In particular, invoking ETR for a recursion on one well-ordered Hahn support is unnecessary; conversely, an iteration of class truth predicates is not made a set recursion by using the same word “ordinal.”

## 6.4 What Kelley–Morse adds, and what it still does not

Kelley–Morse (KM, also MK) permits impredicative class comprehension: class quantifiers may occur in the defining formulas. This supports stronger class recursion and satisfaction constructions, and its proof-theoretic and model-theoretic behaviour differs from GBC’s; it is not a conservative notational extension of ZFC [21, 26]. It is a legitimate ambient theory, especially for a project about classes of structures, iterated truth, or sophisticated class recursions.

But KM is still a theory whose classes have sets as members. It does not make proper classes into ordinary members, does not grant unrestricted power classes of all classes, and does not solve the universal-cut contradiction. It does not automatically provide a *hyperclass* whose elements are all proper-class functions  $\mathbf{SC} \rightarrow \mathbf{SC}$ ; if arbitrary collections of such functions are genuinely required, one must add another level, move to larger universes, or work with codes and definable families. Further strengthening by class collection, class choice, or higher-order objects must also be stated separately; variants of second-order set theory differ in these principles [21]. Nothing in the set-sized ring  $\mathcal{A}_{n,\Gamma}$  requires KM solely because its interpretation lies in the surreal class (M1). The present polynomial workspaces, set-supported Hahn recursions, and restriction of a class function to an ordinary set do not by themselves require KM (M3).

## 6.5 Global choice is optional infrastructure, not a cure for size

Global Choice supplies a class-wide choice function, for example when a construction really asks for simultaneous selections across a proper class of nonempty sets. It is useful for globally chosen representatives and uniformly selected algebraic constructions. It cannot make a proper class into a set, it cannot form  $\{\mathbf{No} \mid \emptyset\}$ , and it cannot create the set of all class functions (M1). It is also *not* needed merely to pass from  $\forall r \exists H_r$  to a proof that fixes one arbitrary  $r$  and one witness for that  $r$ , nor to define a graph whose output is already uniquely characterized (M3). These two observations can substantially reduce an unnecessarily strong axiom ledger.

# 7 Universes, reflection, and bounded surreal fields

## 7.1 Grothendieck universes make smallness relative

A Grothendieck universe  $\mathcal{U}$  is a transitive set closed under the usual set operations and appropriately indexed unions; we assume it contains the ordinary infinite sets needed here. In a familiar model,  $V_\kappa$  for a strongly inaccessible  $\kappa$  is such a universe, and a Tarski–Grothendieck universe principle provides

arbitrarily large ones. A set in  $\mathcal{U}$  is called  $\mathcal{U}$ -small; finite products, power sets of members, and unions indexed by  $\mathcal{U}$ -small families of members remain in  $\mathcal{U}$  [20, §8]. This is additional foundational strength, not a disguised theorem of ZFC.

Define externally

$$\mathbf{No}_{\mathcal{U}} = \{s : \alpha \rightarrow \{-, +\} : \alpha < \kappa\}, \quad (7.1)$$

the sign sequences whose lengths and data belong to  $\mathcal{U}$ . From outside, this is a set; from the small world represented by  $\mathcal{U}$  it is the analogue of a proper class. If  $I \in \mathcal{U}$  indexes a family of its elements, the union of their ordinal lengths is still bounded by an ordinal of  $\mathcal{U}$ , so their small Conway cut can be filled inside  $\mathbf{No}_{\mathcal{U}}$ . An externally arbitrary subset of  $\mathbf{No}_{\mathcal{U}}$  need not be an admissible small option set.

Two of the three audits give a numbered proposition here, in *opposite logical directions*. They are not in conflict: one says what  $\mathbf{No}_{\mathcal{U}}$  does, the other says what it cannot be, and the merge keeps both. M3's is positive and inventorial; it justifies the construction.

**Proposition 7.1** (Why universe relativization avoids the cut contradiction). *For  $\mathcal{U} = V_{\kappa}$  with  $\kappa$  strongly inaccessible,  $\mathbf{No}_{\mathcal{U}}$  contains a copy of every ordinal  $\alpha < \kappa$ , has no upper bound in itself, and fills separated cuts indexed by members of  $\mathcal{U}$ . The external set of all its elements is not an element of  $\mathcal{U}$ .*

*Proof.* The constantly positive sign sequence gives each  $\alpha < \kappa$ . Every  $\mathcal{U}$ -small family of birthdays has supremum below  $\kappa$ , by the closure properties of  $\mathcal{U}$ , so the ordinary small-cut construction remains below  $\kappa$ . If  $\mathbf{No}_{\mathcal{U}}$  were itself in  $\mathcal{U}$ , its family of all birthdays would be  $\mathcal{U}$ -small and hence bounded below  $\kappa$ , contradicting the presence of every  $\alpha < \kappa$ . Finally, appending a plus sign to the sign sequence of a proposed upper bound  $x$  gives a larger element still in  $\mathbf{No}_{\mathcal{U}}$ .  $\square$

M1's is contrapositive and negative; it is the form that functions as an interface regression test.

**Proposition 7.2** (The missing self-cut). *If  $\mathbf{No}_{\mathcal{U}}$  admits surreal cuts of all  $\mathcal{U}$ -small separated option families, then  $\mathbf{No}_{\mathcal{U}}$  is not itself  $\mathcal{U}$ -small.*

*Proof.* Otherwise  $L = \mathbf{No}_{\mathcal{U}}$ ,  $R = \emptyset$  would be an admissible cut, and its value would be strictly larger than every element of its own carrier, contradicting [theorem 3.2](#).  $\square$

M2 makes the same point in prose only. One should not say that  $\mathbf{No}_{\mathcal{U}}$  contains “all surreals” without the qualification relative to  $\mathcal{U}$ : enlarging the universe produces more sign sequences and new scales. When  $\mathcal{U} \subset V$ , a formal development should provide an embedding  $\mathbf{No}_{\mathcal{U}} \rightarrow \mathbf{No}_V$ , preserve canonical sign descriptions, and prove compatibility of order, arithmetic and normal forms, rather than identifying the two carriers definitionally; a series with support merely small relative to  $V$  need not correspond to a member of  $\mathbf{No}_{\mathcal{U}}$  (M1). Arithmetic closure at the smaller level is justified by performing the classical construction internally in the transitive model  $V_{\kappa}$ , whose individual inputs, output codes and small option families belong to  $V_{\kappa}$ ; the interpreted field operations agree with the external ones on common inputs by uniqueness of their recursive definitions, and the same method gives the real-closed-field theorem at that level (M1).

*Remark 7.3* (The cost). The existence of a strongly inaccessible cardinal is not a theorem of ZFC, and arbitrarily many universes is stronger still. None of it is needed merely to discuss the definable proper class  $\mathbf{No}$  in ZFC. The inaccessible bound in (7.1) is a convenient sufficient condition, not a claim of minimality. A semantic model of a type theory is also not a theorem that its exact proof-theoretic strength equals that of one familiar large-cardinal axiom (M2, M3, M1).

## 7.2 Tarski–Grothendieck and Mizar

Tarski–Grothendieck set theory supplies sufficiently large universe-like sets for arbitrary starting data, giving a flexible external setting in which one level’s classes become the next level’s sets. It is the foundation of the Mizar tradition, including the surreal formalization discussed in [section 18.7 \[37\]](#). A completed Mizar proof establishes a result *in that formal environment*; extracting a proof in a weaker theory requires dependency analysis and does not follow simply because the informal theorem is usually stated in ZFC (M2). The extra universe strength is not forced by the elementary assertion that  $\mathbf{No}$  is a definable proper class in ZFC (M1).

## 7.3 Feferman reflection without a blanket universe axiom

M1 is the only audit that treats reflection frameworks, and its treatment survives in full. Feferman’s reflection-based smallness frameworks, often written ZFC/S, provide another way to formalize some large-object reasoning: an added smallness constant satisfies appropriate reflection instances. Their metatheoretic conservativity relies on the fact that any one proof uses only finitely many instances. It is *not* the assertion that ZFC proves the existence of one transitive set model satisfying all of ZFC in a single internal satisfaction statement [20, §11].

For the present application this is a possible way to organize finitely specified mathematical work without postulating arbitrarily many inaccessible cardinals. It is not interchangeable with a Grothendieck universe: closure under arbitrary externally specified small-domain families may fail, or may require a definability condition. A normal-form or workspace theorem quantifying over all such families has to preserve that qualification. Replacing a universe by a reflection constant while retaining all of its closure rules would lose the point of the weaker framework.

## 7.4 Birthday cutoffs: the negative fact and the positive theorem

For an ordinal  $\lambda$  put  $\mathbf{No}_{<\lambda} = \{x : \mathbf{b}(x) < \lambda\}$ , a set in ZFC. All three audits record the negative fact: an arbitrary cutoff is not a field. The finite-birthday surreals are exactly the dyadic rationals; they are closed under addition and multiplication but not inversion, since 3 has finite birthday and  $1/3$  does not. Choosing an ordinal bound and declaring the resulting subtype a field is invalid without a closure theorem, and a formalization must not insert a field instance on a birthday subtype merely because its ambient type is a field (M2).

M3 alone supplies the positive result, and it is the group’s only positive theorem on birthday fragments. The corrected work of van den Dries and Ehrlich identifies the cutoffs that *are* fields as the epsilon numbers, the ordinals satisfying  $\omega^\lambda = \lambda$ , and establishes strong closure and elementary-extension properties for the resulting fields [17, 18]. An epsilon number need not be an inaccessible cardinal:  $\varepsilon_0$  is already one. Hence *no large-cardinal universe axiom is needed to produce useful set-sized real-closed surreal fields*. The accompanying erratum must be taken into account when reusing detailed ordinal support estimates from the original paper — a non-claim recorded independently by M3 and M2, the latter adding that exact ordinal closure bounds for exponential subfields are mathematical theorems from the literature, not consequences of notation, and must be ported together with their hypotheses and the published correction [16].

What a birthday cutoff does *not* automatically provide is closure under every family one calls “small” in an ambient set theory. Its own full set of elements cannot have an upper bound inside itself, and the cofinality of the cutoff and the lengths of the input families become explicit constraints.

## 7.5 Closure under a fixed finitary signature

M2 supplies the following positive closure proposition, together with its essential caveat.

**Proposition 7.4** (Closure under a fixed set of finitary operations). *Let  $A$  be a set of surreal codes and let  $\mathcal{F}$  be a uniformly specified set-indexed family of finitary class operations with set values on their domains — more precisely, one class graph encodes the operations together with their set indices and finite arities, the operations themselves not being members of a set. Then there is a set  $B \supseteq A$  closed under every operation of  $\mathcal{F}$  whenever the inputs belong to  $B$ .*

*Proof.* Put  $A_0 = A$ . Given  $A_n$ , adjoin every value of every operation of  $\mathcal{F}$  on tuples from  $A_n$  lying in its domain, obtaining  $A_{n+1}$ ; finite Cartesian powers, Replacement and Union make this a set. Let  $B = \bigcup_{n < \omega} A_n$ . Every finite tuple from  $B$  already lies in one  $A_n$ , so its output lies in  $A_{n+1} \subseteq B$ .  $\square$

For field operations include  $0, 1, +, -, \cdot$  and inversion away from zero; for canonical exponential and logarithmic closure include those particular operations. *This proposition does not produce closure under all set-indexed Hahn sums:* those operations have unbounded arity and their admissible input families are not one fixed finitary signature. A full Hahn field is the more suitable workspace when such sums are central.

## 7.6 Three distinct closure requirements

For a field or carrier  $F$ , M3 distinguishes three requirements that are easily conflated:

1. closure under finitely many arithmetic operations and finite algebraic extensions;
2. closure under specified set-indexed strongly summable families;
3. the availability of a number above or below every member of each permitted small family.

Real closedness supplies only the first. A fixed Hahn field has a natural summation operation for suitable supports, but its value group may have no element above a given unbounded sequence. A universe-relative surreal construction has the third property only for the lower-universe-small families built into its interface. This three-way distinction is indispensable for translating A's full-scale arguments, and it is exactly what [section 11](#) exhibits in two sharp examples.

The three devices solve different problems. A birthday cutoff gives a concrete algebraic subfield with ordinal closure theorems; a Grothendieck universe gives systematic closure under a broad family of constructions; a Hahn workspace gives a specified value group and transparent supports. Their carriers may overlap substantially, but their universal properties are not the same.

## 7.7 External countability versus internal properness

M2 alone records this warning, and it survives in full. A countable model of set theory can internally believe that its surreal collection is a proper class while externally its domain and its internal surreal codes are countable. There is no contradiction: the external enumeration need not be an internal set or class function available in the model. Similarly, “every set family has a bound” means every family represented by a set *in the relevant model*; it is not an absoluteness claim about every family visible from a stronger metatheory. The same applies to normal forms and supports: set membership, well-foundedness, completeness and the collection of all reals must be interpreted in the stated ambient model.

## 7.8 Universes in Lean are not set-theoretic axioms renamed

Lean’s universe levels are part of its type-formation rules. They are not variables ranging over a Lean datatype of inaccessible cardinals. A set-theoretic model using inaccessible cardinals is a sufficient semantic account of a hierarchy of universes, but it is *not* a theorem that the kernel literally assumes the corresponding ZFC large-cardinal sentences (M1). Exact proof-theoretic comparisons require a specified type theory, universe rules, inductive principles and metatheory. The practical rule is simpler: never collapse a larger universe just to remove an inconvenient type mismatch [40]. [Section 16](#) develops the type-theoretic side in detail.

## 8 Surcomplexification: finite algebra, no new size principle

### 8.1 Pairs with the correct multiplication

For an ordered field  $F$  define  $F[i]$  on pairs by

$$(a, b) + (c, d) = (a + c, b + d), \quad (8.1)$$

$$(a, b)(c, d) = (ac - bd, ad + bc), \quad (8.2)$$

$$\overline{(a, b)} = (a, -b), \quad i = (0, 1), \quad 1 = (1, 0). \quad (8.3)$$

Then  $i^2 = (-1, 0)$ , and if  $(a, b) \neq (0, 0)$  ordered-field positivity gives  $a^2 + b^2 > 0$  and

$$(a, b)^{-1} = \left( \frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right). \quad (8.4)$$

The field construction needs only an *ordered* field, not yet real closedness: the denominator is positive whenever the pair is nonzero. All three audits make this point and prove the following, which we print once.

**Proposition 8.1** (Finite data complexify safely). *The pair construction produces a field from any ordered field  $F$ , and an algebraically closed field when  $F$  is real closed. Its carrier is a set whenever  $F$  is a set and a class of set-coded pairs whenever  $F$  is a class field of set-coded elements. The same coordinate formulas define the surcomplex class field from the surreal class field.*

*Proof.* The ring identities are finite polynomial identities, and (8.4) is verified by direct multiplication using the nonzero denominator. The equivalence between real closedness of  $F$  and algebraic closedness of  $F[i]$  is the standard real-closed-field theorem. For a class carrier, each pair and each polynomial coefficient list is still a set-coded finite object, and the graphs of the operations are class relations defined by the same formulas; pairing and finite products introduce no new proper-class membership operation.  $\square$

Equivalently, in a set-sized workspace one may form  $F[X]/(X^2 + 1)$ , whose irreducibility follows because an ordered field has no square equal to  $-1$ . This is an ordinary ideal quotient; there is no need to form a quotient whose members are proper-class equivalence classes. The quotient method integrates well with generic field-extension libraries; the pair method makes conjugation, real and imaginary parts and the modulus more transparent.

“**SC** is algebraically closed” means: every nonconstant polynomial given by a finite coefficient list in **SC** has a zero in **SC**. It does not assert the existence of a set of all such polynomials (M1). Real closedness of **No** is a substantive construction theorem; once available, adjoining  $i$  introduces no new paradox and no class-level choice.

*Remark 8.2* (Not the product ring). The carrier  $F \times F$  with its *default componentwise multiplication* is a product ring with zero divisors:  $(1, 0)(0, 1) = (0, 0)$ . Its identity is  $(1, 1)$ , not  $(1, 0)$ . It is not  $F[i]$ . A proof-assistant implementation must use a dedicated structure, a type synonym with carefully transported operations, or the quadratic-extension construction, and must not reuse the default product-ring instance. All three audits flag this; M3 states it explicitly as a proof-assistant trap, and it appears as a row in the regression table of [section B](#).

## 8.2 Real closed is not algebraically closed

No nontrivial ordered field is algebraically closed, because  $X^2 + 1$  has no root in it. Thus “**No** is real closed” and “**SC** is algebraically closed” must not be merged. There is likewise no order on  $F[i]$  making it an ordered field with the same multiplication: in an ordered field a nonzero square is positive, whereas  $i^2 = -1$ . Statements about a “real” component of a surcomplex number mean an element of **No** and must not be confused with ordinary  $\mathbb{R}$  (M2).

This is also an audit rule for documentation: an abstract or a comment containing “algebraically closed surreal field” does not override the elementary obstruction, nor does it identify a formally proved theorem. M2 and M3 both flag exactly this wording in the abstract of the published Mizar development; see [section 18.7](#).

## 8.3 The modulus is base-field-valued, not a real norm

For  $z = (a, b)$  over a real closed  $F$  put

$$|z| = \sqrt{a^2 + b^2} \in F_{\geq 0}.$$

Multiplicativity  $|zw| = |z||w|$  follows from the sum-of-two-squares identity. The triangle inequality reduces to

$$(ac + bd)^2 \leq (a^2 + b^2)(c^2 + d^2), \quad (8.5)$$

whose difference is the square  $(ad - bc)^2$  (M1); equivalently it is the two-dimensional Gram identity

$$(ac + bd)^2 + (ad - bc)^2 = (a^2 + b^2)(c^2 + d^2), \quad (8.6)$$

which M2 ships as a Lean one-liner proved by `ring` (uncompiled in the archive; compiled here, depending on `propext` and `Quot.sound` — see [section 2.6](#)). These are order-valued estimates, valid over every real-closed field at infinite as well as infinitesimal scales, and they need no reconstruction from Conway cuts.

For surcomplex numbers the modulus takes *surreal* values. It is not a real-valued norm satisfying the hypotheses of Lean’s ordinary `NormedField` infrastructure, and replacing an infinite modulus by a real number would change the notion of error tolerance. Do not install incompatible order or norm instances merely to reuse classical-analysis lemmas. The geometry and the topology must be formalized using their actual scalar field.

## 8.4 Conjugation, modulus and valuation are three separate structures

M3 alone supplies the separator that makes this precise, and it survives here. In a Hahn workspace the natural additive valuation is

$$v\left(\sum_{\gamma} a_{\gamma} t^{\gamma}\right) = \min \operatorname{supp}(a), \quad v(0) = +\infty,$$

with an ordered-group-valued target, and it is ultrametric. The modulus is  $F$ -valued and satisfies an ordinary triangle inequality. Even the targets differ. Concretely:

$$2t^\rho \text{ lies in the valuation ball } v(z) \geq \rho, \text{ but not in the order disk } |z| \leq t^\rho. \quad (8.7)$$

A surcomplex implementation should therefore expose a conjugation map, a positive quadratic form or modulus, and a valuation as *independent* pieces of structure. An ordinary complex-number type specialized to  $\mathbb{R}$  should not be assumed to support an arbitrary real-closed coefficient field without checking its hypotheses.

## 8.5 The Hahn complexification isomorphism

For a set-sized ordered abelian group  $\Gamma$ , real and imaginary coefficients give a natural isomorphism

$$\mathbb{R}((t^\Gamma))[i] \cong \mathbb{C}((t^\Gamma)). \quad (8.8)$$

The union of two well-ordered supports in a linear order is well ordered, so a pair of real Hahn series determines a complex Hahn series; conversely taking real and imaginary parts only shrinks support. The coefficientwise maps preserve addition, and the finite convolution formulas preserve (8.2). If  $\Gamma$  is divisible, the characteristic-zero Hahn-field theorem gives algebraic closedness of  $\mathbb{C}((t^\Gamma))$  [11], and hence real closedness of  $\mathbb{R}((t^\Gamma))$ .

This is a local theorem about a *set* field. Embedding that field into **SC** requires the surreal normal-form theorem and compatibility with  $t^\gamma = \omega^{-\gamma}$ ; *the mere existence of an abstract Hahn field does not establish the embedding* (M1).

## 9 Set-sized workspaces and theorem transfer

### 9.1 The localization theorem

Write  $t^\gamma = \omega^{-\gamma}$ , as in the supplied manuscripts, so that a Hahn support is well ordered in the *increasing*  $t$ -exponent order — the reverse of the customary decreasing-powers-of- $\omega$  convention. For a set-sized divisible ordered additive subgroup  $\Gamma \subseteq \mathbf{No}$  put

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)).$$

The normal-form theorem embeds these fields in **No** and **SC**, and the classical Hahn-field theorem makes  $K_\Gamma$  algebraically closed and hence  $F_\Gamma$  real closed [8, 9, 11]. These embeddings and closure statements are substantive prerequisites for localization, not consequences of the notation  $((t^\Gamma))$ .

All three audits prove the localization theorem by unioning normal-form supports and taking the  $\mathbb{Q}$ -span. We print the proof once, in M3's three-clause form, which is the strongest statement any of them reaches.

**Theorem 9.1** (Workspace localization). *Assume existence and uniqueness of set-supported surreal normal forms, with evaluation preserving coefficientwise addition and Hahn convolution and sending  $t^\gamma$  to  $\omega^{-\gamma}$ . Assume also the algebraic-closure theorem for Hahn fields with divisible exponent group and algebraically closed coefficient field. Then, in the full-class interpretation:*

1. *every set  $A \subseteq \mathbf{SC}$  is contained in one algebraically closed set-sized field  $K_\Gamma$ , for some nonzero divisible set-sized ordered subgroup  $\Gamma \subseteq \mathbf{No}$ ;*

2. any set of such workspaces is contained in a larger one;
3. a nonzero polynomial that splits in one workspace acquires no new roots in a larger surcomplex extension.

For  $\mathcal{U} = V_\kappa$  as in [theorem 7.1](#), the same statements hold inside  $\mathbf{SC}_\mathcal{U} = \mathbf{No}_\mathcal{U}[i]$  when the input family in (1), and the family of  $\mathcal{U}$ -small workspaces in (2), are indexed by members of  $\mathcal{U}$ . The resulting  $\Gamma$ ,  $F_\Gamma$  and  $K_\Gamma$  are  $\mathcal{U}$ -small. An arbitrary external set of elements of  $\mathbf{SC}_\mathcal{U}$  is not sufficient for this relative conclusion.

*Proof.* For (1), take for each  $a = x + iy \in A$  the union of the exponent supports of the normal forms of  $x$  and  $y$ , with signs reversed to match  $t^\gamma = \omega^{-\gamma}$ . Replacement over  $A$  followed by Union gives a set  $S$  of all required exponents. Inside the additive ordered group of  $\mathbf{No}$  put

$$\Gamma = \text{span}_{\mathbb{Q}}(S \cup \{1\}) = \left\{ \sum_{j=1}^n q_j s_j : n \in \mathbb{N}, q_j \in \mathbb{Q}, s_j \in S \cup \{1\} \right\}.$$

Finite lists from a set form a set, and their images under finite sums form a set, so  $\Gamma$  is a set by Replacement. It is nonzero, divisible, and inherits the order from  $\mathbf{No}$ , and every normal form used for an element of  $A$  has support in  $\Gamma$ . The cited normal-form and Hahn-field theorems give the inclusion  $A \subseteq K_\Gamma \subseteq \mathbf{SC}$  with  $K_\Gamma$  algebraically closed.

For (2), take the union of the exponent groups of the given workspaces and repeat the rational-span construction. The coefficient-extension maps agree with normal-form evaluation, so this gives inclusions of the specified workspaces, not merely abstract isomorphic copies.

For the universe-relative version, each input sign sequence and its normal-form support have codes in  $\mathcal{U}$ . Regularity of  $\kappa$  bounds the ranks of the codes in a  $\mathcal{U}$ -small indexed family strictly below  $\kappa$ ; their union is therefore in  $\mathcal{U}$ . The finite-list and rational-span constructions above stay in  $\mathcal{U}$ . Since  $\Gamma, \mathbb{R}, \mathbb{C} \in \mathcal{U}$ , the coefficient-function sets  $\mathbb{R}^\Gamma$  and  $\mathbb{C}^\Gamma$ , and their Hahn subfields, also belong to  $\mathcal{U}$ . The internal normal-form theorem evaluates them in  $\mathbf{No}_\mathcal{U}$  and  $\mathbf{SC}_\mathcal{U}$ . This is the step that fails for an externally arbitrary family whose support codes can have ranks cofinal in  $\kappa$ .

For (3), if  $P(X) = a \prod_{j=1}^n (X - \alpha_j)$  with  $a \neq 0$  in one field, the same equality holds in every extension field. At a root  $\beta$  the finite product is zero, so  $\beta = \alpha_j$  for some  $j$ . There is no additional root location.  $\square$

*Remark 9.2* (The companion non-claim). M1 attaches the explicit non-claim that [theorem 9.1](#) does not say that a single set field contains all surcomplex numbers, and does not provide a set of workspaces covering the whole class: such a union would be a set by Replacement and Union, contrary to [theorem 3.1](#). The two statements are compatible — clause (2) enlarges a *set* of workspaces, which is a different quantifier from covering a proper class. M2 adds that the proof presupposes the normal-form theorem and is not a new elementary proof of it, and that an abstract ordered group  $\Gamma$  has no canonical embedding into  $\mathbf{No}$  merely because the notation  $t^\Gamma$  is written; the canonical embedding above uses the actual inclusion  $\Gamma \subseteq \mathbf{No}$ .

[Theorem 9.1](#) is the foundational content of P's workspace proposition (M3), made explicit as a dependency theorem, and it already supports finite polynomial factorization, finite matrices, Hermite interpolation and the residue-pairing constructions without passing a proper-class ring to an ordinary algebraic-geometry functor.

## 9.2 Enlargement as a system of compatible maps

For  $\Gamma \subseteq \Delta$ , extend a coefficient function by zero outside  $\Gamma$  to obtain an injective field homomorphism

$$j_{\Gamma\Delta} : K_{\Gamma} \hookrightarrow K_{\Delta}.$$

These maps satisfy identity and composition laws and preserve addition, multiplication, order on the real part, valuation with the corresponding target embedding, and coefficient extraction. Strong sums are preserved when the same summability certificate is transported. The same coefficientwise construction gives maps on  $\mathcal{A}_{n,\Gamma}$ , formal-series coefficient rings, and finite polynomial or matrix data (M1).

*Remark 9.3* (Not a small diagram). The collection of all set-sized exponent subgroups of the full surreal class is not a set-indexed diagram. Its “direct limit” is therefore not automatically an instance of a library theorem about colimits of small diagrams (M3). Either the union is definable in class theory, or one fixes a larger universe in which the index category is an actual type, or one states compatible local theorems without forming the colimit as a single object (M1). This prevents a well-intended categorical reformulation from reintroducing the original size problem.

The safe pattern for globalizing a local theorem is: gather the parameters into a workspace; apply the theorem there; if a conclusion quantifies over a proposed additional surcomplex point, enlarge again to contain that point and the original parameters; and use a proved compatibility or descent theorem to compare the two conclusions. Mere inclusion of carriers is not a proof of such compatibility.

## 9.3 A sound localization argument has two halves

All three audits converge on the same warning, stated most crisply by M2: to reduce a theorem to a workspace one must first prove that all input data fit in it, and *then* prove that every construction used by the proof stays there or specify a controlled enlargement. [Theorem 9.1](#) supplies the first half only.

Finite identities descend particularly well: a polynomial has finitely many coefficients, a matrix finitely many entries, a quotient-algebra computation finitely many structure constants, and one common workspace contains all of them. A support recursion can also be set-sized even if it produces infinitely many coefficients, provided its entire input and recursion support is a set. Polynomial factorization stays in a divisible algebraically closed Hahn workspace.

What does not descend automatically: a statement about *every* point of **SC** is not equivalent to its restriction to one  $K_{\Gamma}$ ; a positive-dimensional algebraic variety can acquire new points over a larger field even when its defining equations do not change; a function defined separately on different infinitesimal monads can behave outside the chosen workspace in ways its restriction does not determine; and a radius required to dominate every member of an unbounded subset of  $\Gamma$  may lie outside  $\Gamma$  altogether. A rescaling that dominates every exponent in a coefficient sequence need not stay in the workspace — which is exactly [theorem 11.1](#).

A safe descent proof therefore names the mechanism: persistence of a polynomial identity; invariance of a finite root multiset; base change of a finite free algebra; coefficientwise uniqueness; or an explicitly applicable model-theoretic transfer theorem. “The problem has finitely many symbols” is not a replacement for that argument when those symbols include quantification over a full function class. Statements about all sequences, all new analytic functions or all neighbourhoods should never be globalized by deleting a subscript  $\Gamma$  from a theorem.

Embedding compatibility is algebraic but not topological (M2): in  $\mathbb{R}((t^{\mathbb{Q}}))$  the sequence  $t^n$  tends to 0 in the intrinsic valuation topology, while after embedding into the full surreal field the same sequence does not converge in the full fine topology, whose radii include scales below all its terms. A library must never synthesize the latter property from the former.

#### 9.4 Real-closed-field quantifier elimination as a restricted tool

M3 alone develops this, and it survives in full. For a fixed finite degree bound, complex polynomial identities and inequalities can be encoded by pairs of real coordinates, and root counts with multiplicity can be expressed using a finite factorization list and finite case distinctions. Quantifier elimination for real-closed fields then permits transfer of such first-order statements between real-closed workspaces [19]. P’s order-circle polynomial Rouché theorem is a scheme indexed by a fixed ordinary degree bound; its hypotheses quantify over pairs of field coordinates on an algebraically specified circle, and the count is encoded by a finite root list rather than by a quantifier over arbitrary functions or subsets. That makes the transfer legitimate in  $F_{\Gamma}$  once coefficients, centre and radius have been localized (M3).

The inventory of what the field language does *not* contain is the point. It does not distinguish the ordinary integer subset from arbitrary field elements, and it contains no symbol for

- birthday,
- simplicity,
- Hahn support,
- a particular exponential,
- a predicate for ordinary holomorphic functions.

It therefore cannot automatically transfer an assertion about all Taylor coefficients, all ordinal stages, or all coherent charts. Adding such symbols changes the theory and requires new arguments. In particular, pure real-closed-field axioms cannot characterize the canonical surreal simplicity hierarchy — the same point made from the axiomatic side in [section 15](#).

#### 9.5 Finite algebra over an algebraically closed base

The following is M1’s, and it is the only substantial commutative-algebra theorem in the group. It explains why finite algebraic zero sets do not acquire new points at larger surreal scales.

**Theorem 9.4** (Finite algebra over an algebraically closed base). *Let  $K \subseteq L$  be fields with  $K$  algebraically closed, and let  $B$  be a finite-dimensional commutative unital  $K$ -algebra. Then every  $K$ -algebra homomorphism  $\phi : B \rightarrow L$  has image in  $K$ , and the finite local lengths of  $B$  are preserved after extension to  $L$ .*

*Proof.* For  $b \in B$ , multiplication by  $b$  is a linear operator on the finite-dimensional space  $B$ . Its characteristic polynomial  $p_b \in K[T]$  annihilates  $b$ , by Cayley–Hamilton applied to multiplication and then evaluated at 1. Hence  $p_b(\phi(b)) = 0$ . The polynomial splits over  $K$ , and since  $L$  is a field,  $\phi(b)$  must equal one of its roots in  $K$ .

For the length statement, decompose the Artinian algebra as  $B = \prod_j B_j$  with each  $B_j$  local. Its residue field is a finite extension of  $K$ , hence  $K$ . If  $\mathfrak{m}_j$  is its nilpotent maximal ideal, then  $\mathfrak{m}_j \otimes_K L$

is nilpotent and the quotient of  $B_j \otimes_K L$  by it is  $L$ ; thus the base-changed algebra is still local. Moreover  $\dim_L(B_j \otimes_K L) = \dim_K B_j$ . Since both residue fields equal the corresponding base field, these dimensions are the local algebraic lengths. The zero algebra is a harmless empty-product case with no points.  $\square$

*Remark 9.5* (Finite base change is not global tensor equality). To apply [theorem 9.4](#) to AG (M1) one must *first* establish the specific base-change isomorphism

$$\mathcal{A}_{n,\Delta}/(F) \cong K_\Delta \otimes_{K_\Gamma} (\mathcal{A}_{n,\Gamma}/(F)) \quad (9.1)$$

for the finite complete-intersection deformation under consideration. AG obtains it by using the same finite ordinary basis and support-controlled multiplication constants in both workspaces. *Neither theorem 9.4 nor AG's theorem asserts  $\mathcal{A}_{n,\Delta} = K_\Delta \otimes_{K_\Gamma} \mathcal{A}_{n,\Gamma}$  for the entire infinite-dimensional analytic ring.* Finite quotient compatibility is a much more specific statement, and the distinction matters when choosing a Lean abstraction for workspace extension.

Once (9.1) is available, [theorem 9.4](#) explains why any proposed full-surcomplex zero already has coordinates in  $K_\Gamma$ . The theorem needs no global surreal metric space and no theory of fine-continuous paths; it is generic algebra, and [section 19.8](#) recommends implementing it before any analytic topology.

## 9.6 Characters and actual analytic fibers

R (M3) explicitly avoids assuming that an arbitrary algebra homomorphism commutes with an infinite Hahn sum. Instead it proves finite identities such as

$$H(w) - H(\xi) = \sum_j (w_j - \xi_j) G_j(w) \quad (9.2)$$

inside the coherent ring; a character sending  $w_j$  to  $\xi_j$  must then send  $H$  to  $H(\xi)$  by an ordinary finite homomorphism calculation. This is a model formalization pattern: *prove a finite identity that carries the intended semantic information* instead of silently assigning continuity or strong-sum preservation to every homomorphism. It appears as a row in the regression table of [section B](#).

Similarly, a finite algebra  $K[X_1, \dots, X_d]/I$  can be treated with ordinary ideal quotients, finite-dimensional modules and localization; a statement about Spec should use that set-sized ring and should not treat the full class field as an object of a small category of commutative rings without a universe specification.

## 9.7 The residue pairing

M3 alone records the following, and it is a strong candidate for a generic reusable theorem before any full-class analytic layer exists. For monic  $P$  of degree  $n$  over a field  $K$ , the perfect residue pairing on  $K[X]/(P)$  uses only finite remainders and a finite Gram matrix. With

$$\lambda_P(H) = [X^{n-1}](H \bmod P), \quad (9.3)$$

the matrix in the power basis has ones on its antidiagonal and zeros on one side of that antidiagonal; reversing columns makes it triangular, and its determinant is  $(-1)^{n(n-1)/2}$ . This argument depends neither on surreal representations, nor on a valuation, nor even on  $K$  being algebraically closed.

By contrast, an initial-polynomial root count needs a splitting field, the valuation and residue maps, and compatibility of those maps with finite polynomial operations. Those assumptions must not be silently discharged by a bare `Field Surreal` instance; P's explicitly algebraically closed Hahn workspace is the intended source of the stronger hypotheses (M3).

## 10 Hahn support as a logical interface

### 10.1 Three independent certificates

A Hahn construction requires separate answers to three questions (M3). Is the data indexed by a permitted set or small type? Is the resulting support well ordered? Does each coefficient receive only finitely many contributions? None of these follows from the other two in the generality needed here.

For a set-indexed family  $(A_j)_{j \in J}$  the manuscripts define *strong summability* by

$$S = \bigcup_{j \in J} \text{supp}(A_j) \text{ is well ordered,} \quad \{j : \gamma \in \text{supp}(A_j)\} \text{ is finite for every } \gamma, \quad (10.1)$$

and the sum has coefficient  $\sum_j [t^\gamma] A_j$ , a finite sum after the second condition is applied. Zero coefficients can be discarded, so the actual support of the result is a subset of  $S$ . All three audits state both clauses and stress that neither implies the other. The second clause cannot be replaced by a lower valuation bound, and positivity alone cannot replace well-ordering: the positive set  $\{1/n : n \geq 1\} \subset \mathbb{Q}$  is not well ordered (M1).

**Example 10.1** (Two failed simplifications). Each series  $t^{1/n}$ ,  $n \geq 1$ , has singleton support, but the family is not strongly summable because  $\{1/n : n \geq 1\}$  is not well ordered in the increasing exponent order. Conversely the constant family  $A_n = 1$  has well-ordered union of supports  $\{0\}$ , but infinitely many terms contribute at exponent zero. It is not a formal Hahn sum either.

An ordinary convergent series of complex constants may have a perfectly good complex sum while failing the second condition in (10.1); that operation belongs to the coefficient field's analysis, not to formal Hahn summation, and mixing the two requires an explicit additional definition.

### 10.2 The positive-support mechanism and $E^*$

If  $E$  is a well-ordered set of strictly positive elements of an ordered abelian group, let

$$E^* = \{e_1 + \cdots + e_k : k \in \mathbb{N}, e_j \in E\},$$

including the empty sum. The Hahn–Neumann lemma says that  $E^*$  is well ordered and that each exponent has only finitely many representations by nondecreasing finite words in  $E$  [12]. Combined with a well-ordered input support  $S$ , this gives a controlled support  $S + E^*$ .

M3 isolates *two different uses* of this lemma, and the distinction is load-bearing. It proves that the output expression denotes a Hahn series; and it proves that recursive coefficient equations really involve finite sums. A proof that checks only well-ordering has not yet justified the coefficient calculation; a pointwise finite formula does not guarantee that the union of its output exponents is well ordered.

For instance, when  $B$  has positive support the formal geometric identity  $(1 + B)^{-1} = \sum_{k \geq 0} (-B)^k$  is justified by the support lemma and coefficientwise cancellation, with no metric convergence

theorem applied. For AG's positive-support operator  $T$  (M1), the useful certificate is not that valuations tend to increase but that the expansion of  $T^r G$  has contributions indexed by

$$s + e_1 + \cdots + e_r = \gamma, \quad s \in S, \quad e_j \in E \subset \Gamma_{>0},$$

with  $S$  well ordered and  $E$  positive and well ordered; the Hahn–Neumann lemma gives the well-ordered union  $S + E^*$  and finitely many contributing words at each exponent. This licenses coefficientwise telescoping for  $\sum_{r \geq 0} (-T)^r G$ , rearrangement of residues, and comparison with a finite ordinary parameter family. A proof of a trace identity at a fixed coefficient should retain the finite contributing data explicitly; it must *not* claim that there are finitely many exponents below that coefficient, which in a higher-rank Hahn group can be false.

### 10.3 A support-local nonlinear recursion

M3 alone proves the following, which isolates the common mechanism in P's Hensel factorization and R's preparation and matrix division. The isolation is useful because it removes both surreal birthdays and ordinary complex topology from the recursion proof.

**Proposition 10.2** (Support-local nonlinear recursion). *Let  $A$  be a commutative coefficient ring,  $m \in \mathbb{N}$ ,  $V = A^m$ , and let  $\Gamma$  be an ordered abelian group. Fix a set  $E \subseteq \Gamma_{>0}$  that is well ordered in the increasing exponent order. Let  $L : V \rightarrow V$  be an invertible  $A$ -linear map, and let  $N : V \rightarrow V$  be given by finitely many polynomials over  $A$ , each with no terms of total degree zero or one. Extend these maps coefficientwise and by finite Hahn convolution, respectively. If  $X$  is a  $V$ -valued Hahn series with  $\text{supp } X \subseteq E$ , then*

$$LY + N(Y) = X$$

*has a unique positive-support Hahn solution  $Y$ , and  $\text{supp } Y \subseteq E^* \setminus \{0\}$ .*

*Proof.* Use well-founded recursion on  $E^* \setminus \{0\}$ . Its well-ordering follows from the stated hypotheses on  $E$  and the Hahn–Neumann lemma. For each polynomial monomial of degree  $d \geq 2$ , a contribution at exponent  $\nu$  uses positive exponents  $\nu_1, \dots, \nu_d \in E^*$  with  $\nu_1 + \cdots + \nu_d = \nu$ . Each  $\nu_j$  is strictly smaller than  $\nu$ . There are finitely many such tuples: expand their entries into words in  $E$ ; the Hahn–Neumann lemma gives finitely many nondecreasing words with total  $\nu$ , and each finite word has finitely many permutations and partitions into  $d$  nonempty groups. There are also only finitely many polynomial monomials and coordinates. Thus the following coefficient is a finite sum of already constructed data:

$$Y_\nu = L^{-1}(X_\nu - [t^\nu]N(Y_{<\nu})).$$

The constructed family is a Hahn family because its support is a subset of the well-ordered set  $E^*$ , and exponents outside that set receive no input and no contribution from the constructed nonlinear products; thus the equation holds coefficientwise.

For uniqueness among positive-support solutions, take the least exponent where two candidate vectors differ; their finitely many component supports have well-ordered union. A difference in a nonlinear coefficient at that exponent would require a difference at a strictly smaller exponent, so the nonlinear differences vanish there, and injectivity of  $L$  forces the alleged first difference to vanish as well.  $\square$

**Corollary 10.3** (Fixed domain). *If  $A$  is a ring of holomorphic functions on one ordinary complex domain and  $L^{-1}$  preserves that ring, the same proof keeps every coefficient on the original domain.*

That is the crucial fixed-domain property in R (M3). The proof uses set recursion on  $E^*$ , finite polynomial convolution and a supplied inverse  $L^{-1}$ ; it requires no Noetherian completion theorem, no fine-topological limit, and no ETR axiom. It is the concrete instance of [theorem 6.4](#).

## 10.4 Formal summability is not tsum

The two source examples are retained below. A third shows that rank one alone does not identify strong summation with topological summation.

**Example 10.4** (The geometric series that never enters the neighbourhood). Let  $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$  and consider  $1 + t + t^2 + \dots$ . It is a valid Hahn sum, equal to  $(1 - t)^{-1}$ . But the tail after the  $N$ -th partial sum has valuation  $N + 1 < \omega$ , so the partial sums never enter the valuation neighbourhood demanding an error of valuation greater than  $\omega$ ; they do not converge to that sum in this higher-rank valuation topology. AG uses this example explicitly to rule out unjustified numerical-contraction arguments (M1).

**Example 10.5** (Natural-number iteration need not reach the required scale). Let the exponent group contain 1 and an element  $\Omega$  greater than every integer. The strongly summable series  $H = \sum_{n \geq 0} t^n$  has a remainder beginning at  $t^{N+1}$  after  $N$  terms, and that remainder is *not* smaller than  $t^\Omega$  for any ordinary  $N$ . So finite partial sums need not converge in a topology whose neighbourhoods include that scale (M3).

**Example 10.6** (Strong summation can fail to be a limit even in rank one). In  $\mathbb{C}((t^\mathbb{Q}))$  the family  $A_n = t^{n/(n+1)}$ ,  $n \geq 0$ , is strongly summable: its union of supports is a strictly increasing sequence, hence well ordered, and every exponent occurs once. Let  $H$  be its formal sum. For  $P_N = \sum_{n=0}^N A_n$  one has

$$v(H - P_N) = \frac{N+1}{N+2} < 1.$$

Thus no partial sum lies in the neighbourhood  $\{z : v(H - z) > 1\}$ . Even the net of all finite subsums misses that neighbourhood, since some exponent below 1 is always omitted. Rank one alone therefore does not identify arbitrary strong sums with topological sums. What matters for a particular family is whether its remaining exponents eventually escape every valuation bound.

*Related (batch 32).* These examples concern particular valuation topologies. The companion report *Omnific-Preserving Automorphisms*, held in this repository at `docs/surreal/omnific-preserving-automorphisms/`, proves the general statement for full Hahn fields in its part on topological collapse: for a field  $k$  and a set-sized ordered abelian group  $\Gamma$ , the field  $k((t^\Gamma))$  admits a Hausdorff ring topology in which every Hahn-summable family converges to its sum along the net of its finite partial sums if and only if  $\Gamma = 0$  or  $\Gamma \cong \mathbb{Z}$ ; and on **No** and on **No**[ $z$ ] the only such ring topology is the indiscrete one. [Theorem 10.6](#) shows the failure for the valuation topology with  $\Gamma = \mathbb{Q}$ ; that theorem excludes every Hausdorff ring topology.

*Remark 10.7* (The Archimedean-argument warning). An argument asserting that  $n\delta$  eventually exceeds every group element is an *Archimedean* argument and fails in [theorem 10.5](#) (M3). This is why P's coprime factor lifting is naturally formulated as well-founded recursion through a support monoid rather than as an unspecified limit of Newton iterates, and it is why formal Hahn summability and topological summability need different interfaces.

A formal implementation should therefore have a dedicated admissible-family structure and a coefficientwise sum operation. Mathlib's `HahnSeries.SummableFamily` supplies exactly this kind of

infrastructure, separately from topological summability; see [section 18.6](#). Choosing a default value for a sum outside its admissible domain, as some APIs do for totality, does not justify the intended sum equation without its support or positive-order hypothesis (M1).

## 10.5 The full surreal exponent type needs a further restriction

At a fixed type-theoretic level, a coefficient function on the entire surreal type can have a well-ordered support that is too large to be a normal form for a surreal at that same level. The support must be *small relative to the universe used to construct those surreals*, not merely a subtype of their larger carrier. The direct set-theoretic analogue is the proper-class expression

$$\sum_{\alpha \in \mathbf{Ord}} t^\alpha, \quad (10.2)$$

which has a well-ordered class of exponents, each receiving one coefficient, and is nevertheless not a surreal normal form because its support is not a set. Well ordering and finite coefficient dependence do not repair the missing size condition.

M2 states the type-theoretic form of the same obstruction: a generic `HahnSeries Surreal R` with no small-support restriction admits series too large to represent a surreal at the chosen level, since the family of singleton monomials indexed by every ordinal of the smaller universe has individually small disjoint supports whose union is not small. *A proof that the union is well ordered does not prove its sethood*. That is exactly why a small-support subtype is needed, and the inspected Lean library does introduce one; see [section 18.4](#).

## 11 Two series over $\mathbb{C}((t^{\mathbb{Q}}))$ with opposite signs

### 11.1 The two series, side by side

This section contains the two sharpest examples in the report. They live over the *same* field  $K = \mathbb{C}((t^{\mathbb{Q}}))$ , with ordered real part  $F = \mathbb{R}((t^{\mathbb{Q}}))$ , and they differ only in the sign of an exponent:

$$A(X) = \sum_{n \geq 0} t^{-n^2} X^n, \quad (11.1)$$

$$f(z) = \sum_{n \geq 0} t^{+n^2} z^n. \quad (11.2)$$

Their conclusions are opposite in every respect. [Equation \(11.1\)](#) is strongly summable at *no* nonzero argument of  $K$ , because its term valuations  $-n^2 + nq$  are unbounded *below* in  $\mathbb{Q}$ ; it becomes summable only after the value group is enlarged. [Equation \(11.2\)](#) is strongly summable everywhere in  $K$ , is Hahn-coherent at every positive radius  $r \in F$ , and is nonpolynomial.

|                               | $A(X) = \sum t^{-n^2} X^n$                          | $f(z) = \sum t^{+n^2} z^n$                  |
|-------------------------------|-----------------------------------------------------|---------------------------------------------|
| Field                         | $K = \mathbb{C}((t^{\mathbb{Q}}))$                  | $K = \mathbb{C}((t^{\mathbb{Q}}))$          |
| Exponent sign                 | minus                                               | plus                                        |
| Term valuations at $v(h) = q$ | $-n^2 + nq$ , unbounded <i>below</i>                | $n^2 + n\sigma$ , unbounded <i>above</i>    |
| Strongly summable in $K$ ?    | at no nonzero argument                              | at every argument                           |
| Repair                        | enlarge to $\Delta = \mathbb{Q} + \mathbb{Q}\omega$ | none needed                                 |
| Conclusion                    | fixed workspace too poor to <i>evaluate</i>         | fixed workspace too poor to <i>rule out</i> |
| Audits                        | M1 and M2                                           | M3                                          |

**Read this before reading the proofs.** These are *two different series*, not two variants of one example. The sign of the exponent is the whole difference, and it reverses every conclusion. A reader who conflates them will come away believing that this report contradicts itself, or will quote the wrong one. [Section 11.6](#) states the relation between them explicitly.

## 11.2 The rescaling obstruction, with the minus sign

M1 and M2 both prove the following, by the same valuation computation. We print it once.

**Proposition 11.1** (Universal rescaling fails in a fixed Hahn field). *Over  $K = \mathbb{C}((t^{\mathbb{Q}}))$  the formal series  $A(X) = \sum_{n \geq 0} t^{-n^2} X^n \in K[[X]]$  cannot be evaluated by strong summation at any nonzero  $h \in K$ .*

*Proof.* Let  $q = v(h) \in \mathbb{Q}$ . The leading exponent of  $t^{-n^2} h^n$  is  $-n^2 + nq$ , and the successive difference is  $q - 2n - 1$ , which is eventually negative. So for all sufficiently large  $n$  these exponents strictly decrease, without a lower bound in  $\mathbb{Q}$ . Each lies in the support of its term, so the union of term supports contains an infinite strictly decreasing sequence and is not well ordered. This violates the first clause of (10.1).  $\square$

The conclusion is *negative*, and it is about A: the small-scale evaluation theorem of the analysis manuscript does not survive restriction to a fixed Hahn workspace, and no nonzero radius in this field has the universal small-scale property asserted for the full class. In the full surreal field one may instead choose an exponent  $\Delta$  dominating the set of coefficient supports and use  $\rho = t^{\Delta}$ ; the separated-band proof in A uses precisely that additional freedom (M2). A formal port to  $K_{\Gamma}$  must either assume an appropriate dominating exponent in  $\Gamma$  or enlarge the workspace. It must not assert the full-class theorem uniformly inside every fixed Hahn field, and no formalization should weaken A's rescaling theorem to the assertion that every series has a usable nonzero argument in whatever field happened to contain its coefficients (M1).

M1 alone supplies the *positive half*, and it survives here in full.

**Proposition 11.2** (The enlarged group repairs it). *After enlarging the exponent group to  $\Delta = \mathbb{Q} + \mathbb{Q}\omega \subset \mathbf{No}$ , evaluation of  $A$  at  $h = t^{\omega}$  is strongly summable.*

*Proof.* The exponents are  $n\omega - n^2$ , with consecutive difference  $\omega - (2n + 1) > 0$  since  $\omega$  exceeds every integer. They therefore strictly increase, and exactly one term sits at each exponent, so both clauses of (10.1) hold.  $\square$

M2 proves only the negative half. [Theorem 11.2](#) is what makes [theorem 11.1](#) a genuine *obstruction* justifying A's permission to *enlarge the workspace*, rather than a defect in A.

### 11.3 All-scale polynomial rigidity, in support-theoretic form

M3 reconstructs the core of A’s polynomial-rigidity argument in a form small enough to audit. Suppose a function has Taylor coefficients  $a_n$  at zero and a coherent chart at every positive surreal radius  $r$ ; for each  $r$  the supports of  $a_n r^n$  lie in one well-ordered chart support. If infinitely many  $a_n$  are nonzero their valuations form a *set*, so one may choose  $\Delta > 0$  with

$$|v(a_n)| < \Delta/4 \quad (a_n \neq 0). \quad (11.3)$$

At  $r = t^{-\Delta}$ , successive nonzero coefficients indexed by  $m > n$  satisfy

$$v(a_m r^m) - v(a_n r^n) = v(a_m) - v(a_n) - (m - n)\Delta < 0. \quad (11.4)$$

Their valuations form an infinite strictly decreasing sequence inside one well-ordered support: a contradiction. Coefficientwise identity then identifies the function with its finite Taylor polynomial on every chart.

The foundational requirement is the ability to dominate this *whole set* of valuations. The proof does not need a global choice of charts for every radius at once — it fixes the one radius constructed from the hypothetical infinite Taylor support — but it does need that radius to be available in the same ambient full-scale structure. A universe-relative formulation retains the argument for small coefficient families. *A fixed arbitrary Hahn field cannot supply it*, and the next subsection shows why not.

*Remark 11.3* (Two unrelated theorems are called “rigidity”). The word “rigidity” names two different things in the merged literature, and this report never uses the bare word. The result reconstructed here is *all-scale polynomial rigidity*, an analytic statement taken from A. A different result, *sign-tree categoricity* ([theorem 15.1](#)), is a foundational statement due to M2 about the uniqueness of realizations of the sign-tree specification. They share no content. Each occurrence of one carries a pointer to the other.

### 11.4 The nonpolynomial internally coherent example, with the plus sign

M3 alone supplies the following, and it is the sharpest mathematical asset in the group.

**Example 11.4** (Internal all-scale coherence need not force a polynomial). Let  $K = \mathbb{C}((t^{\mathbb{Q}}))$ , with ordered real part  $F = \mathbb{R}((t^{\mathbb{Q}}))$ . Define, using strong Hahn summation,

$$f(z) = \sum_{n \geq 0} t^{n^2} z^n \quad (z \in K). \quad (11.5)$$

This is a *nonpolynomial* function possessing a Hahn-coherent chart on the whole ordinary complex plane at every positive radius  $r \in F$ .

*Support proof.* Write a nonzero input as  $z = ct^{\sigma}(1 + \eta)$  with  $\sigma \in \mathbb{Q}$ ,  $c \in \mathbb{C}^{\times}$ , and  $\eta$  of positive support. The base exponents  $n^2 + n\sigma$  tend upward and are strictly increasing for all sufficiently large  $n$ ; the finite initial portion causes no obstruction. They therefore form a well-ordered set. Moreover each base exponent comes from at most *two* indices, since  $n^2 + n\sigma = m^2 + m\sigma$  with  $n \neq m$  forces  $n + m = -\sigma$ . Finite binomial expansion of  $(1 + \eta)^n$  has support contained in the positive monoid generated by  $\text{supp}(\eta)$ . The support lemma of [section 10.2](#) therefore proves strong summability of (11.5); the value at zero is 1.

Apply the same argument with  $z = r$  to see that the coefficient family  $t^{n^2} r^n$  is strongly summable. Regrouping

$$\sum_{n \geq 0} t^{n^2} r^n w^n$$

by Hahn exponent gives an *ordinary polynomial in  $w$*  at each exponent, because only finitely many indices contribute there. These entire coefficient polynomials have one well-ordered support and define the claimed chart, and evaluation at finite  $w$  agrees with  $f(rw)$  by the same finite-word support calculation.

Finally, at  $r = 1$  the coefficient at exponent  $n^2$  is  $w^n$ . A finite-degree polynomial over  $K$  has a uniform bound on the degrees of all of its ordinary coefficient polynomials, so coefficient uniqueness on ordinary points rules out equality with  $f$ . Hence  $f$  is nonpolynomial.  $\square$

*Remark 11.5* (Why the rigidity proof cannot run here). The argument of [section 11.3](#) needs a  $\Delta$  in the value group satisfying (11.3) for *all*  $n$ . Here  $v(a_n) = n^2$ , which is unbounded in  $\mathbb{Q}$ . No such  $\Delta$  exists in  $\mathbb{Q}$ . The value group  $\mathbb{Q}$  simply contains no exponent dominating all the values  $n^2$ .

## 11.5 Exact scope of the plus-sign example: a sharpness result, not a refutation

This point must not be compressed. [Theorem 11.4](#) is a **sharpness result**. It does *not* refute the all-scale polynomial rigidity theorem of the companion report *Surcomplex Analysis: germs, coherent Hahn sections, and infinitesimal zero geometry*, held in this repository at `docs/surcomplex/analysis/` [6]. The two statements have different hypotheses and both are true:

- That companion theorem is stated for **class functions**  $f : \mathbf{SC} \rightarrow \mathbf{SC}$  on the whole of  $\mathbf{No}[i]$ . Its hypothesis quantifies over a *proper class* of multiplicative scales, and that proper class is precisely what supplies the dominating exponent  $\Delta$  that (11.3) requires. Over the full surreal scale system the dominating exponent always exists.
- [Theorem 11.4](#) is stated over a **fixed value group**, namely  $\mathbb{Q}$ . There the dominating exponent does not exist ([theorem 11.5](#)), and the conclusion genuinely fails: a nonpolynomial function satisfies the internal all-scale condition.

So the example shows that the all-scale rigidity theorem is *specific to the full-class scale system* and does not transfer to a fixed Hahn field. That is its entire content. M3 states this itself, in its own words: the example “does not refute A: the value group  $\mathbb{Q}$  has no exponent dominating all the values  $n^2$ , and A explicitly quantifies over the larger surreal scale system.” M3 further records, and we carry verbatim, that the example “is an elementary comparison supplied here,” that it “is not attributed to an unpublished theorem in the attached manuscripts,” and that *no priority claim and no machine-verification claim is made for it*. Nothing in the present report attaches a priority claim to it either, and nothing here should be read as a correction to the companion report; that report is cited by title and repository path, and none of its internal labels is referenced.

## 11.6 Why the two are complementary, not contradictory

[Theorem 11.1](#) and [theorem 11.4](#) concern *different series*. One has exponent  $-n^2$  and the other  $+n^2$ , over the same field, and the sign of the exponent is the whole of the difference between them. They are not variants of one example, and neither is a strengthening or weakening of the other.

They are two *different non-transfer theorems* about the same fixed field, pointing in complementary directions:

- [Theorem 11.1](#) says a fixed workspace may be too poor to **evaluate** a given series. The exponents run downward out of the group, and the series has no admissible argument at all. The repair ([theorem 11.2](#)) is to enlarge the value group.
- [Theorem 11.4](#) says a fixed workspace may be too poor to **rule out** a nonpolynomial all-scale-coherent function. The exponents run upward out of reach of any single dominating element of the group, so the rigidity argument has no  $\Delta$  to work with, and the classification theorem that holds over the full class fails over  $\mathbb{Q}$ .

Both are consequences of one fact about  $\mathbb{Q}$ : it is a value group in which neither an unbounded-below family nor an unbounded-above family can be dominated. Both are negative results about *transfer*, not about the source manuscript. Neither refutes A. Together they are the sharpest available evidence for the report’s recommendation that summability and ambient scale system be separate explicit hypotheses, never connotations of the word “surreal.”

## 12 Topology: full-class, intrinsic, coefficientwise

### 12.1 A class neighbourhood relation is enough

A’s fine topology uses balls

$$B(a, r) = \{z \in \mathbf{SC} : |z - a| < r\}, \quad r \in \mathbf{No}_{>0},$$

with radii ranging over positive *surreals*. An individual ball is generally a proper class. In class set theory one defines openness of a given class  $V$  by a formula — for every  $a \in V$  there is  $r > 0$  with  $B(a, r) \subseteq V$  — so there is no need to form a set, or a class with class members, of all open subclasses. Continuity and local analytic statements are formulated through the same neighbourhood relation. On a set-sized workspace, of course, an ordinary topology is a set and the usual definition applies (M2).

In Lean a universe-relative surreal carrier is an actual type, so an ordinary `TopologicalSpace` on that type is available in principle, and its subsets are Lean predicates. This does *not* mean that every such subset is small in the lower universe used for the cut construction. The apparent discrepancy disappears once the two meanings of “set” are distinguished (M3).

### 12.2 The degeneracy theorem, in its strongest form

All three audits prove that small subsets are discrete. M1’s version has two clauses (subsets, convergent nets); M2 and M3 both add a third clause about Cauchy nets, proved by collecting the set of nonzero *pairwise* distances rather than distances to a limit. We print the strongest statement, with the Cauchy clause credited to M2 and M3, and their common proof once.

**Theorem 12.1** (Small subsets and small-index nets are discrete). *Assume [theorem 3.3](#) for the permitted small families, and give the carrier the ball topology with radii in the positive part of its own scalar field. Then:*

1. *every set-sized (respectively permitted-small) subset is closed and discrete;*
2. *every convergent net with a set-sized (permitted-small) directed index set is eventually equal to its limit;*

3. every Cauchy net with such an index set is eventually constant.

*Proof.* Let  $A$  be such a subset and  $p$  a point. The nonzero distances  $D_p = \{|p - a| : a \in A, a \neq p\}$  form a set of positive surreals, so by [theorem 3.3](#) choose  $\delta > 0$  smaller than each of them. Then  $B(p, \delta) \cap A$  is empty if  $p \notin A$  and is  $\{p\}$  otherwise. This proves closedness and discreteness.

For a convergent net, its range is a set; apply the same construction to the nonzero distances of its terms from the proposed limit. Eventual membership in the resulting ball forces eventual equality.

For a Cauchy net, instead collect the nonzero distances  $|z_j - z_k|$  between all pairs of its terms; that is still a set. Choosing  $\delta$  below all of them, the Cauchy condition for tolerance  $\delta$  makes a tail have all pairwise distances zero, hence constant.  $\square$

This is the precise size-dependent argument underlying A’s topological degeneracy theorem, with its essential size quantifier retained (all three audits). A Lean theorem quantifying over arbitrary `Set Surreal` without a smallness hypothesis would say something stronger and false (M2). It does *not* make the whole full-class topology discrete: every ball contains a point distinct from its centre, for example  $a + r/2$ , and the carrier’s own point set is not one of the sets to which the theorem applies. At a universe-relative level the correct statement is: every *lower-universe-small* subset is discrete in the full topology of the larger carrier. It is false with the smallness qualifier removed, since the carrier is a subset of itself and has no isolated points. This is one of the most important statement changes required in a faithful Lean translation (M3).

*Related (batch 90).* Part XII of the report `Algebra/SurrealNumbers/docs/foundations-and-computation/polish-models-of-omnific-arithmetic` (its source 18, batch-90 manuscript 03; AI-assisted, unrefereed) restates clauses 1 and 2 for sets of surreals in the native order topology of the proper class `No` (its Proposition 163.4(ii)–(iii), `pma:gcm:prop:fine`), crediting the Lean file `SignSequenceTopology.lean`, and adds three observations: every nonempty open interval is a proper class, every continuous map from a connected set space into `Nom` is constant (its Proposition 163.4(i) and (iv); compare Corollary [12.6](#) below), and `Nom` is totally separated by clopen subclasses (its Proposition 163.5, `pma:gcm:prop:fineseparated`). Those three are not formalized.

## 12.3 Two sharp boundaries: the index and the ambient

[Theorem 12.1](#) has two independent sharpness statements, from two different audits. Neither is implied by the other, and both are needed.

**Example 12.2** (The larger-index net counterexample). Let  $D = F_{>0}$  and direct it by *reverse* numerical order:  $r \preceq s$  means  $s \leq r$ . Define  $x_r = r$ . For any  $\epsilon > 0$ , once  $r \preceq s$  with  $r = \epsilon/2$ , one has  $0 < x_s = s \leq \epsilon/2 < \epsilon$ . Hence this net converges to zero and is never zero.

In class theory  $D$  is a proper class and the net is class-indexed. In Lean, with  $F = \mathbf{No}_u : \mathbf{Type}_{u+1}$ ,  $D$  is a perfectly legitimate type in the larger universe but is not a permitted  $u$ -small index type. There is no conflict with [theorem 12.1](#). The engineering consequence is stated by M2 exactly as it should be carried: *replacing “sequence” by “net” does not repair a missing size annotation unless the index universe is also specified*. This example is the exact complement to the discreteness theorem on the index side.

**Proposition 12.3** (Intrinsic and full-class subspace topologies differ). *Let  $\Gamma$  be nonzero, divisible and set-sized, and let  $K_\Gamma \subset \mathbf{SC}$  be its workspace. Its subspace topology induced by the full-class fine neighbourhoods is discrete. Its intrinsic field-valued ball topology, with radii in  $(F_\Gamma)_{>0}$ , is not discrete.*

*Proof.* The first conclusion follows from [theorem 12.1](#), since  $K_\Gamma$  is a set. For the second, write  $F_\Gamma = \mathbb{R}((t^\Gamma))$ , so  $K_\Gamma = F_\Gamma[i]$ ; every intrinsic ball  $B(0, r)$  with  $r \in (F_\Gamma)_{>0}$  contains the nonzero element  $r/2$ . Hence zero is not isolated intrinsically.  $\square$

This is not a contradiction about subspace topology. The intrinsic field does not contain the arbitrarily small *external* radii used to isolate its points — which is the concrete content of the earlier remark that the  $\delta$  of [theorem 3.3](#) need not lie in any previously specified subfield. The intrinsic valuation topology is likewise nondiscrete for a nontrivial ordered exponent group, since there are monomials of arbitrarily high valuation *within that group*; indeed  $t^n \rightarrow 0$  intrinsically in  $\mathbb{R}((t^\mathbb{Q}))$  but not finely (M2). The canonical coefficient-extension embeddings of workspaces preserve algebra and strong sums but need not preserve the intended ambient topology in the naive subspace sense. Formal theorems about continuity, limits or derivatives must specify which topology is meant before invoking a library result.

**Which intrinsic topology?** For these nontrivial workspaces, intrinsic order-modulus balls and intrinsic valuation balls generate the *same* topology, although the individual balls are different as [equation \(8.7\)](#) shows. Write  $V_\gamma = \{z : v(z) > \gamma\}$ . If  $r \in (F_\Gamma)_{>0}$  and  $\rho = v(r)$ , then  $V_\rho \subseteq B(0, r)$ : higher valuation means a smaller positive leading magnitude. Conversely, given  $\gamma \in \Gamma$ , choose  $\eta > \gamma$ , possible because  $\Gamma \neq 0$ . Then  $B(0, t^\eta) \subseteq V_\gamma$ , since  $|z| < t^\eta$  implies  $v(z) \geq \eta > \gamma$ . On the complex field the same argument uses  $v(|z|) = v(z)$ : in  $|z|^2 = x^2 + y^2$  the first nonzero real coefficients are squares, so the leading terms cannot cancel. Translation gives the comparison at every centre. None of these comparisons identifies the topology with the *external* fine subspace topology.

For a geometric family this gives an exact convergence criterion. If  $\delta \in \Gamma_{>0}$ , the strong sum  $H = \sum_{n \geq 0} t^{n\delta} = (1 - t^\delta)^{-1}$  always exists, and

$$v\left(H - \sum_{n=0}^N t^{n\delta}\right) = (N+1)\delta.$$

Consequently its partial sums converge intrinsically if and only if the multiples  $n\delta$  are cofinal in  $\Gamma$ : for every  $\gamma \in \Gamma$  some  $n$  satisfies  $n\delta > \gamma$ . This holds for every  $\delta > 0$  in an Archimedean ordered group such as  $\mathbb{Q}$ , but fails for  $\delta = 1$  in  $\mathbb{Q} + \mathbb{Q}\omega$ . The criterion concerns this geometric family; [theorem 10.6](#) explains why rank one does not suffice for an arbitrary strongly summable family.

The consequence for Lean is direct (M1): if a native type is used as the whole carrier and every Lean subset is declared to satisfy the small-subset theorem, taking the subset to be the whole carrier forces discreteness. To express the full-class theorem at level  $u$  one must restrict it to subsets with  $u$ -small underlying type while the ambient  $\mathbf{No}_u$  or  $\mathbf{SC}_u$  lives at level  $u+1$ . Alternatively one can express the theorem inside  $\mathbf{ZFSet}$ , where an internal set and a Lean predicate class are already distinguished.

## 12.4 No countable ball basis, hence no real metric

M2 alone supplies the following, and its engineering consequence is the most directly actionable statement in this section.

**Lemma 12.4** (No countable ball basis is coinital). *The full fine topology is not discrete — every ball around zero contains half its radius — yet its small subsets are discrete. No countable ball basis at zero is coinital.*

*Proof.* A countable local basis would have a positive radius  $r_n$  inside each of its members. The radii  $\{r_n : n \in \mathbb{N}\}$  form a set, so by [theorem 3.3](#) choose  $\delta$  with  $\delta < r_n/2$  for every  $n$ . Then  $r_n/2$  lies in the  $n$ -th basic ball but  $r_n/2 \notin B(0, \delta)$ , so no basic ball is contained in  $B(0, \delta)$ .  $\square$

Ordinary real metrics always have a countable local basis, so *no ordinary real metric represents the fine topology*. The precise engineering consequence: Mathlib’s metric typeclasses are the wrong target, while `TopologicalSpace` and filters are correct, since they are not restricted to real metrics. This also prevents importing the ordinary intuition that pointwise isolation on all small samples makes the whole space discrete.

## 12.5 The fine derivative, over the actual scalar field

M2 gives the derivative in the only form that is correct here:

$$\forall \epsilon \in F_{>0} \exists \delta \in F_{>0} \forall h, \quad 0 < |h| < \delta \implies \left| \frac{f(z+h) - f(z)}{h} - d \right| < \epsilon, \quad (12.1)$$

where  $\epsilon$  and  $\delta$  range over the *actual scalar field*  $F_{>0}$ , not over  $\mathbb{R}_{>0}$ ,  $f$  is defined on a neighbourhood of  $z$ , and the chosen ball of increments is required to stay in the domain. The derivative is unique by ordered-field estimates. This definition requires no real-valued norm, no sequence limit and no derivation on the scalar field.

*Remark 12.5* (Four notions that must be distinct types). A function constant on each infinitesimal coset can be globally nonconstant and have derivative zero everywhere, so the usual implication “zero derivative on a connected domain implies constant” cannot be reused with an ordinary connected base domain and a class-sized surreal thickening. A supplied analysis manuscript imposes common-domain Hahn coherence precisely to obtain stronger global conclusions. In a formalization, `FineHasDerivAt`, a formal-series germ, a Hahn-coherent datum and a scalar-field derivation must be *distinct types or predicates* (M2). Proving implications among them is useful; identifying them by notation is not.

## 12.6 Ordinary complex contours live in a third structure

The ordinary complex parameter domain  $U \subseteq \mathbb{C}^d$  retains its usual topology and analytic structure. A Hahn-coherent datum is a family of ordinary holomorphic functions on this common domain indexed by a well-ordered support; its contour integral in A or R integrates ordinary coefficient functions and then forms a Hahn series. It is not an integral along a nonconstant fine-continuous map from a real interval into the full surcomplex class.

M3 makes the reason a corollary of [theorem 12.1](#), and this survives as a distinctive item.

**Corollary 12.6** (No nonconstant fine-continuous paths). *The range of any real-parameter map into SC is a set, hence discrete by [theorem 12.1](#). A continuous image of a connected real interval in a discrete space has at most one point. Therefore there are no nonconstant fine-continuous paths, and the contour formalism necessarily lives in a third structure, distinct from both the full fine topology and the intrinsic  $K_\Gamma$  topology.*

Similarly, the global fine-analytic logarithm and the nonunique fine-analytic exponentials described in  $\mathbf{A}$  do not contradict classical complex analysis; their local neighbourhoods and global coherence conditions differ from the ordinary ones. A foundational formalization should preserve this distinction rather than silently replacing  $\mathbf{A}$ 's functions with ordinary holomorphic maps into a normed field.

## 12.7 Set-coded germs versus quotients of class-function graphs

M1 alone treats the representation question for germs.  $\mathbf{A}$  describes fine-analytic germs using equivalence of functions on sufficiently small fine balls. As a semantic description this is natural; taken literally as a quotient whose representatives are arbitrary class-function graphs, it is not automatically an ordinary NBG class quotient of sets.

A safe implementation uses *descriptions*. A formal coefficient sequence  $(a_n)$ , a centre, and any needed admissibility data form a set code. Evaluation is a definable class relation. Equality of two descriptions is eventual equality of their evaluations on a smaller ball, expressed by a formula quantifying over points and radii. One then uses canonical formal series if uniqueness of expansion has been proved, or Scott codes ([theorem 4.1](#)) if a more general quotient is needed. No equivalence class of arbitrary proper-class graphs is used as an element.

$\mathbf{A}$ 's theorem identifying fine-analytic germs with  $\mathbf{SC}[[X]]$  can support this representation, but it must not be used circularly to justify the objects needed to prove that very theorem: begin with formal-series codes and their evaluation relation, prove the local uniqueness statement, then identify them with the intended semantic germs. The resulting  $\mathbf{SC}[[X]]$  is a *class* of set-coded sequences unless a workspace or universe has been fixed.

For a fixed class function  $f : \mathbf{SC} \rightarrow \mathbf{SC}$ , the assertion

$$\forall r \in \mathbf{No}_{>0} \exists s [s \text{ codes a suitable chart for } f(r \cdot)]$$

quantifies over set-valued radii and set-coded charts with  $f$  as a class parameter. This is compatible with elementary class comprehension when the chart condition has only set quantifiers; it needs no set of all radii or all charts. If a proof needs a simultaneously selected class function  $r \mapsto s_r$ , that is a further choice or definability question.  $\mathbf{A}$ 's all-scale argument can reason about a chart at a specified radius without first choosing all charts globally, and a formal development should retain that economical existential formulation.

## 13 Coefficient rings, sheaves, and Hartogs coherence

### 13.1 Four rings that must not be merged

M1 alone distinguishes the following four objects, and the distinction survives in full with both of its worked separators.

$$\mathcal{H}_\Gamma(U) = \mathcal{O}(U)((t^\Gamma)), \tag{13.1}$$

$$\mathcal{A}_{n,\Gamma} = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma)), \tag{13.2}$$

$$\mathcal{F}_{n,\Gamma} = \mathbb{C}[[z_1, \dots, z_n]]((t^\Gamma)), \tag{13.3}$$

$$K_\Gamma[[z]] = \text{formal power series over the Hahn field.} \tag{13.4}$$

In (13.1) all coefficient functions are defined on *one specified* ordinary open set  $U$ . In (13.2) every coefficient is an ordinary convergent germ, but different coefficients need *not* have a common radius. In (13.3) no ordinary convergence is required at all. All three are set-sized for fixed set  $\Gamma$ .

*First separator.* Taking germs of common-domain families does not produce (13.2). AG's example

$$\sum_{r \geq 1} \frac{t^{r\eta}}{1 - rz}, \quad \eta > 0,$$

has coefficient poles approaching zero, so there is no common ordinary disk on which all coefficients are holomorphic; it nevertheless belongs to  $\mathcal{A}_{1,\Gamma}$ . Formalizing AG by imposing a single uniform analytic norm on every coefficient would strengthen its hypotheses and lose this example.

*Second separator.* Rearrangement of a double family does not identify (13.4) with the preceding rings. The identity

$$-\sum_{r \geq 0} t^{-(r+1)\eta} z^r = (z - t^\eta)^{-1}$$

is valid in  $K_\Gamma[[z]]$ , but its combined Hahn support is not well ordered, so it is *not* an element of  $\mathcal{A}_{1,\Gamma}$ , where  $z - t^\eta$  has an actual infinitesimal zero. A proof assistant should expose this distinction in the types, not erase it by a convenient coercion.

### 13.2 Sheaves: two different questions, two answers

Two audits treat the sheaf question, and they are answering different questions. Both answers survive, and the framing below is the point.

**M1: the size of the totality of sections.** A describes a class-valued sheaf over the ordinary complex analytic base. This can be read in GBC without a set of all classes. Let  $U$  range over ordinary open subsets of  $\mathbb{C}$  and let  $s$  be a *set code* for an admissible Hahn section. Define a single class relation  $\text{Section}(U, s)$ , and another class relation coding restrictions; the sheaf axioms are then expressed using set-indexed covers and compatible set-coded families. Each section code has set-sized support, while the totality of section codes over a fixed  $U$  may be a proper class. One must not instead take an arbitrary class function as a *member* of a section set. M1 attaches the non-claim explicitly: this encoding is *a clarification of a workable reading of A, not a proof of its analytic gluing theorem*; the latter still requires the coefficientwise overlap argument and A's exact coherence hypotheses.

**M3: the shape of an individual section.** A's gluing theorem on a *connected* ordinary domain obtains a common support by a strong identity argument: on overlapping connected open sets a holomorphic coefficient that is nonzero cannot vanish identically on a nonempty open overlap, so exact supports agree along the connected overlap graph. On a *disconnected* open set A deliberately defines sections componentwise, and M3 shows this convention is *necessary*, not a convenience.

**Example 13.1** (The disjoint-disks necessity example). Let  $U$  be a union of pairwise disjoint ordinary disks  $D_n$ , and prescribe the constant section  $t^{1/n}$  on  $D_n$ . Each local datum has singleton support, and all overlap conditions are vacuous. A definition insisting on *one* globally well-ordered support for the whole disconnected  $U$  would not admit this glue, since  $\{1/n : n \geq 1\}$  is decreasing. The componentwise definition does admit it.

For fixed  $\Gamma$  one formalizes this as a set- or type-valued sheaf with

$$\mathcal{H}_\Gamma(U) = \prod_{V \in \pi_0(U)} \mathcal{H}_\Gamma(V), \quad (13.5)$$

the factors on connected components using the global-support convention; the component index is an ordinary set. Alternatively one may use an appropriate sheafification and prove agreement on connected domains. For the full surreal exponent class, use set-coded individual sections and a schema of gluing assertions, or move to a specified higher universe. One should not invoke an ordinary category whose objects are literally all proper classes without supplying its foundations.

**The relation between the two.** M1’s class-relation encoding addresses the *size* of the collection of sections; M3’s componentwise convention addresses the *shape* of an individual section. M1 says nothing about disconnected bases; M3 says nothing about the size of the totality of sections. They are compatible, and M3’s necessity example is a constraint binding on any encoding *including* M1’s: an implementer who adopts the  $\text{Section}(U, s)$  relation must still define sections componentwise over  $\pi_0(U)$ .

### 13.3 Hartogs coherence: where class replacement matters

M3 alone audits R’s automatic Hahn–Hartogs theorem, and the analysis survives in three parts.

*The decisive size step.* R begins with a class function on a full product of thickenings. Its decisive step is

$$A = \bigcup_{c \in U} \text{supp}(F(c)), \quad (13.6)$$

where  $U$  is an ordinary complex domain and therefore a set. Class replacement makes the family of values  $F(c)$  a set; another replacement and union make  $A$  a set. This is exactly the use of class parameters supported by GBC. In ZFC the same reasoning is a *scheme* for definable  $F$  — see [section 5.3](#).

*The analytic step.* Once that is justified, the coefficient family  $(f_\gamma)_{\gamma \in A}$  is set-indexed, and the classical separate-holomorphy theorem supplies joint holomorphicity. The automatic-support argument then excludes a decreasing sequence of globally active exponents by choosing a point where countably many nonzero holomorphic coefficient functions are simultaneously nonzero. That step uses a Baire-category argument and an appropriate classical choice principle to extract the decreasing sequence.

*The step that is not another replacement argument.* The final extension from ordinary tuples to arbitrary *surcomplex* tuples is *not* another replacement argument. It proceeds one coordinate at a time using slice coherence and coefficientwise identity, and its hypothesis must hold when all the other fixed coordinates are surcomplex, not merely ordinary complex. Weakening this hypothesis changes the theorem: arbitrary values at tuples with several nonordinary coordinates would otherwise remain unconstrained.

A Lean specification should therefore expose a function on the relevant universe-relative type, prove that its values on the small ordinary domain have collectively small support, and then state the ordinary Hartogs and Baire lemmas independently. *Availability of those exact ordinary analytic lemmas in the chosen dependency revision must be independently checked*; the existence of a Hahn-series type does not provide them.

### 13.4 Support certificates are the key reusable interfaces

M1’s summary of this section is the right implementation advice. The support lemmas of [section 10](#) are ideal formalization targets: they are independent of global surreal topology, and once proved they support many later constructions. The proof pattern they enable is: replace the finitely many

retained coefficient germs by a finite ordinary analytic family; use the classical identity for that family; compare finitely many coefficients; conclude the Hahn coefficient identity. This is the formal counterpart of AG's finite-parameter testing argument, and it does not require a single ordinary contour for the original infinite family.

## 14 Trigonometry, angles, and global phases

*This entire section rests on a single audit and a single source. Only M2 was given the trigonometry manuscript T [3], and no other audit in this merge read it. Every statement below is tagged to that one reading.*

### 14.1 The finite-angle coset group passes the size check

The finite-angle group  $\mathcal{O}_{\mathbb{R}}/(2\pi\mathbb{Z})$  of T deserves a separate size check (M2), and it passes for a reason that the raw-name equivalence classes of [section 4.3](#) do not: *each coset  $\theta + 2\pi\mathbb{Z}$  is a set, because it is indexed by  $\mathbb{Z}$* . The totality of these cosets can therefore be a class of sets. This is the clean contrast with the class-quotient trap: there, zero had presentations of unbounded rank and its equivalence class was a proper class; here every fibre is countable.

One may avoid the quotient altogether by using the unit-circle group, or by choosing the unique representative in  $[0, 2\pi)$ . Two warnings travel with the latter choice. First, *the interval  $[0, 2\pi)$  is itself a proper class in the full surreal universe*: boundedness by ordinary reals does not imply sethood, since even the interval of infinitesimals contains too many distinct scales to be a set. Second, the chosen interval convention has an actual *branch jump*; it must not be mistaken for a globally analytic argument function.

### 14.2 Omnific integers, the finite-angle character, and its kernel

In the full surreal field the omnific integers have the normal-form description

$$\mathbf{Oz} = \Pi \oplus \mathbb{Z}, \quad (14.1)$$

where  $\Pi$  consists of the purely infinite normal forms. For the finite-angle cis, put

$$U_0(p + u) = \text{cis}(u), \quad p \in \Pi, \quad u \in \mathcal{O}_{\mathbb{R}}. \quad (14.2)$$

Its kernel is  $\Pi \oplus 2\pi\mathbb{Z} = 2\pi\mathbf{Oz}$ , because multiplication by the nonzero *ordinary real*  $2\pi$  preserves  $\Pi$ . Hence

$$E_0(x + iy) = \exp_{\mathbf{No}}(x) U_0(y) \quad (14.3)$$

has kernel  $2\pi i\mathbf{Oz}$ .

**Proposition 14.1** (Why an infinite period is not a paradox). *Let  $U : (\mathbf{No}, +) \rightarrow \mathbb{T}(\mathbf{No})$  be a homomorphism extending the finite-angle map, and assume the latter is already onto the full unit circle. Then every coset of the finite additive subgroup contains an element of  $\ker U$ .*

*Proof.* Given  $x \in \mathbf{No}$ , choose finite  $a$  with  $\text{cis}(a) = U(x)^{-1}$ . Then  $U(x + a) = 1$ . If  $x$  is outside the finite subgroup,  $x + a$  is still infinite.  $\square$

### 14.3 The reconciliation with a canonical exponential

T correctly separates finite-angle trigonometry from extensions determined only by weak global group and local differential axioms, and proves an unavoidable infinite period. Outside literature supplies the reconciling qualification (M2): Ehrlich and Kaplan construct canonical surcomplex exponentials using an *initial integer part* [16]. That is *extra structural information*, not a contradiction of nonuniqueness under weaker hypotheses. Specifying the integer part kills the free purely infinite phase component, and the kernel  $2\pi i \mathbf{Oz}$  of (14.3) is the normal-form description consistent with that canonical extension.

A formal library should expose both levels: finite-angle identities requiring no infinite-phase choice, and the extra integer-part or character data used by a selected global extension. *Neither the foundation nor the bare field axioms determine that data for free* (M2). The companion report *Trigonometry on the Surcomplex Plane*, held in this repository at `docs/surcomplex/trigonometry/` [7], develops the trigonometric theory itself; it is cited here by title and path only.

### 14.4 The rational unit-circle parametrization is pure field algebra

Much of T’s geometric algebra — determinant identities, the cosine-law polynomial identity, Heron’s factorization, circumcenter equations, Ptolemy’s inequality once a modulus is available — is finite algebra over an ordered or real-closed field. So is the rational parametrization of the unit circle:

$$t \mapsto \frac{1-t^2}{1+t^2} + i \frac{2t}{1+t^2}. \quad (14.4)$$

Representing an abstract direction by a circle point does not yet assert that a chosen sine function parametrizes it by numerical finite angles; that is a later theorem (M2). Much of the useful geometry is finite algebra even when the lengths are infinite, and the computational complexity of evaluating a particular surreal coefficient does not affect the logical validity of a polynomial identity — which makes generic algebra an unusually high-return first formalization target.

*Remark 14.2* (Symbolic checks are not proofs). Validating finitely many Taylor coefficients cannot verify a support theorem, a transfinite recursion, or an assertion about every scale. The symbolic checks reported inside T are useful sanity checks of specific identities, *not* substitutes for these proofs (M2). The same limitation applies to the illustrative Lean files in the M1 and M2 archives: they are intentionally much narrower than the complete analysis project, and none of them was run (section 2).

## 15 Axiomatizing the structure rather than constructing it

### 15.1 Real closedness is an interface, not a characterization

A type equipped with a real-closed-field structure supports much of finite polynomial geometry after complexification. It does not thereby become the canonical surreal field:  $\mathbb{R}$ , the real algebraic numbers and many non-Archimedean fields satisfy the same first-order ordered-field sentences. Adding an infinite element only rules out Archimedean examples; adding a specified family of infinitesimals still does not recover Conway’s simplicity structure. The distinguished birthday function, simplicity tree, ordinal embedding, normal-form monomials and small-cut filling need separate data and laws. The same warning applies to an abstract algebraically closed field advertised

as “surcomplex”: it may suffice for polynomial factorization, but it does not supply a distinguished conjugation, an ordered fixed field, a Conway monomial map, or an ordinary complex subfield (M2).

This suggests two legitimate axiomatic interfaces (M3). A *generic algebra interface* deliberately assumes only the field, order, valuation or closure needed by a theorem. A *surreal structure interface* also records the simplicity and size data and has a proved model given by a construction. The former is excellent for reusable mathematics; the latter is needed when one wants to justify that the generic results apply to the intended surreal object.

## 15.2 A safe abstract package

Merging M2’s four ingredients with M3’s interface clauses gives the following specification. There is a carrier  $S$  with:

1. an ordered-field structure, an ordinal-valued birthday map, a simplicity relation with set-sized predecessor chains, and explicitly sized bounded birthday strata;
2. a *full sign-tree representation*: each element has a unique ordinal-length sign sequence; every set-sized (or small) such sequence is represented; numerical order is the termination-sensitive lexicographic order;
3. a cut operation for separated *small* families, whose output lies between its options, is characterized as the simplest eligible element, and carries a rank bound; together with a compatibility theorem identifying the designated simplicity relation with initial truncation in the sign presentation;
4. arithmetic obeying the recursive Conway rules and respecting equality — not merely arbitrary field operations on the same carrier.

This is a structural specification, *not* a proposed minimal finite axiom system (M2), and *not* a claimed minimal or independent axiomatization (M3). To establish categoricity one must state the precise notion of morphism and prove an isomorphism to a canonical construction. Pure numerical order and field operations are not enough, and adding an arbitrary rank label does not force it to be the true birthday; conversely, specifying every cut by all earlier numerical neighbours without a bound can make the options improperly large. In a class theory the quantification over sign sequences is quantification over set codes; in a universe-indexed theory it is restricted to the stated universe. In a Lean project such an interface is best packaged as a structure with fields and laws and then *inhabited* by the constructed model; merely declaring an uninhabited interface is harmless, but declaring a global inhabitant as a new axiom shifts the entire consistency burden to that assumption (M3).

## 15.3 Sign-tree categoricity

The following is M2’s, and it is a *foundational categoricity* statement. It is renamed here from the source’s “rigidity” to avoid collision with the analytic all-scale polynomial rigidity of [section 11.3](#), which is a completely different theorem about a completely different subject; see [theorem 11.3](#).

**Proposition 15.1** (Sign-tree categoricity). *Two realizations of the full sign-tree specification over the same ordinal and set universe admit a unique sign-sequence-preserving isomorphism. If both use the recursive Conway arithmetic, that isomorphism also preserves the arithmetic.*

*Proof.* Map each element to the unique element of the second realization with the same sign sequence. Existence and uniqueness of represented sequences make this a bijection, and the order and simplicity relations are preserved by their definitions. Compatibility with addition and multiplication follows by well-founded induction on the input presentations, since the map identifies corresponding option values and corresponding simplest cuts. No other sign-preserving map can differ on any element.  $\square$

Ehrlich’s work on simplicity hierarchies supplies more conceptual axiomatizations and initial-embedding results [13, 14, 15]. Their strength comes from the extra tree structure and the specified notion of initial embedding, not from field axioms plus a vague maximality condition. The 2001 paper must be cited together with its published 2005 correction, which is listed here as a separate bibliography item for exactly that reason (M1, M2).

## 15.4 Universality caveats

Universality claims need a specified class of inputs and embeddings (M3). Embedding every *set-sized* ordered field into the surreal class is different from asserting a final object among all class-sized ordered fields. Uniqueness of an initial simplicity-preserving embedding is different from uniqueness of an arbitrary ordered-field embedding. The adjective “absolute” in a structural account does not remove the need to say which parameters, presentations and morphisms are preserved, and maximality in a structured category must not be confused with the absence of every possible ordered-field extension (M2). For the supplied manuscripts an all-purpose universality theorem is not required: the concrete normal-form embedding of the selected Hahn workspaces, plus the compatibility laws needed for each theorem, is both more precise and more directly useful.

## 15.5 A recent first-order program, at slide level, with its status boundary

All three audits cite the March 2026 announcement by Joel David Hamkins, reporting joint work with Junhong Chen and Ruizhi Yang, of a first-order theory of surreal arithmetic enriched by a birthday-precedence relation, with a reported bi-interpretability with set theory [38]. M1 reports the lecture announcement posted February 4 2026 for a March 13 2026 CUNY Logic Workshop talk; M2 reports the same and adds that it is not used as a proved dependency. **Only M3 inspected the slides**, and its slide-level report is therefore printed here as the substantive content, since it is the only part actually read rather than inferred from an abstract [39].

From the slides (M3): the proposed theory uses *schemes controlling birthday-bounded definable cuts and definable families*. These restrictions are the important foundational feature — the proposed theory deliberately does *not* fill unrestricted definable cuts such as the entire carrier against the empty side. That is precisely the restriction [theorem 3.2](#) forces on any consistent axiomatization.

The union of all three status boundaries applies, and every item survives:

- the slides explicitly describe the work as new, with details still being checked; it is a promising research program, not a completed published axiomatization;
- it is not a verified consistency proof and not a substitute for a proved model of the interface of [section 15.2](#);
- it is *not used as a proved dependency* of this report;

- it is not a statement about the pure ordered-field theory or the pure language of real-closed ordered fields; an expanded first-order structure can encode information that the real-closed-field reduct cannot, and statements about their theories must not be conflated;
- the birthday-precedence relation of that program must be distinguished from sign-prefix simplicity ([theorem 4.2](#)) — the one technical warning M1 and M2 do not carry.

Its importance here is conceptual: enriching the field by construction history can drastically change what its first-order language expresses.

*Related (batch 95, 4 October 2026).* Part IX of the report `Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets` (its sources 13 and 14, batch-95 manuscripts 01 and 02, merged; AI-assisted, unrefereed, not formalized) gives explicit recursively axiomatized first-order theories whose only individuals are surreal numbers, or omnific integers, in the ordered-ring language enriched by birthday precedence or by the birthday function, each parameter-free bi-interpretable with ZFC (its Theorems 73.4 and 73.6, `hset:sf:thm:theories`, `hset:sf:thm:surrealtheory`). They are built from the interpretation by a round-trip recipe, not from the announced intrinsic axioms, which remain unverified there (its Question 79.2, `hset:sf:q:intrinsic`). The same Part proves, assuming  $\text{Con}(\text{ZFC})$ , that GBC, equivalently NBG, is not uniformly interpretable in ZFC, so that no theory bi-interpretable with ZFC interprets it (its Theorem 75.2, `hset:sf:thm:noint`), and gives a two-sorted class lift, with classes of numbers as the second sort, bi-interpretable with GBC (its Theorem 76.5, `hset:sf:thm:classlift`). Its GBC, with Global Choice, is the convention of Subsection 6.1.

## 16 Dependent type theory and the universe boundary

### 16.1 The universe shift is the construction, not an inconvenience

In dependent type theory one constructs a family of surreal types relative to a universe of small indexing types. Schematically,

$$\mathbf{No}_u : \text{Type}_{u+1}, \quad I, J : \text{Type}_u,$$

with cut options  $I \rightarrow \mathbf{No}_u$  and  $J \rightarrow \mathbf{No}_u$ : the larger universe contains the carrier, while the smaller universe controls permissible branching. The illegal cut  $(\mathbf{No}_u, \emptyset)$  is excluded because its left index would generally require the larger level. If all possible option indexing types were drawn from the same size as the resulting carrier, one could try to use the whole carrier as the left-option family and obtain [theorem 3.2](#) again. The constructor's arity and the carrier size must therefore be kept distinct. This is the type-theoretic analogue of [theorem 7.1](#), and it is explicitly *not* a claim that types are literally sets in a particular chosen model, nor an exact lower-bound theorem for the consistency strength of any type theory or of Lean itself (M3).

An illustrative raw-game constructor has the shape

```
universe u

inductive RawGame : Type (u + 1) where
  | cut (L R : Type u)
    (left : L -> RawGame)
    (right : R -> RawGame) : RawGame
```

and all three audits print some form of it. The recursive occurrences are strictly positive: they occur as function *codomains*. Each branch is structurally well founded even though the branching can be infinite; “inductive” does not mean “finite birthday,” and well-founded trees of transfinite rank are represented by such inductive or W-type constructions (M2). This is an illustration of the size discipline, not the current repository’s exact implementation (M1, M3).

*Remark 16.1* (The **Set RawGame** positivity and size warning). M3 alone records the following. A tempting alternative constructor taking **Set RawGame** on each side changes both positivity and size. A predicate has the form **RawGame**  $\rightarrow$  **Prop**, so the recursive type occurs in a function *domain*: it is not the small positive family representation above, and positivity is broken. Moreover a predicate on the entire larger carrier need not have lower-universe-small support, so the size discipline is broken as well. *A constructor is not validated merely because its surface notation resembles a mathematical set of options.*

One should not try to put a not-yet-defined numerical comparison into the constructor’s hypotheses and expect an ordinary inductive declaration to solve simultaneous order, quotient and numericity problems (M1).

## 16.2 An ordinary-inductive route avoids primitive induction-recursion

M2 gives the five-step route that removes the circular appearance of the cut presentation:

1. define all raw games with small branching by an ordinary inductive construction;
2. define the game comparison relation by well-founded recursion on pairs of game trees;
3. define the numeric predicate recursively, requiring numeric options and separation;
4. prove totality of comparison on numeric games, and prove the arithmetic congruences;
5. quotient numeric games by mutual comparison.

The comparison relation on arbitrary games need not be a total order; totality is a theorem restricted to the numeric subtype. This restriction is indispensable: games and surreal numbers are related objects, not synonymous types. The alternative is a direct sign-sequence type indexed by a universe of ordinals, which gives a canonical carrier and a straightforward simplicity relation but still requires substantial recursion for the field operations. Neither approach eliminates the mathematical work; it changes where that work is concentrated. The Mizar development is evidence that no primitive induction-recursion constructor is required (M2, M3) [37].

## 16.3 Quotient descent is a proof obligation

A function on raw games descends to numeric equivalence only after congruence is proved; for addition this means

$$x \sim x', y \sim y' \implies x + y \sim x' + y'.$$

For a map selecting representatives, choice can supply an output, but the mathematical result must not depend on which representative was selected. A quotient recursor is the proper boundary: its respectfulness proof records exactly this invariance.

Two different notions of equality are in play. Structural equality compares raw presentations; quotient equality compares numerical values. A proof about the options of a selected raw representative is not automatically a theorem about arbitrary canonical options of the number, and a

statement about the canonical birthday of a number must not accidentally use the birthday of an arbitrarily chosen noncanonical representative. The distinction matters especially for birthday minimization and simplicity.

In a universe-relative quotient the collection of raw games is an actual type, so the quotient constructor is a legitimate type-theoretic operation; this avoids the literal proper-class-quotient problem of [theorem 4.1](#), but it does not make all option families small and does not remove the need to prove congruence (M3). Quotienting a large type usually leaves it large (M1). Classical Lean has native quotients and a large quotient-based algebraic library, so quotienting numeric games is a natural fit; the cost is that reductions involving a representative chosen by choice are generally not computational algorithms for surreal arithmetic (M2).

## 16.4 Universe polymorphism is a schema, not a universal carrier

A declaration parameterized by universe  $u$  can be instantiated at many levels. It does not define a term  $\sum_u \mathbf{No}_u$  over “all universes,” because universe levels are not ordinary term-level elements of a single universal universe. There is no top universe containing every universe at its own level, and any purported final surreal type obtained by collapsing all levels must face [theorem 3.2](#) again. Lean has  $\text{Type}_u : \text{Type}_{u+1}$  and is not cumulative in the sense of automatic inclusion: explicit lifting constructions such as `ULift` and universe-polymorphic declarations handle changes of level [\[40\]](#). When comparing  $\mathbf{No}_u$  with  $\mathbf{No}_v$  one should construct the embedding by lifting or re-encoding the underlying data, not assume a definitional equality of carriers (M3).

*Remark 16.2* (`Small` is evidence to be proved). Lifting a small type to a larger universe is harmless. *Shrinking is different.* A `Small` hypothesis is evidence that a type has an equivalent representative in a specified smaller universe; Mathlib’s mechanism records exactly that, and a `shrink` construction uses such a witness. The proof may be nonconstructive, but it is still a theorem to be supplied. It cannot be manufactured to convert  $\mathbf{No}_u$  into a legal index type for all of its own options: the whole  $\mathbf{No}_u$  cannot be declared  $u$ -small consistently with its small-cut filling property and its linear order. A `Small` hypothesis on a support should be an explicit constructor argument or theorem assumption; it should not be inferred from the fact that the values happen to be surreal (M1, M3).

For a fixed-level surreal type the admissible cut interface should therefore resemble

$$L, R : \text{Type}_u, \quad \ell : L \rightarrow \mathbf{No}_u, \quad r : R \rightarrow \mathbf{No}_u, \quad \forall a, b \quad \ell(a) < r(b),$$

with  $\mathbf{No}_u : \text{Type}_{u+1}$ . An interface using *subsets* of  $\mathbf{No}_u$  must explicitly require their supports to be  $u$ -small. The same rule applies to Hahn supports. [Section 19.4](#) makes this concrete.

## 16.5 Impredicative Prop is not Type : Type

Lean’s `Prop` is impredicative: a proposition may quantify over arbitrary types without moving to a larger `Prop`, its proofs are definitionally proof-irrelevant, and elimination from propositions into data is restricted. These features are not an unrestricted universe of data containing itself [\[41, 40\]](#). The statement “every surreal in this universe has property  $P$ ” is well formed even when its quantifier ranges over a large type; this does not produce a smaller type enumerating those surreals. Logical quantification and construction of a data collection are separate operations.

A predicate  $P : \mathbf{No}_u \rightarrow \text{Prop}$  defines a subtype at the existing data level; it does not imply that the subtype is  $u$ -small (M2). This rules out the tempting error of using the small universe of truth values to encode all surreals as small data: the *output* of a predicate is a proposition, but

the domain, graph, subtype and family of all predicates retain their appropriate universe levels. Propositional resizing principles, in foundations that adopt them, address proposition universes; they do not compress the ordinal-indexed surreal carrier or its arbitrary supports, and resizing must be a stated principle with a suitable scope rather than a consequence of impredicativity.

## 16.6 Classical and computational content

A **noncomputable** definition in Lean is *not* an unproved axiom (M1). It can be fully kernel-checked while depending on classical choice. Arbitrary real coefficients, arbitrary well-ordered supports and quotient representatives naturally lead to such definitions. Efficient computation requires additional presentations — finite games, effective exponent arithmetic, an enumeration procedure for coefficient dependencies — and the abstract existence of a finite contributing set is not automatically an algorithm for finding it. Arbitrary well-ordered supports need not have a computable enumeration or decidable coefficient equality; exact finite-prefix computation requires a representation and a theorem that the prefix depends only on the supplied finite data (M3). A formalization can first verify the noncomputable mathematics and later refine effective subclasses.

Choice at a fixed universe level can select representatives from the types under discussion. It does not form a single function quantified internally over all universe levels, and identifying such an axiom with set-theoretic global choice requires a specified interpretation of sets and classes: the common word “choice” alone is not a proof that the foundational principles are equivalent (M3). “Lean choice” and “NBG global choice” are not literally the same axiom in the same language (M2).

## 16.7 Internal ZFSet as an alternative route

One can also formalize a set-theoretic universe inside Lean and define surreals as sets in that model. Mathlib provides pre-sets **PSet** in  $\text{Type}_{u+1}$ , quotients them by extensional equivalence to obtain **ZFSet**, and represents internal classes by **Set ZFSet** [48, 49, 50]. The distinction

$$\text{Set (ZFSet.\{u\})} \quad \text{versus} \quad \text{ZFSet.\{u\}}$$

separates an external Lean predicate class from an internal set, and Mathlib documents it explicitly. This route permits translating a conventional set-theoretic argument almost literally, and it is well suited to interpretation or conservativity results — internal properness of **No**, interpretations of class arithmetic, or the dependence of a construction on a collection principle.

The cost is a bridge to ordinary algebraic types. An internal field described by set-coded operations is not definitionally a Lean **Field** instance on its external element type: one must extract the carrier, transport the operations, and establish the relation between internal membership and external predicates. Each ordinary field, polynomial, function and set operation may need an internal-to-native translation. The existence of a ZFC model in Lean also does not mean that Lean is ZFC under another spelling; the construction uses Lean’s ambient type theory and universe hierarchy.

*Remark 16.3* (The **Definable** warning). Mathlib’s technical predicate **ZFSet.Definable**, and its theorem **Classical.allZFSetDefinable**, must *not* be read as saying that every external function is definable by a first-order formula in the language of set theory. The API concerns representative-level liftability of functions through pre-set representatives; the documentation expressly notes the distinction. One must not use that identifier as evidence that a native Lean function has been translated into a ZFC formula. First-order definability for a metamathematical interpretation must

be formalized with actual syntax and satisfaction, which is a further layer. M1 and M3 record the warning for `ZFSet.Definable`; **M2 alone also names `Classical.allZFSetDefinable`**.

To compare GBC and KM inside Lean one must specify a two-sorted language, formulas, models or interpretations, and the exact comprehension and replacement schemas (M1). If classes are interpreted by all external Lean predicates on an internal universe, one must prove which schemas are satisfied and at what metatheoretic strength; the fact that the predicate type exists is not a proof that the intended minimal class theory and its exact consistency strength have been reproduced. Unrestricted external quantification over predicates can provide more semantic resources than elementary class comprehension, so a development claiming conservativity or precise axiom calibration must reason about formal proof systems or explicitly restricted model expansions.

## 17 Constructive set theory and univalent foundations

### 17.1 IZF, CZF, and type-theoretic interpretations

IZF retains a power-set style set theory with intuitionistic logic; CZF is designed for predicative constructive mathematics and replaces some classical set-existence principles with constructive collection principles. Type-theoretic interpretations of constructive set theories provide a well-developed connection with dependent types and W-type methods [30], and Aczel’s type-theoretic interpretation with its treatment of inductive definitions is the standard source [28]; regular extension axioms supply additional closure resources in some treatments [29]. In weak theories without power set, Replacement and Collection can differ in ways that are immaterial in ordinary ZFC but material to formalization [27].

These are genuine foundational options, but a classical sign-sequence argument cannot be moved into them by deleting the words “by excluded middle” (M2). One cannot obtain a constructive formalization simply by removing the word “classical” from a classical surreal proof (M3).

### 17.2 Why the constructive order needs positive information

Classically, strict comparison can be recovered from a suitable negated non-strict comparison. Constructively, absence of a proof of  $y \leq x$  need not give positive evidence that  $x < y$ , and a definition based on that negation can also violate the positivity requirements of the intended inductive construction. The equivalence

$$x < y \iff \neg(y \leq x)$$

must not be assumed without the relevant order principle. An *apartness* relation —  $x \# y$  meaning  $x < y$  or  $y < x$  — carries positive information absent from mere inequality. Even familiar definitions of a field differ constructively: inversion from positive apartness from zero is stronger evidence than mere negation of equality, and ordinary Cauchy and Dedekind reals need not coincide without additional principles. Classical ordinal comparability, deciding whether an option lies above a point, and turning negated universal statements into witnesses can all fail constructively.

### 17.3 The higher inductive-inductive construction

The HoTT book develops surreal numbers by a higher inductive-inductive definition [31, §11.6]: one simultaneously defines the carrier, its non-strict order and its strict order. Small separated left and

right families create a number, and an antisymmetry path constructor identifies  $x$  and  $y$  when both inequalities hold. The positive comparison pattern reads

$$\begin{aligned} [\forall x^L (x^L < y)] \wedge [\forall y^R (x < y^R)] &\Longrightarrow x \leq y, \\ [\exists x^R (x^R \leq y)] \vee [\exists y^L (x \leq y^L)] &\Longrightarrow x < y, \end{aligned}$$

replacing the classical negative test (4.1). Positive constructors for the order relations support recursion without requiring an arbitrary lift of quotient-valued option families to raw representatives.

This is not the classical quotient construction with different syntax. In a constructive quotient approach a family of equivalence classes need not have a chosen family of representatives, and the higher inductive-inductive construction addresses that in its elimination principle. The numerical and order components depend on each other, so a standard W-type alone does not present the whole construction. A formalization must specify the elimination principle, the coherence of the resulting operations, and the status of the order relations as propositions. It *is not licensed merely by writing mutually recursive symbols* (M1), and there is no blanket theorem that every informal recursive signature is sound: general semantic work on higher inductive types is important background but does not certify a particular signature [33, 34]. A proof assistant supporting some higher inductive types does not thereby support every higher inductive-inductive signature with all its required computation rules.

## 17.4 Plump ordinals and constructive comparison

Constructively, classical well-orders, trichotomous ordinal notions, and the ordinal objects naturally living inside surreals need not coincide without further hypotheses. Shulman’s discussion of *plump ordinals* illustrates the distinction and explains why the constructive definition treats strict and non-strict comparison separately [32]. A constructive comparison theorem must therefore name its ordinal notion; the slogan “the surreals contain all ordinals” needs more interpretation than in classical ZFC. Joseph develops an ordered commutative-ring structure with apartness in univalent foundations [35]. These sources establish that the constructive approach is mathematically substantial; they do *not* by themselves constitute a Lean 4 implementation of the complete surcomplex analytic theory (M2), and the constructive route is not claimed to already supply the normal-form, algebraic-closure and analytic package of the supplied manuscripts (M3).

## 17.5 Why full univalence cannot simply be added to Lean’s equality

Lean’s ordinary equality is proposition-valued and proof-irrelevant, so for fixed types  $A, B$  the identity type  $A = B$  has at most one proof. Full univalence for a universe with nontrivial automorphisms has a different homotopical equality structure, in which distinct self-equivalences correspond to distinct equality paths. M2 and M3 both prove the incompatibility as a numbered proposition by the same two-element-type argument; M1 states it in prose only. We print the proof once.

**Proposition 17.1** (The elementary incompatibility test). *Suppose a type theory has proof-irrelevant equality between types, contains a two-element type  $B$ , and asserts full univalence for a universe containing  $B$ . These assumptions are incompatible.*

*Proof.* Proof irrelevance makes the type of equality proofs  $B = B$  have at most one element. Univalence makes its map to the type of self-equivalences  $B \simeq B$  an equivalence, so  $B \simeq B$  would also be a subsingleton and all self-equivalences of  $B$  would be equal. But the identity map and the map exchanging the two elements are not equal, as evaluation at either element of  $B$  shows.  $\square$

The conclusion is drawn carefully, and in exactly this form: *it is not that univalent mathematics is inconsistent*. It is that incompatible equality mechanisms cannot be superimposed on Lean’s native proof-irrelevant type equality without changing the foundation. Propositional extensionality is a different, much weaker and compatible assertion [42].

M1 names the three legitimate alternatives, and they survive: translate the set-level result to quotient-based mathematics; encode the syntax and semantics of an HoTT object theory *inside* Lean; or conduct the higher-inductive development in a system designed for it. None of these changes Lean’s kernel equality. For the classical algebra and analysis of the supplied manuscripts, ordinary quotient-based Lean mathematics is a better fit than rebuilding the analysis on a new homotopical notion of equality.

## 17.6 Categorical and algebraic-set-theoretic perspectives

An elementary topos with a natural numbers object supplies a useful environment for much ordinary mathematics, but does not by itself guarantee the unrestricted cumulative hierarchy or the class of all classical ordinals. Structural set theories can be strengthened with replacement-like principles, universes, or classes of small maps; in such a framework a surreal construction must specify which well-founded trees are small and why the relevant recursive constructions preserve smallness [20, §§13–15]. Algebraic approaches to surreal structures have been developed in a class-set-theoretic setting [36].

Such frameworks do *not* remove the diagonal obstruction (M1, M2). Internally in any logic where  $<$  is irreflexive, a rule producing a strict upper bound for every subobject of the same carrier fails on its maximal subobject. A categorical account must restrict its small maps or its admissible cuts, and must state its semantics, quotient existence and inductive constructions explicitly rather than hiding them behind the word “universal.” Likewise a categorical description of **No** as a class-directed union of substructures is useful, but one must not use an ordinary small-diagram colimit constructor on a proper-class index category (theorem 9.3).

## 17.7 Choosing the constructive route for the right purpose

A constructive development is attractive when the computational meaning of order, choice-free construction, or the elimination principle is central; a classical quotient-based Lean development is attractive when the immediate goal is to reuse algebra, ordinary complex analysis and established classical libraries. For a constructive version of the supplied analytic theory the axiom ledger would have to revisit the extraction of decreasing sequences, Baire arguments, holomorphic zero-set reasoning, and the choice of roots or representatives. The aim need not be to recover the classical statements unchanged: one can prove a constructive theorem with explicit apartness, locatedness or supplied enumeration hypotheses, and then recover a classical specialization when excluded middle and suitable choice are added. A constructive project should specify its ordinal notion, its inductive or W-type existence principles, its quotient policy, and its use of choice or propositional resizing. The two routes can inform each other through explicit interpretation theorems; neither is a drop-in implementation of the other.

## 18 The Lean ecosystem in the source audits: one inspection, three readers

### 18.1 The aggregation warning, in plain language

Everything in this section comes from **one act of source reading, performed three times**. The three audits inspected the same repository at the same immutable revision on the same day, through the public web interface, reading source only. They agree. That agreement is evidence about what the source *says*, and about nothing else.

It is not evidence that the repository builds. It is not evidence that its theorems mean what their names suggest. It is not evidence about its axiom closure. Three concurring readings of one web page corroborate each other only about the contents of that page. No claim in this section was kernel-checked, and the findings below are printed once, not three times, precisely so that their repetition cannot be mistaken for replication. See [section 2](#).

### 18.2 The pinned revision

| Item               | Recorded value                                                                                                        |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| Repository         | vihdzp/combinatorial-games (a separate Lean 4 project, <i>not</i> upstream Mathlib)                                   |
| Revision inspected | 02b4a908ea2ecfeffecb438f691951a814a5264                                                                               |
| Commit timestamp   | September 19, 2026, 10:25:40 UTC                                                                                      |
| Inspection date    | September 21, 2026                                                                                                    |
| Lean toolchain     | leanprover/lean4:v4.35.0-rc2                                                                                          |
| Mathlib revision   | dec5b2b780537b6eaf7f5e5f000c12f7387fb24d                                                                              |
| Method             | public GitHub interface, source reading only; no clone, no build, no comprehensive placeholder search, no axiom audit |

All three audits record the repository revision and the toolchain; all three record the Mathlib revision, each reading it from the `lake-manifest.json` at that pinned commit rather than from a search-engine snippet or an unrelated cached branch page [51, 52, 53]. Files inspected at that revision, with pinned blob URLs in the bibliography: `CombinatorialGames.lean` (umbrella module inventory, M3); `CombinatorialGames/Surreal/Basic.lean`; `CombinatorialGames/Surreal/Division.lean`; `CombinatorialGames/Surreal/HahnSeries/Basic.lean`; `CombinatorialGames/Surreal/Cut.lean` (M3 only); plus `lean-toolchain` and `lake-manifest.json`.

*Remark 18.1* (Historical caution). Old Lean 3 surreal documentation describes an earlier stage of the development and is not a reliable inventory of the current separate Lean 4 project. Conversely, finding a theorem in that project does not make it available by importing upstream Mathlib alone (M3).

### 18.3 The surreal carrier and the field instance

At the pinned revision, `CombinatorialGames.Surreal.Basic` defines the carrier in universe  $u + 1$  by antisymmetrizing the subtype of numeric games:

```
def Surreal : Type (u + 1) :=
  Antisymmetrization (Subtype Numeric) (...)
```

where the ellipsis stands schematically for the comparison relation. The same module supplies the quotient map, numerical equality of representatives, a linear order, additive structure, a choice-based representative operation (`Surreal.out`, using quotient choice), and a set-based cut interface carrying smallness conditions on the option families. This is the universe-sensitive pattern developed in [section 16](#), not a single small type accepting arbitrary subsets of itself [\[55\]](#).

The inspected `CombinatorialGames.Surreal.Division` contains an *actual* declaration

```
noncomputable instance : Field Surreal where
...
```

whose inverse descends from numeric-game inversion by a quotient map and a congruence theorem, with the field cancellation proof transported through numerical equivalence and inverse correctness established by well-founded induction over numeric games [\[56\]](#). It is therefore incorrect to describe surreal multiplication or inversion as wholly unimplemented in the current inspected ecosystem, or to describe current Lean surreal work solely by older Lean 3 documentation.

*Remark 18.2* (The “complete ordered field” prose). Prose in the basic module uses the phrase “complete ordered field.” It must *not* be read as a proved Dedekind-completeness statement; [theorem 3.4](#) gives the elementary obstruction. The field structure and the actual theorem signatures are what matter. This is a useful example of why a source audit cannot stop at a module’s introductory paragraph.

## 18.4 The small-support surreal Hahn layer

The inspected `CombinatorialGames.Surreal.HahnSeries.Basic` defines `SurrealHahnSeries` in universe  $u + 1$  as a small-support subfield of lexicographically ordered real Hahn series on the *order dual* of the surreal exponent type. Its `mk` constructor requires *both* a `Small.{u}` support witness *and* reverse well-foundedness of the support. It provides field, linear-order and strict-ordered-ring infrastructure, coefficient extraction, support, truncation, and ordinal indexing of the small support [\[57\]](#).

Two details matter. First, it lives at `Typeu+1`, matching the surreal carrier. Second, support is required to be  $u$ -small although the exponent carrier is larger. This is precisely the relative-smallness discipline of [section 10.5](#), and it is an important *implemented* safeguard: the library does not identify the unrestricted Hahn-series type with the surreal carrier.

The module presents itself as a translation layer. In the inspected file, including its end, *no completed ordered-field isomorphism between the surreal carrier and this Hahn type was verified*. A constructor for the target type, or a comment describing an intended future isomorphism, is not that isomorphism: normal-form existence, surjectivity of evaluation and compatibility with multiplication remain proof obligations unless a separately located theorem discharges them. A finished bridge needs maps both ways, inverse laws, compatibility with arithmetic, and support preservation.

Likewise, *no real-closedness instance for the concrete game-based carrier, no surcomplex algebraic-closure specialization, and no Gonshor exponential was verified in the inspected modules*. These must be checked or supplied before specializing the algebraically closed-field parts of the supplied manuscripts.

*Remark 18.3* (Scope of the negative findings). Every negative finding in this subsection is *scope-limited by construction*: it is a statement about what these particular files, read on this particular date, were observed to contain. It is **not** a claim that these results are absent from any other file,

branch, repository, project or later revision. All three audits state this qualification, and it applies to every sentence of the form “was not verified” in this section.

## 18.5 What `Surreal/Cut.lean` actually contains

**M3 alone inspected this file**, and it is the module most likely to be misread. The inspected `CombinatorialGames.Surreal.Cut` concerns complementary lower and upper *sections of the entire universe-relative surreal order*. It builds a complete linear order of such sections via concept-lattice machinery, and a number corresponds to two adjacent sections according to which side contains it [58].

This is *not* the same object as a small pair of Conway options. A section of the full carrier may be large and need not be filled by a number in that carrier. Therefore its completeness yields *neither* an unrestricted Conway cut constructor *nor* Dedekind completeness of the surreal field, and it does not contradict [theorem 3.2](#) or [theorem 3.4](#). M3’s own instruction stands: *a formal API should name the option-pair interface and the section interface distinctly, even though the literature calls both “cuts.”* [Section 19.4](#) adopts that instruction.

## 18.6 What generic Mathlib supplies

Mathlib’s Hahn-series development already provides coefficient functions with support restrictions, convolution operations, and formal summable families. In the inspected documentation, `HahnSeries.SummableFamily` carries *both* a partially well-ordered union-of-supports condition *and* finite coefficient co-supports; the order hypotheses specialize that support condition to the usual well-ordered condition used here, and its `hsum` operation is the *formal* sum, not an unqualified topological series [44, 45]. Mathlib’s documentation explicitly distinguishes this from topological summability, so inventing a new use of ordinary `tsum` would obscure an already-supported distinction (M2). A surreal specialization must prove that the general partially-well-ordered hypotheses yield the required well-ordered supports in the chosen linear order, and must establish smallness of the resulting support, not only its order-theoretic shape (M2, M3).

**M1 alone** inspected the `HEval` module. It supplies one-variable `PowerSeries.heval`, evaluation of a formal power series at a positive-order Hahn series, and its retrieved documentation explicitly lists the finite-variable `MvPowerSeries.heval` as a **TODO** [46]. This is the single concrete extension target named anywhere in the group. The observation concerns the inspected module and its documentation; it is *not* a proof that no other project contains a multivariable evaluation theorem.

**M2 alone** inspected the real-closed-field page. Mathlib has an `IsRealClosed` *interface* [47]. An interface name is not an instance for a particular type. Before depending on real closedness of the actual game-based `Surreal`, a project must locate or prove the relevant instance and check its dependencies; similar care is needed when deriving algebraic closedness of its quadratic extension.

Mathlib also supplies `PSet`, `ZFSet` and class infrastructure ([section 16.7](#)).

*Remark 18.4* (Rolling documentation). All Mathlib generated documentation pages and all official Lean “latest” reference pages cited in this report are *moving* sources consulted on September 21, 2026. They are **not** asserted to be generated from the pinned dependency commit, and they are not substitutes for the repository’s dependency lock. Exact module imports and instance hypotheses must be re-tested against the pinned Mathlib revision, and exact typeclass names should be taken

from the pinned dependency source rather than invented to match a mathematical phrase. All three audits record this.

## 18.7 Mizar as an independent completed milestone

Pał and Kaliszyk’s ITP 2024 paper describes a substantial Mizar formalization combining complementary presentations of surreal numbers and reaching Conway normal form, discussing how ordinary transfinite methods replace an otherwise troublesome inductive-recursive presentation and how sign-based and Conway-style reasoning support different parts of the development [37]. It demonstrates that the representation bridge is a realistic, substantial formalization task, not an optional piece of notation.

Mizar’s Tarski–Grothendieck foundation and its set-theoretic proof style differ from Lean’s native dependent type theory. The development is therefore a valuable mathematical and formal-engineering reference, *not* a source of Lean proof terms that can be imported unchanged: theorems do not transfer between libraries merely because their informal names agree. No Mizar run was performed here; no formal article in that development was inspected individually; its axiom closure was not reconstructed.

M2 and M3 both flag one further terminological point: the abstract’s description of the surreal field as “algebraically closed” must be distinguished from the actual real-versus-complex field statements. The ordered surreal field cannot have a root of  $X^2 + 1$ ; algebraic closedness belongs to  $\mathbf{No}[i]$  (section 8.2).

## 18.8 Terminological warnings, collected

Four naming traps recur, and all four are recorded here in one place.

1. “**complete ordered field**” in source prose is not a Dedekind-completeness theorem (theorem 18.2).
2. “**cut**” names at least two different implemented objects: a small Conway option pair, and a complementary section of the whole order (section 18.5).
3. **IsRealClosed** is an interface, not an instance (section 18.6).
4. **ZFSet.Definable** and **Classical.allZFSetDefinable** are representative-level liftability conditions, not first-order definability by a formula (theorem 16.3).

To these add the Mizar abstract’s “algebraically closed” wording (section 18.7).

## 18.9 The evidence table, with its third column

The following is M1’s evidence table, reproduced with its third column **verbatim**. That column — “What is not claimed” — is the group’s only artifact built specifically to prevent a reader from converting inspected declarations into verified mathematics, and it is used here as the organizing device of this whole section.

| Component               | Evidence inspected                                                                       | What is not claimed                                             |
|-------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Numeric-game quotient   | Current <code>Surreal.Basic</code> definition                                            | A universe-free type of all surreals                            |
| Surreal field           | Explicit <code>Field Surreal</code> declaration                                          | Independent local rebuild and complete dependency audit         |
| Surreal Hahn layer      | Small-support subfield, coefficients, truncations, ordinal indexing                      | Finished normal-form equivalence inferred from the module title |
| Formal Hahn sums        | <code>SummableFamily</code> and <code>hsum</code> APIs                                   | Topological convergence of every formal sum                     |
| Power-series evaluation | One-variable <code>PowerSeries.heval</code> ; multivariable TODO in the inspected module | A proof that multivariable evaluation exists nowhere else       |
| Internal set theory     | <code>PSet</code> , <code>ZFSet</code> , and class infrastructure                        | An absolute consistency proof of ZFC or Lean                    |
| The supplied analysis   | Mathematical proofs in the supplied manuscripts                                          | Already machine-verified analytic geometry and residues         |

Two rows must be read with [section 1.3](#) in hand: “the supplied analysis” in M1’s original table meant the two manuscripts M1 read, and in this merged report the row covers five manuscripts, of which only one was read by all three audits.

## 18.10 Per-archive Lean inventory

The per-archive inventory — which files ship, in what state, with what import footprint — is [section 2.4](#), in the verification-boundary section, where it belongs. Its single most important line bears repeating here: M1 ships three uncompiled Lean files, one of which imports `Init` only; M2 ships two uncompiled Lean files, one of which imports all of `Mathlib`; and M3 ships *no Lean files at all*, its five code blocks existing only as typeset listings inside its PDF. None of the five files was compiled by the audit that shipped it; three of the five have since been compiled for this report ([section 2.6](#)).

## 19 A layered Lean architecture

*All module names, layers, architectures, completion criteria and implementation sequences in this section are design suggestions. None of them is a claim that files with those paths exist.* The implementation has progressed since the source audits; see [section 2.1](#) and the repository ledger for the current correspondence. Historical expressions such as “the next target” below describe the proposal’s dependency order.

### 19.1 Separate the global number type from the local algebra

M1’s organizing decision is to run two tracks. The *foundational track* uses the existing numeric-game quotient, sign-expansion interfaces and small-support normal forms to justify a universe-indexed

surreal interpretation. The *analytic track* works over arbitrary set-sized Hahn fields and ordinary complex germ rings. They meet at explicit embedding and compatibility theorems.

This avoids making the entire analytic program depend on finishing every global surreal theorem first. An abstract theorem for  $\mathbb{C}((t^\Gamma))$  is already genuine mathematics, and its eventual application to  $\mathbf{No}[i]$  has a clearly identified dependency: the normal-form embedding and its preservation of the constructions used. Conversely the local track must not claim that an arbitrary ordered abelian group  $\Gamma$  is already a subgroup of  $\mathbf{No}$ ; it can prove theorems for abstract  $\Gamma$ , and the embedding is additional data at the application boundary ([theorem 9.2](#)).

M3's complementary decomposition is by *kind of definition*, in four layers: *representation* (raw games, numerical equivalence, canonical sign data, birthdays, universe lifts); *algebra* (ordered-field laws, real closedness, complexification, finite polynomial and matrix structures); *Hahn structure* (monomials, small normal forms, support-preserving maps, valuation, residue, strong sums); and *analysis* (ordinary holomorphic coefficient rings, coherent evaluation, finite-domain division, contours, full-scale hypotheses). Dependencies must point in that direction: a proof of a residue determinant should import finite algebra, not full surreal topology; a proof that a support recursion stays in  $S + E^*$  should import ordered-group support theory, not Hartogs. The full-scale rigidity theorem may then combine the small-cut bound from the first layer with the common-support fact from the fourth.

## 19.2 Layers A–E, with completion tests

M2's lettered layering is the most operational of the three and is adopted here as the working spine.

**Layer A: reusable finite algebra.** Start with an arbitrary ordered field  $F$ , not a concrete surreal datatype. Define its quadratic extension with the correct cross-term multiplication; prove conjugation, norm-square identities, inverses and the Gram identity ([8.6](#)). Add real closedness only where positive square roots or polynomial factorization are genuinely required. This layer already carries most of the geometric algebra of [section 14](#), including the rational circle parametrization ([14.4](#)).

**Layer B: set-sized Hahn analysis.** Use  $F_\Gamma$  and  $K_\Gamma$  as ordinary types, with the ordered value group, its divisibility when needed, and the coefficient field as explicit parameters. Reuse the existing formal-summability API. Establish standard part for finite series from the coefficient at exponent zero, and distinguish finite elements from positive-valuation infinitesimals. Implement evaluation as an operation whose input includes an admissibility proof, or as a total operation with an accompanying theorem guaranteeing its meaning on admissible inputs; the former is clearer:

$$\text{evaluate}(A, h, \text{proof that } (a_n h^n)_n \text{ is strongly summable}).$$

An ordinary polynomial needs no infinite-sum proof; a formal power series does. Keeping these constructors distinct prevents accidental evaluation outside the proven domain. The next targets are the positive-support geometric series, the binomial series, the infinitesimal exponential and logarithm, and Taylor lifting of ordinary analytic germs.

*Remark 19.1* (The Layer B recentering warning). M2 alone records this, and it is easy to get wrong. At a finite point  $r + \epsilon$ , first evaluate the ordinary function at  $r$  and *then* substitute  $\epsilon$  into its Taylor series. Do *not* replace this by the ordinary Maclaurin series at the nonzero real  $r$ : infinitely many of those terms can occupy the same Hahn exponent, and the second clause of ([10.1](#)) fails.

**Layer C: support-controlled coherent functions.** For an ordinary open  $U \subseteq \mathbb{C}$ , represent a coherent datum as a Hahn series with coefficients in the ring  $\mathcal{O}(U)$  of ordinary holomorphic sections. Equality of sections is equality on  $U$ ; values outside  $U$  must not become part of the object. Coefficientwise differentiation does not enlarge the common support, and integration along a fixed ordinary admissible contour also acts coefficientwise and does not enlarge it — which is why ordinary complex-analytic theorems can be applied to the coefficients and then reassembled. Arbitrary composition does *not* follow: for that one must control the supports of the inner perturbations and verify that the ordinary part maps into the common coefficient domain. A useful composition interface has an ordinary base map  $g_0 : V \rightarrow U$  and an infinitesimal perturbation whose coefficient support lies in a common positive well-ordered set  $T$ ; Taylor expansion then produces supports in  $S + T^*$ , and the formalization must prove local finiteness of coefficient contributions as well as well-ordering.

**Layer D: the universe-indexed surreal bridge.** Establish explicit maps among numeric games, canonical sign sequences and small-support normal forms. Deliverables: preservation of order, field operations, ordinary real coefficients, Conway monomials and admissible sums, with the support universe visible in the statement of the strongest comparison theorem. The reason a small-support subtype is required is [section 10.5](#).

**Layer E: topology, functions, and enrichment.** Define the fine topology and the derivative ([12.1](#)) using all radii of the chosen carrier. State the discreteness and eventual-constancy theorems *only* for small subsets or small index types, and ship the larger-index counterexample [theorem 12.2](#) alongside them. Workspace embeddings get separate algebraic, order, summability and topological compatibility theorems; the last is not automatic. After the normal-form bridge, add canonical finite sine and cosine, the finite-angle quotient and the circle logarithm; add a real exponential as a separately proved structure; for a global exponential, require a specified character or the stronger initial-integer-part structure of [section 14.3](#). Do not package all of this into one enormous typeclass: a theorem about Heron’s identity should not acquire an accidental dependency on a global exponential or a class-choice axiom.

| Proposed module group (M2) | A meaningful completion test                                            |
|----------------------------|-------------------------------------------------------------------------|
| Algebra/Quadratic          | Pair multiplication, conjugation, inverse, norm-square identities.      |
| Hahn/Support               | Strong sum API, positive-support substitutions, smallness preservation. |
| Hahn/Germs                 | Formal evaluation and recentering with explicit domain witnesses.       |
| Hahn/Coherent              | Coefficientwise identity, differentiation, and contour theorems.        |
| Surreal/Bridge             | Normal-form comparison with algebra and order compatibility.            |
| Surreal/FineTopology       | Small-index statements plus the larger-index counterexample.            |
| Surcomplex/Angles          | Full unit-circle parametrization and finite-angle kernel.               |
| Surcomplex/Enrichment      | Real exponential, strip functions, specified global phases.             |

### 19.3 Proposed module layers with contracts

M1’s nine-row table gives the complementary view: each proposed layer with the contract it must establish and its typical dependencies. Each contract should state its universe parameters, its hypotheses, and whether it concerns a formal sum or a topological limit. A theorem using ordinary residue duality should take the exact proved structure as input, rather than a loosely named “analytic field” typeclass that silently includes every desired conclusion.

| Proposed layer (M1)    | Contract to establish                                                        | Typical dependencies                                     |
|------------------------|------------------------------------------------------------------------------|----------------------------------------------------------|
| Complexify             | Pair or quadratic-quotient field;<br>conjugation; squared modulus            | Ordered-field algebra                                    |
| Hahn.Support           | Admissible-family certificates;<br>reindexing; finite coefficient dependency | Mathlib Hahn and<br>well-order APIs                      |
| Hahn.MultivariateEval  | Evaluation at finite tuples of<br>positive-order series                      | Finite-word support lemma;<br>multivariate formal series |
| Analytic.CoeffGerm     | A precise convergent complex germ ring<br>and derivative maps                | Ordinary complex analysis                                |
| Analytic.HahnGerm      | $\mathcal{A}_{n,\Gamma}$ , evaluation, translation, finite jets              | Previous two layers                                      |
| Analytic.Division      | Support-controlled division and<br>preparation                               | Ordinary Weierstrass<br>division; operator inversion     |
| Analytic.FiniteAlgebra | Finite bases, residue pairing, trace,<br>local lengths                       | Division; ordinary residue<br>inputs; linear algebra     |
| Workspace.Extension    | Embeddings and finite-quotient base<br>change                                | Field and algebra maps;<br>support compatibility         |
| Surreal.Interpretation | Small-support normal forms and<br>class/universe interpretation              | Existing surreal<br>development; normal-form<br>bridge   |

## 19.4 Encode the permitted size of a cut in its data

Two of the three audits present a `SmallCutData` structure, and they differ only in whether the value universe is left general or already fixed. We print M1's two-universe form as the *general* form, and record M3's as the intended instantiation; M1 itself prescribes  $v = u + 1$ , so this is a specialization, not a rival.

```
universe u v

-- General form (M1): indices in u, values in a separate universe v.
structure SmallCutData (X : Type v) (lt : X -> X -> Prop) where
  Left : Type u
  Right : Type u
  left : Left -> X
  right : Right -> X
  separated : forall l r, lt (left l) (right r)
```

The intended instantiation takes  $v = u + 1$  and a typeclass order, which is exactly M3's form:

```
universe u

-- Intended instantiation (M3): the value universe is fixed at u+1.
structure SmallCutData' (A : Type (u + 1)) [LT A] where
  Left : Type u
  Right : Type u
  left : Left -> A
  right : Right -> A
  separated : forall l r, left l < right r
```

Either structure records cut *input* data; neither asserts that every such cut is realized. A constructor consuming it can fill exactly the lower-universe-small families. What must *not* be exposed is a filler for every unrestricted **Set**  $A$ , and it would be invalid to introduce a realization axiom uniformly for all relations and carriers, or to instantiate the interface at an illicit same-size self-family. One may also work with a predicate-based set of options plus a **Small** witness for its subtype; the two interfaces should be related by a proved reindexing theorem.

M2 ships no such structure. Instead it proves, over **Set**  $F$  with **[Preorder F]**, a subset-based negative guard — the version it actually delivered as Lean source, uncompiled:

```
-- Subset-based variant (M2), shipped as Lean source; compiled for this report, see \S2
.5.
theorem noUnrestrictedUpperBounds (F : Type u) [Preorder F] :
  Not (forall s : Set F, exists x : F, forall y in s, y < x) := by
  intro h
  obtain (x, hx) := h Set.univ
  exact (lt_irrefl x) (hx x (Set.mem_univ x))
```

M1’s corresponding guard uses *predicates* rather than subsets of a set, and imports **Init** only:

```
-- Predicate-based variant (M1), shipped as Lean source; compiled for this report, see
\S2.5.
theorem noUniversalStrictBound
  {X : Type u} (lt : X -> X -> Prop)
  (irrefl : forall x, Not (lt x x))
  (bound : (X -> Prop) -> X)
  (above : forall A x, A x -> lt x (bound A)) : False := by
  let all : X -> Prop := fun _ => True
  exact irrefl (bound all) (above all (bound all) True.intro)
```

Both are the machine-level form of [theorem 3.2](#). Neither had been run when the archives were prepared; both have since been compiled and axiom-audited for this report, and each depends on no axioms whatever ([section 2.6](#)). A smallness test belongs in the statement of every analogue of [theorem 3.3](#) and [theorem 12.1](#): for a function from a small indexing type the range is small, but for a subtype of  $\mathbf{No}_u$  smallness is an additional assertion, not a fact inferred from the word **Set**.

*Remark 19.2* (Name the two cut interfaces distinctly). By [section 18.5](#), “cut” already names two different implemented objects. A formal API should give the option-pair interface and the whole-order-section interface distinct names, even though the literature calls both “cuts.” Add to this [theorem 16.1](#): a **Set RawGame** constructor breaks positivity, because the predicate puts the recursive type in a function domain.

## 19.5 Complex pairs as independently reusable algebra

The coordinate data can be introduced with no surreal-specific assumptions:

```
universe u
structure ComplexPair (F : Type u) where
  re : F
  im : F
namespace ComplexPair
variable {F : Type u} [CommRing F]
def multiply (z w : ComplexPair F) : ComplexPair F :=
```

```

{ re := z.re * w.re - z.im * w.im
  im := z.re * w.im + z.im * w.re }
end ComplexPair

```

The next steps are coordinate extensionality, ring laws, the real embedding, conjugation, and the quadratic norm  $x^2 + y^2$ ; prove positivity over an ordered field, then implement inversion and its laws. A quadratic quotient is an alternative with different API advantages. *Neither design should inherit the coordinatewise product-ring instance* (theorem 8.2). Real closedness and algebraic closedness should enter only where needed: construct the field before introducing a square-root-valued modulus, and prove polynomial algebra generically over the algebraically closed extension only after the real-closedness bridge is available.

## 19.6 Formal sums carried by certified data

Reuse `HahnSeries.SummableFamily` when its hypotheses match the desired operation. For a full surreal-support interface, add the lower-universe-smallness certificate where it is not already implied by the input types. If  $I$  is small and each  $A_i$  has small support, prove smallness of the dependent union as a *separate lemma*; do not leave it to an informal cardinality argument at each use (M3).

The central transport results should have these meanings: coefficients of a sum are finite coefficient sums; embeddings commute with the certified sum; a positive correction has inverse supported in its generated monoid; and reindexing preserves the sum under the stated finiteness conditions. These are stronger and more precise than declaring a generic infinite-sum notation and hoping elaboration will choose the intended convergence theory.

M1's finite-coefficient-dependency interface is the reusable lemma to aim for: for prescribed source support  $S$ , positive perturbation support  $E$  and exponent  $\gamma$ , there is a *finite* set of words  $(s, e_1, \dots, e_r)$  with total  $\gamma$ , and that finite set controls all allowed operator expansions contributing to the coefficient. The statement should make the finiteness witness reusable, supporting the proof pattern of section 13.4. A separate theorem should handle  $K_\Gamma$ -linearity of an operator initially extended coefficientwise from a complex-linear map; it follows from finite convolution and admissible rearrangements, not from the map's being called coefficientwise.

## 19.7 Representing ordinary analytic germs without a global radius field

M1 compares two representations. One uses ordinary function germs at zero, for instance a germ construction on the neighbourhood filter, restricted to germs with an analytic representative; equality is eventual equality on an ordinary neighbourhood, and differentiation and multiplication must be proved representative-independent. The other represents a convergent germ by a multivariate formal power series together with an existential positive convergence radius, making coefficient extraction, finite jets and formal differentiation transparent, but still requiring a theorem identifying these codes with ordinary holomorphic germs in the desired number of variables.

The design rule is the important part: for  $\mathcal{A}_{n,\Gamma}$ , use a Hahn series with this germ ring as coefficient type, and *do not* make the radius a global field of the Hahn-series structure. Such a field would enforce the excluded common-radius hypothesis and destroy the first separator of section 13.1. A radius can instead be selected locally when a finite list of coefficient germs is being compared with an ordinary analytic theorem.

## 19.8 Finite-dimensional algebra before analytic topology

Once a finite basis of a quotient has been obtained, ordinary matrix APIs express multiplication, characteristic polynomials, traces and nondegenerate bilinear forms, and the zero-count descent of [theorem 9.4](#) becomes a generic algebraic theorem needing no global surreal metric space and no theory of fine-continuous paths. The effective order of attack: verify the support-controlled division operator; prove spanning of the quotient; prove independence using the residue Gram matrix (9.3); then establish trace and base-change results. AG explicitly separates spanning from independence, and preserving that separation in Lean avoids accidentally assuming the constant dimension one is trying to prove (M1).

## 19.9 The normal-form bridge has several independent obligations

A usable normal-form bridge needs more than a map between two carriers (M3). It must prove injectivity and surjectivity, agreement with canonical monomials, preservation of addition and multiplication, order compatibility on real coefficients, and preservation of the specified strong sums. To support the analytic manuscripts one also needs coefficient extraction, standard part and Taylor evaluation to commute with the bridge. If the bridge is first established at a fixed exponent group, prove its coherence under group enlargement; if at a universe-relative small-support Hahn type, prove the corresponding smallness and universe-lift compatibility. These are mathematical obligations, not cosmetic coercion lemmas: a noncanonical field isomorphism may fail to preserve the distinguished monomials or the strong summation.

### 19.10 Three implementation strategies, four implementation strategies, and the recommendation to combine

The two audits that surveyed implementation strategy carved the space differently, and both survive.

M1 compares three:

- **Game-first** follows the existing Lean representation most directly. Strengths: recursive arithmetic and compatibility with current game theory. Costs: representative management, and the later normal-form bridge.
- **Sign-first** makes canonical equality, birthdays and simplicity transparent. Costs: defining arithmetic efficiently, and proving agreement with the well-developed cut interface. Especially appropriate for foundational questions about construction history.
- **Hahn-first** is strongest for the analytic goals. Strengths: coefficientwise algebra, explicit support, valuation, finite-dimensional methods. Limitation: an abstract Hahn field is not by itself the full surreal field.

M1's conclusion is that the recommended project *combines* game and sign interfaces for interpretation with Hahn-first proofs for analysis, rather than choosing one representation exclusively.

M3 tabulates four:

| Strategy (M3)                        | Best use                                                        | Principal proof burden                                                                                |
|--------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Extend the current surreal library   | Canonical numbers, birthdays, small cuts, full-scale theorems   | Validate the current APIs; locate or prove real-closedness and normal-form bridges.                   |
| Develop over generic Hahn fields     | Polynomial valuations, support recursion, coherent coefficients | Prove the needed closure hypotheses and later embed the workspace into the intended surreal carrier.  |
| Use a bounded surreal field          | A concrete set-sized target with strong algebraic closure       | Prove the chosen birthday cutoff and all operations stay inside it; expose its scale limitations.     |
| Internal set-theoretic model in Lean | Formal foundational interpretation and comparison of set axioms | Bridge modeled sets and operations to external algebraic types; maintain internal/external semantics. |

M3's recommendation is a *hybrid of the first two*: the polynomial and support lemmas can progress over generic fields while the full surreal bridge is developed. A statement whose proof really needs all small cuts must wait for the corresponding carrier theorem rather than being smuggled into a generic Hahn interface as an axiom. The two recommendations agree in substance: do not commit the entire project to one representation.

### 19.11 A theorem-sized implementation sequence

M3's sequence, with its useful failure mode. First lock the toolchain and rebuild the existing carrier and field imports. Next prove or locate the needed small-cut and universe-lift laws, and independently implement generic complexification. Then establish the real-closedness or algebraic-closure input, followed by the normal-form and support transport bridge. In parallel, develop generic polynomial identities and the finite residue pairing. After that, prove support-local recursion, construct ordinary holomorphic coefficient rings, and formalize coherent evaluation and fixed-domain division. Only then combine the layers for automatic Hartogs coherence and all-scale rigidity.

The useful failure mode: missing normal-form or closure theorems do not block generic algebra, but they do prevent incorrectly advertised surreal specializations. Each milestone should produce named theorems with audited hypotheses rather than a growing collection of unproved global instances. M1's ordering agrees: begin with multivariable positive-order evaluation and the support-controlled operator calculus, which directly address the analytic needs at the set-field level; then formalize germ division, finite-dimensional deformation, residues and finite base change; and let the full-class interpretation preserve only the properties for which compatibility has been proved.

*Remark 19.3* (The closing rule). A dependency gate that has not yet been proved must remain a clearly stated hypothesis in downstream work. It must never be replaced by a new unexplained axiom called a theorem. Declaring one axiom named after each desired theorem makes later deductions *conditional*; it does not turn a manuscript into a verified result. Conversely, an explicitly quantified hypothesis — such as a proved ordinary Weierstrass-division structure — is not an inconsistency: it makes the theorem conditional until that structure is instantiated. The final analytic theorems remain conditional until their ordinary complex-analytic dependencies, in particular Hartogs and Baire, have also been checked in the chosen dependency revision.

## 20 Trust, relative consistency, and reproducible verification

### 20.1 What a construction can guarantee

If a base theory  $T$  proves that a definable structure satisfies a collection of axioms  $A$ , the interpretation is a *relative* consistency justification: a contradiction obtained from the interpreted mathematics would yield a contradiction in  $T$ . Definitional extensions add no new mathematical existence assumptions beyond what their definitions and existence proofs establish. A construction in ZFC shows that the added surreal notation can be interpreted in that foundation without an additional surreal-specific axiom; a universe-relative type construction gives a corresponding interpretation relative to the chosen type theory and any semantic universe assumptions used to justify it; and an axiomatic surreal interface becomes safer when one proves that a known construction inhabits it.

None of these is an absolute proof that ZFC, GBC, KM, a universe principle or Lean’s full foundations are consistent. A strong effectively axiomatized theory cannot in general prove its own consistency under the usual hypotheses of the second incompleteness theorem, and a Lean proof of a model or interpretation carries its ambient foundation and universe assumptions. It would also be incorrect to infer a specific minimal consistency strength merely from the use of universe levels in source code: a set-theoretic model with inaccessibles is a sufficient *semantic* account of Lean universes, but not a theorem that the kernel assumes the corresponding large-cardinal sentences (section 7.8). These qualifications are not reasons to distrust surreal arithmetic; they are the same qualifications that apply to ordinary formalized mathematics.

A checked construction of an internal model can yield a relative consistency implication *after* the appropriate syntactic soundness theorem has been formalized. It does not imply that Lean’s own foundation is absolutely consistent, and there is no conflict with incompleteness in proving a weaker theory’s consistency in a stronger metatheory. The relevant statement records the metatheory and the model interpretation instead of presenting a circular guarantee (M1).

### 20.2 Kernel checking and mathematical faithfulness are separate

M2 states the point that no amount of kernel checking addresses. Lean’s kernel checks proof terms against definitions, inference rules and admitted axioms, and tactics that synthesize such terms are not substitutes for the kernel. But a final theorem *can be kernel-checked and still mathematically misidentified* if its formal statement uses the wrong notion of summability, omits a smallness hypothesis, strengthens the base assumptions, or proves the result only for a restricted surrogate field.

A reliable verification report therefore records both a successful build and a statement-level correspondence with the intended theorem. It should say whether the carrier is  $\mathbf{No}_u$ , a birthday fragment, a Hahn workspace or an abstract real-closed field, and whether a function is fine-local, Hahn-coherent, or equipped with a chosen global phase (theorem 12.5).

### 20.3 Axioms should be audited transitively, not hidden

Lean’s official documentation distinguishes its logical axioms — choice, propositional extensionality, quotient soundness — from computational trust mechanisms and from unfinished proofs, and explains how to inspect a theorem’s axiom dependencies [42, 43]. A declaration accepted by the parser is not a proof, and a theorem depending on `sorryAx` is not a completed result: that axiom can prove arbitrary statements and does not belong in a finished proof. The relevant pattern is

```
#print axioms theoremName
```

with `theoremName` the actual exported theorem being audited.

The audit must extend through imports and must inspect the *transitive* dependency set, not merely search the final file for the word `axiom`. A trusted library theorem, an imported assumed instance, or a computationally generated axiom can carry a dependency invisible in the local proof script. *Absence of the token `sorry` in one short file is not enough if a dependency contains a placeholder or custom postulate.* Tactics that produce ordinary proof terms leave those terms to the kernel; native evaluation can involve additional trust in the evaluation or compilation mechanism, and a project should state whether its certification target permits such dependencies rather than treating all successful tactic invocations as identical trust events.

*Remark 20.1* (What an axiom listing is not). An axiom listing is *not* a complete description of the kernel’s rules or of the semantic strength of its universe hierarchy, and an empty list does not establish the consistency of the kernel. It is, however, a valuable way to detect unproved placeholders or unexpected project-specific axioms in a purported theorem (M2).

## 20.4 What to record for a reproducible formal result

M1’s checklist. A publishable verification record should contain: the exact repository and dependency revisions; the toolchain; a clean build command and its exit status; theorem names and types; the transitive axiom report; and the precise interpretation theorem linking the implementation to the mathematical statement. It should distinguish kernel-checked proof terms from numerical experiments, tactic benchmarks and external computation.

Useful regression tests for the supplied manuscripts include a finite zero cluster, a higher-rank positive-support geometric series, the non-common-radius coefficient example of [section 13.1](#), the rescaling obstruction [theorem 11.1](#), and a negative test rejecting an unrestricted cut. These test different layers. *Passing a finite example does not prove a general Hahn-support lemma; rejecting a self-cut does not prove real closedness.* The full merged regression suite is [section B](#), and it consists of *specification checks*, not randomized evidence for any general theorem.

## 20.5 This report’s own verification status

The September 21 source inspections, the original merge’s three illustrative Lean checks, and the later repository implementation are separated in [section 2](#). The last is tracked by `docs/FORMALIZATION.md`; it does not verify every claim in this report. The historical per-audit build evidence and per-archive inventories remain in [sections 2.3](#) and [2.4](#). In particular, checking a changed dependency graph after `lake update` would not be a check of the original pinned revision.

# 21 Comparison and recommendations

## 21.1 One reconciled foundation table

The three audits produced three overlapping foundation-comparison tables — nine rows (M1), ten rows (M2), seven rows (M3). Printing all three would be a third kind of duplication. They are reconciled here into one wider table, with a final column recording which audits rated each foundation.

| Foundation                          | Principal advantage here                                                                                                                                         | Principal qualification to keep explicit                                                                                                                                                                                     | Rated by   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| ZF / ZFC with definable classes     | Economical classical construction; canonical set-coded numbers; full <b>No</b> as a definable class; local set recursion; no inaccessible required               | Class functions and global assertions are formula-based; quantification over arbitrary classes becomes a definability schema; every recursive graph and set-bounding step must be justified; choice use depends on the proof | M1, M2, M3 |
| GB / GBC / specified NBG            | Direct language for <b>No</b> , <b>SC</b> and fixed global maps; explicit class-function parameters; replacement on ordinary domains; closest to the manuscripts | Classes cannot be elements; predicative comprehension only; no unrestricted class-member quotients or collection of all classes; arbitrary ETR and hyperclasses are not automatic; the choice convention must be specified   | M1, M2, M3 |
| KM / MK and stronger class theories | Impredicative class comprehension; stronger class recursion and satisfaction constructions                                                                       | Greater foundational commitment than the present local algebra requires; still no unrestricted cuts, no naive hyperclasses, no class of all proper classes; class collection and other variants must be named                | M1, M2, M3 |
| Grothendieck universes / TG         | Treat one level's classes as larger sets; externally set-sized but internally large surreal systems; systematic size closure; rich set operations                | Additional universe assumptions beyond ZFC; smallness must survive every change of level; no single small carrier fills all of its external cuts; relative totality is not absolute totality                                 | M1, M2, M3 |
| Feferman reflection (ZFC/S)         | Some large-object reasoning without blanket inaccessibles                                                                                                        | Definability and family-closure conditions are not those of a full universe; conservativity rests on each proof using finitely many instances                                                                                | M1 only    |
| Birthday fragments                  | Concrete set-sized construction stages; clear ordinal bookkeeping; explicit real-closed fields at suitable epsilon bounds                                        | An arbitrary stage need not be a field (dyadic rationals); analytic closure and closure under arbitrary families require specific theorems; full surreal scales are not automatic                                            | M2, M3     |
| Set-sized Hahn fields               | Direct, modular access to support-controlled algebra and analysis; specified value group; transparent supports                                                   | Not the entire surreal universe; full-class rescaling and topology do not automatically survive restriction ( <a href="#">section 11</a> )                                                                                   | M2         |

| Foundation                          | Principal advantage here                                                                                                                       | Principal qualification to keep explicit                                                                                                                | Rated by   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Classical dependent type theory     | Native arithmetic libraries, quotients, well-founded recursion, explicit universe polymorphism; small-branching raw games and numeric quotient | Smallness certificates, quotient congruence and universe maps are mathematical obligations; quotient representatives are not computational normal forms | M1, M2, M3 |
| Constructive set / type theory      | Exposes witnesses, apartness and computational content                                                                                         | Classical trichotomy, choice and ordinal identifications need explicit hypotheses; classical order arguments must be replaced or added explicitly       | M1, M2, M3 |
| Univalent HoTT / higher induction   | Values and extensional comparisons generated together; numerical equivalence can be built into identity                                        | Requires the relevant higher inductive-inductive principles with their computation rules; not native Lean equality ( <a href="#">theorem 17.1</a> )     | M1, M2, M3 |
| Native Lean with universes          | Existing surreal quotient; strong reusable local algebra                                                                                       | Small supports, lifts and interpretation theorems remain essential                                                                                      | M1         |
| Internal <code>ZFSet</code> in Lean | Formalizes the foundational comparison itself; suitable for internal models and conservativity results                                         | More encoding and transport work; bridges to native algebra and analysis; still relative to Lean’s metatheory; not needed for every application         | M1, M2     |

## 21.2 The split recommendation

The spine of the recommendation is M2’s *split*, adopted here because it is the only framing among the three that names the two questions separately.

**For the mathematics.** A precisely specified class theory, or an explicitly definable-class ZFC presentation, organized by M1’s three-part architecture (1.1): set-sized algebra and analysis, plus explicit size-aware comparison maps, plus a class or universe interpretation. The first component carries most proofs; the second prevents accidental changes of meaning; the third supplies the unbounded interpretation. GBC with explicitly stated Global Choice is equally acceptable. More powerful class or universe systems are useful alternatives, not mandatory cures for a paradox, and a change of foundation should be motivated by an operation actually needed — such as manipulating arbitrary families of class functions — rather than by the sheer size of a surreal exponent.

**For the implementation.** A universe-indexed classical Lean development with a set-sized Hahn analytic core. Use classical Lean unless constructive content is itself a research goal. Build on the current separate surreal library rather than an outdated snapshot. Keep finite polynomial algebra and Hahn support calculus *generic*, so they remain useful independently of the full surreal model. Treat the small-support normal-form comparison as a named bridge. Establish the surcomplex pair

or quadratic-field layer generically so it applies both to abstract real-closed fields and to the chosen surreal realization. Adopt M3’s hybrid of the first two strategies together with M1’s conclusion that game, sign and Hahn strategies should be combined rather than chosen between (section 19.10).

**Specific requirements.** Require a proved normal-form bridge, not just a Hahn-series datatype. Require a proved real-closedness or algebraic-closure input where root existence is used, not just a field instance. Represent coherent data on ordinary complex domains with explicit support certificates and the componentwise sheaf convention (13.5). State full-class topological and all-scale results as lower-universe-small-family theorems, or as precise class-theoretic schemes. Keep the entire-carrier cut outside the constructor’s domain by design.

**These recommendations are compatible, not competing.** One is a convenient mathematical language; the other is a proof-engineering representation with a documented interpretation.

### 21.3 What all three independently agreed on

Three points are common to all three audits:

1. class theory, or definable-class ZFC, is the right language for the written mathematics;
2. universe-indexed classical Lean with a set-sized Hahn analytic core is the right shape for the code;
3. *no large-cardinal axiom is required merely because **No** is a proper class.*

This convergence should be read for exactly what it is: **three independent judgements that happen to coincide, not a single argued position reached by one author and endorsed twice.** The three audits carved the recommendation three different ways — M1 as a default organizing language plus a separate remark about ordinary classical Lean for the local layers, M2 as an explicit and load-bearing split, M3 as a layered construction plus a choice among four strategies. This report adopts M2’s framing for its clarity, and does not claim that any single consensus text existed.

### 21.4 The hierarchy of completion claims

M1’s five milestones must not be relabelled as one another:

1. a legal representation;
2. an ordered field with an interpretation theorem;
3. a normal-form and workspace bridge;
4. the local analytic theorems;
5. the full-class or universe-coherent interpretation of those theorems.

Each is meaningful progress. None should be promoted to the next merely because a notation or a typeclass name suggests it.

## 21.5 The five items that must stay visible

M3's closing discipline. The most robust formalization keeps five items visible in every theorem statement:

1. the carrier's universe or class status;
2. the permissible index sizes;
3. the exact algebraic closure available;
4. the summability certificate;
5. the topology or coefficient structure used.

With those stated, the foundational choices become transparent alternatives rather than competing informal interpretations of a single ambiguous notation.

## 21.6 The unifying rule, and the three meanings of “all”

M2's unifying rule is simple and has substantial consequences:

*an individual object can have small data without the totality of all such objects being small.*

Applied consistently it explains why surreal cuts, surcomplex fields, normal forms and strong sums are legitimate; why full-class analysis differs from analysis in one field; and why a formalization must preserve the size annotations rather than erase them.

The surreal numbers do not force unrestricted comprehension (M3). They force clarity about what is small. Once every cut, quotient, support, family and function has an explicit size boundary, the apparent paradoxes become either short contradictions to an overstrong specification or ordinary theorems about different ambient structures, and surcomplexification adds a finite algebraic layer rather than a new foundational difficulty. The architecture proposed here does not weaken the interesting mathematics: it explains why finite polynomial theory can be developed in an ordinary set-sized field, why positive-support preparation needs only set recursion, why the Hartogs support argument needs replacement on the ordinary domain, and why all-scale rigidity uses a resource unavailable in an arbitrary fixed Hahn field.

The central lesson, in M1's formulation, is not to avoid proper classes. It is to prevent three different meanings of “all” from being confused:

1. all members of a specified *set*;
2. all values at a specified *universe level*, subject to smaller-index rules;
3. the entire *definable class*.

With those meanings separated, surreal and surcomplex mathematics fits into rigorous foundations without new set-theoretic paradoxes, while retaining the scales and support phenomena that motivate the theory.

## A The source-manuscript ledger

Three audits were merged into this report. They were given different source manuscripts, and this appendix records exactly which. It is referenced from [section 1.3](#), and it is the authority for the per-audit tagging convention used throughout the body.

- A** *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory.*  
 FILENAMES USED: `surcomplex_analysis(3).tex` by M1 and M3; `surcomplex_analysis(2).tex` by M2.  
 RECORDED SHA-256: `e6e2ef4a8b8d920fffa72564 90f98aa312023078fec966c33b3f204cb17748f4`, recorded identically by M2 and by M3; none recorded by M1. 105,194 bytes (M3).  
 READ BY: **M1, M2 and M3.**
- AG** *Infinitesimal Analytic Geometry over the Surcomplex Numbers: Radius-Free Hahn Germs, a Nullstellensatz, and Conservation of Singular Zeros.*  
 FILENAME USED: `surcomplex_analytic_geometry.tex`.  
 RECORDED SHA-256: none.  
 READ BY: **M1 only.**
- T** *Trigonometry on the Surcomplex Plane: Canonical Angles, Infinite Triangles, Infinitesimal Contact, and Hahn-Analytic Oscillation.*  
 FILENAME USED: `surcomplex_trigonometry.tex`.  
 RECORDED SHA-256: `e61577fffc23c1dabe60a22 57ab2e490a30ab274315c1e6edaeab7639ad4600` (M2).  
 READ BY: **M2 only.**
- P** *Surcomplex Polynomial Algebra: Order Geometry, Newton Profiles, and Scale-Resolved Stability.*  
 FILENAME USED: `surcomplex_polynomial_algebra(2).tex`.  
 RECORDED SHA-256: `9d2ad437715ab5a31804a282 6911e5f9138fa394aa330c3735ea3dfbfe516005`; 112,766 bytes (M3).  
 READ BY: **M3 only.**
- R** *Hartogs Coherence, Finite Maps, and Residue Duality over the Surcomplex Numbers.*  
 FILENAME USED: `surcomplex_research.tex`.  
 RECORDED SHA-256: `9130327331d5635a6e95bb2b b3f972f92f18fe0dbea9889a8d32f121c857c712`; 98,818 bytes (M3).  
 READ BY: **M3 only.**

### The (2)/(3) filename divergence

M2 labels the Analysis manuscript `surcomplex_analysis(2).tex`, while M1 and M3 label it `surcomplex_analysis(3).tex`. The SHA-256 values recorded independently by M2 and by M3 are *identical*, so it is one file under two labels. This report does not silently pick a filename, and asserts nothing about which of the two labels is the correct one. M3 additionally records the file as 105,194 bytes.

### Consequences of the ledger

- **A is the only manuscript read by all three audits.** Claims derived from A may be attributed jointly.
- **Every claim derived from AG, T, P or R rests on a single audit.** In the body, each such claim carries the tag of that single audit: (M1) for AG, (M2) for T, (M3) for P and R.

- No sentence in this report says “the supplied manuscripts” when it means a text that only one member saw.
- **A fourth, unsupplied manuscript.** P refers to a separate *global-divisors* manuscript that was not supplied for any of the three tasks. Its contents are not treated as inspected evidence and no independent assessment of it is offered (M3).
- **Other archives were not examined.** Archives named in the bibliographies of the supplied manuscripts, in particular in T’s bibliography, were not part of the materials examined (M2); the same holds for M1 and M3.
- All five are user-supplied research expositions, not independently certified libraries or refereed publications. Their original files are not duplicated in any of the three archives, and none of them was machine-verified.

## The three merged audits

For completeness, the three audit archives merged into this report are recorded in this repository’s provenance material as `01-size-ledger-and-universe-audit.tex` (M1, 110,177 bytes, 1073 lines), `03-set-and-type-theory-comparison.tex` (M2, 100,299 bytes, 923 lines) and `06-classes-universes-and-lean-audit.tex` (M3, 125,084 bytes, 1109 lines), together with their READMEs and audit records. M3 also recorded SHA-256 values for its own `.tex` (`0f51c217e7b090089c245c46b0f8985cc16587d94b3d0796d5ddb361fcb31e17`) and its own `.pdf` (`aa7ea0641f50b40de9fe4a5146aefa6291cef0c7e5f53273ac0c83f62b6fc3d6`); it is the only audit that hashed its own outputs. None of those files is redistributed with this report; the disposition of every result they contain is recorded in the provenance material that remains.

## B Forbidden shortcuts and regression tests

Everything in this appendix consists of **specification checks**. None of it is randomized evidence for a general theorem. These lists are especially effective immediately before a mathematically strong typeclass instance is about to be introduced. Passing a finite example does not prove a general Hahn-support lemma; rejecting a self-cut does not prove real closedness.

### B.1 Twelve forbidden shortcuts

M2’s list, reproduced in full. Each item records a specific size, domain, algebraic or verification obligation established in the body; none is an ad hoc prohibition.

1. Do not apply the cut operation to all surreals at its own operative size level. ([theorem 3.2](#))
2. Do not treat a proper-class equivalence class as an element of an NBG class. ([section 4.3](#))
3. Do not identify surreal cut interpolation with Dedekind completeness. ([theorem 3.4](#))
4. Do not use an arbitrary birthday bound as a field instance without closure proofs. ([section 7.4](#))
5. Do not identify the self-Hahn normal-form theorem with an unrestricted function-space definition. ([section 4.5](#))
6. Do not confuse well-ordered support with small support, or either with finite coefficient incidence. ([section 10.1](#))

7. Do not replace strong Hahn summation by a topological  $\mathbf{tsum}$ . (section 10.4)
8. Do not infer workspace stability from the fact that the inputs lie in a workspace. (section 9.3)
9. Do not remove the small-index qualification from the fine-net theorem. (theorem 12.2)
10. Do not substitute a real-valued norm for the actual surreal-valued modulus. (section 8.3)
11. Do not derive global analytic uniqueness from a local differential equation alone. (theorem 12.5)
12. Do not call illustrative source, symbolic checks, or a module comment a machine-checked proof of the intended theorem. (section 2, theorem 14.2)

## B.2 A ten-row diagnostic table

M3's table pairs a proposed statement with the exact diagnostic input that refutes it.

| Proposed statement                                            | Diagnostic input or contradiction                                                                               |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Every pair of subsets of the carrier has a Conway separator   | The carrier itself on the left and the empty set on the right (theorem 3.2).                                    |
| The ordered surreal field is algebraically closed             | The polynomial $X^2 + 1$ (section 8.2).                                                                         |
| The surreal field is Dedekind complete                        | The set of natural numbers; subtract one from a proposed least upper bound (theorem 3.4).                       |
| Every well-ordered support defines a same-level surreal       | The too-large ordinal-indexed support (10.2); require relative smallness.                                       |
| Individual legal supports imply a legal family sum            | The singleton supports $\{1/n\}$ (theorem 10.1).                                                                |
| Well-ordered union suffices for a Hahn family sum             | Infinitely many copies of the constant series one (theorem 10.1).                                               |
| Every internal all-scale coherent function is polynomial      | Theorem 11.4 in $\mathbb{C}((t^{\mathbb{Q}}))$ — note the <i>plus</i> sign, and see section 11.6.               |
| Every subset of a universe-relative surreal type is discrete  | The entire carrier; only the lower-universe-small subsets satisfy theorem 12.1.                                 |
| The product ring on two real-field copies is surcomplex       | $(1,0)(0,1) = 0$ in the product ring, unlike complex multiplication (theorem 8.2).                              |
| Every algebra homomorphism preserves infinite Hahn evaluation | Require a strong-sum preservation theorem, or replace the step by the finite divided-difference identity (9.2). |

## B.3 A layered regression suite

M1's suite tests different layers, and passing one says nothing about another:

1. a finite zero cluster (finite algebra layer);
2. a higher-rank positive-support geometric series (support layer);
3. the non-common-radius coefficient example  $\sum_{r \geq 1} t^{r\eta}/(1 - rz)$ , which lies in  $\mathcal{A}_{1,\Gamma}$  but has no common ordinary radius (coefficient-ring layer, section 13.1);
4. the rescaling obstruction theorem 11.1, with its *minus* sign (workspace layer);

5. a negative test rejecting an unrestricted cut (interface layer).

To these the merge adds two further separators that were not in any single suite: the valuation-versus-modulus separator (8.7), and the disjoint-disks sheaf necessity example [theorem 13.1](#).

## C Dependency ledgers: three complementary views

The three audits each produced a dependency view of the same graph, from a different direction. All three are printed, because none subsumes another.

### C.1 Construction, input, and what it does not supply

M3's eleven-row ledger.

| Construction or theorem        | Explicit input                                                                                        | What it does not supply                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Canonical sign carrier         | Ordinal-indexed two-sign functions; bounded function sets                                             | Field operations or normal form.                                                                  |
| Numeric-game quotient          | Well-founded game relation, numericity, equivalence, congruence                                       | Smallness of arbitrary subsets of the quotient.                                                   |
| Small-cut filling              | Small separated option families; birthday control                                                     | Filling arbitrary proper-class or larger-universe cuts.                                           |
| Complexification               | Ordered-field laws; pair multiplication                                                               | Algebraic closedness without real closedness.                                                     |
| Hahn field                     | Ordered exponent group and coefficient field; support calculus                                        | Normal-form equivalence with the surreal carrier.                                                 |
| Algebraically closed workspace | Divisible group; algebraically closed coefficients; Hahn closure theorem                              | Every full-scale analytic theorem.                                                                |
| Strong Hahn evaluation         | Well-ordered union; finite coefficient dependence; small support                                      | Topological convergence of partial sums.                                                          |
| Positive-support recursion     | The set $E^*$ , finite convolution, an invertible linear step                                         | General class-valued ETR.                                                                         |
| Automatic Hartogs support      | Set-indexed holomorphic coefficients, Baire and choice; class replacement for the initial restriction | Coherence when only ordinary slices have been specified.                                          |
| Full-scale polynomial rigidity | One chart support per scale; a bound for the entire small valuation family                            | The same conclusion in a fixed value group such as $\mathbb{Q}$ ( <a href="#">theorem 11.4</a> ). |
| Kernel-verified specialization | All required instances constructed, dependency audit, pinned build                                    | Absolute consistency of the foundational kernel and axioms.                                       |

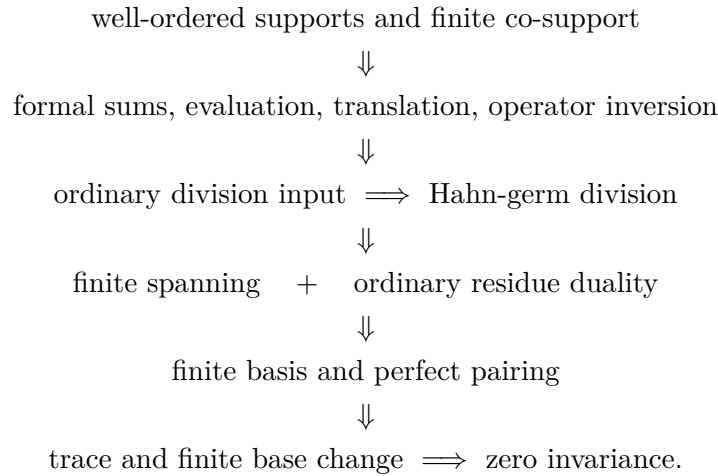
## C.2 Theorem-by-theorem audit of the source manuscripts

M2’s eleven-row audit. It concerns foundational and representation requirements; it does *not* certify all analytic proof details in the original texts. Its rows were drawn from the two manuscripts M2 read (A and T); the rows are retained with that scope.

| Source result family                          | Sufficient working layer                                                  | Obligation that must remain explicit                                                                                 |
|-----------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Finite vector and triangle identities         | Generic ordered or real-closed field; quadratic extension                 | Positive square roots and nonzero denominators; genuine angle interpretation added separately.                       |
| Strong sums and infinitesimal Taylor lifts    | A set-sized Hahn workspace                                                | Well-ordered union, finite coefficient incidence, and correct ordinary base value.                                   |
| Polar decomposition by finite angles          | Hahn infinitesimal exponential and logarithm, plus the ordinary circle    | Surjectivity on every unit point, exact kernel, and quotient size.                                                   |
| Arbitrary-coefficient small-radius evaluation | Full class, or universe-relative small-family boundedness                 | A dominating exponent may lie outside the initial workspace; <a href="#">theorem 11.1</a> .                          |
| Fine-topology degeneracy                      | The small positive lower-bound principle                                  | Smallness of the subset or net index, not only the existence of a Lean type.                                         |
| Coherent Cauchy and identity theorems         | Ordinary holomorphic coefficients on one common domain                    | Support compatibility, ordinary domain hypotheses, and coefficientwise contour definition.                           |
| Preparation and root-cluster conservation     | Finite polynomial and algebraic data plus support recursion               | Algebraic closure; finite coefficient equations; root supports may enlarge beyond factor supports.                   |
| Finite Fourier and Fejér–Riesz results        | Finite sums and real-closed-field algebra, with circle interpretation     | Ordinary finite degree; positivity on the entire relevant circle, not only its real shadow.                          |
| Strip exponential and hyperbolic geometry     | A Gonshor exponential plus finite phases                                  | The real exponential structure is additional to the bare field interface.                                            |
| Global phase classification                   | Additive normal-form splitting and a specified class or type homomorphism | Quantify over characters at the right sort; do not form an illegal class of classes ( <a href="#">theorem 6.1</a> ). |
| All-scale rigidity                            | A full class, or universe-relative quantification over scales             | No inference from a single fixed workspace; each coefficient family must be small at the operative level.            |

### C.3 A minimal theorem-dependency diagram

M1's recommended logical order for the analytic-geometry programme:



The normal-form embedding into **SC** supplies the interpretation boundary; it does not replace any of these local implications. The topological and all-scale theorems form a separate branch with additional smallness arguments.

### C.4 Two source-specific checklists

M1's checklists, retained with their single-audit scope.

**For A (read by all three; the checklist is M1's).** Represent individual surcomplex values by set codes; specify normal-form support and the sign of the Hahn exponent convention; give class replacement or a universe-smallness theorem for each family of outputs; encode formal coefficient sequences as sets; allow the scale chosen by the small-rescaling theorem to enlarge the initial workspace; and state the fine-topology degeneracy theorem only for small sets and small nets. Keep fine-local series, common-domain holomorphic coefficients and all-scale chart conditions distinct. Encode a global class function by its graph, not as an element of an unconstructed function space. Give a set-code interpretation to analytic germs before quotienting. Define the contour functional on its coefficientwise data and prove its laws without silently replacing it by a path integral in the full fine topology. *These conditions clarify the intended objects; the substantive theorems still need their own proof terms.*

**For AG (M1 only).** Fix the exponent group before constructing the germ ring, its ideals and finite quotients. Preserve the “one ordinary convergence radius per coefficient germ” condition rather than imposing a uniform radius. Make both clauses of strong summability explicit. Prove that coefficientwise maps preserve support and interact correctly with Hahn scalar multiplication. Formalize the positive-support inverse independently of convergence of its partial sums. For finite deformation, prove spanning and independence separately; define the perturbed residue before using the roots to interpret it; state ordinary residue duality and finite-parameter trace identities with exactly the hypotheses needed; prove (9.1) in the finite situation rather than assuming global tensor-product equality; and only then invoke [theorem 9.4](#) to exclude zeros at new surreal scales. Distinguish the formal coefficient ring from completion at an infinitesimal point.

## D Notation and implementation vocabulary

M1’s glossary, extended with items from M2 and M3.

| Symbol or term                                  | Meaning in this report                                                                                                                                 |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No</b> , <b>SC</b>                           | The full definable class surreal field and its complexification<br><b>SC</b> = <b>No</b> [ $i$ ]                                                       |
| <b>No</b> <sub>U</sub> , <b>No</b> <sub>u</sub> | A universe-relative realization; the latter is schematic type-theoretic notation                                                                       |
| <b>No</b> <sub>&lt;λ</sub>                      | The birthday cutoff $\{x : \mathbf{b}(x) < \lambda\}$ , a set in ZFC                                                                                   |
| $\mathbf{b}(x)$                                 | Birthday: the ordinal length of the sign sequence                                                                                                      |
| $\prec_s, \prec_b$                              | Simplicity (proper initial segment) and birthday precedence; the second does not imply the first                                                       |
| #                                               | Hessenberg natural sum, used as a pair-recursion measure                                                                                               |
| $\Gamma, K_\Gamma, F_\Gamma$                    | A set-sized ordered exponent group; its complex Hahn field $\mathbb{C}((t^\Gamma))$ ; its real Hahn field $\mathbb{R}((t^\Gamma))$                     |
| $t^\gamma$                                      | $\omega^{-\gamma}$ after the surreal normal-form interpretation; supports are well ordered in the <i>increasing</i> $t$ -exponent order                |
| $R_n, \mathcal{A}_{n,\Gamma}$                   | Ordinary convergent complex germs $\mathbb{C}\{z_1, \dots, z_n\}$ , and Hahn series of those germs (radius-free)                                       |
| $\mathcal{F}_{n,\Gamma}$                        | Hahn series with arbitrary formal complex coefficient series                                                                                           |
| $\mathcal{H}_\Gamma(U)$                         | $\mathcal{O}(U)((t^\Gamma))$ : Hahn series whose coefficient functions are holomorphic on <i>one</i> specified ordinary open $U$                       |
| $E^*$                                           | The additive monoid of finite sums of members of a positive well-ordered support $E$                                                                   |
| Strong sum                                      | A coefficientwise sum satisfying <i>both</i> conditions in (10.1)                                                                                      |
| Coherent datum                                  | A Hahn series with coefficients in $\mathcal{O}(U)$ for one ordinary open $U$ ; a third structure, distinct from both topologies below                 |
| Fine topology                                   | The full-surreal-radius neighbourhood language, unless a universe is specified                                                                         |
| Intrinsic topology                              | A workspace’s own neighbourhood structure, using only its own tolerances $(F_\Gamma)_{>0}$                                                             |
| <b>Oz</b> , $\Pi$                               | The omnific integers <b>Oz</b> = $\Pi \oplus \mathbb{Z}$ ; $\Pi$ the purely infinite normal forms                                                      |
| <b>SmallCutData</b>                             | A structure recording cut <i>input</i> data with the index universe separated from the value universe; it records an input, and asserts no realization |
| option-cut vs. section-cut                      | A small pair of Conway options, versus a complementary section of the whole order; the literature calls both “cuts”                                    |
| <b>Small</b>                                    | Evidence that a type is equivalent to one in a specified smaller universe; it must be <i>proved</i> , never manufactured                               |
| <b>Set X</b>                                    | A Lean predicate on <b>X</b> ; not an internal ZFC set by default                                                                                      |
| <b>ZFSet</b>                                    | Mathlib’s internal extensional set universe, relative to a universe level                                                                              |
| <b>ZFSet.Definable</b>                          | A representative-level liftability condition, <i>not</i> first-order definability by a formula                                                         |

| Symbol or term       | Meaning in this report                                                                                                                    |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>noncomputable</b> | May use noncomputational classical choices; <i>not</i> synonymous with an unproved axiom                                                  |
| <b>sorryAx</b>       | A placeholder dependency that prevents any claim of a completed proof                                                                     |
| ETR                  | Elementary transfinite recursion: recursions producing a whole <i>class</i> at each stage; genuinely stronger than GBC in its usual forms |
| GB / GBC / NBG       | See <a href="#">section 6.1</a> ; the convention is fixed before the letters are used                                                     |
| “Paradox-free”       | Only the explicit exclusion of the specific invalid constructions analysed here; expressly not an absolute consistency claim              |

## E Precise limitations of the present verification

This appendix preserves the three source audits’ limitation lists, deduplicated, and the original merge’s verification boundary. Statements about what was compiled refer to that historical work; subsequent repository formalization is tracked separately in [section 2.1](#). The mathematical and size qualifications continue to apply to the revised exposition.

### E.1 On verification and build status

1. No Lean or Lake executable was available in any of the three preparation environments.
2. No Lean file in any archive was compiled or kernel-checked *by the three audits*. Three of the five were compiled for the merged report; see [section 2.6](#), and item 10 below for what that does and does not cover.
3. The external Lean repository was not rebuilt.
4. No transitive `#print axioms` audit was executed. The `#print axioms` commands present in M1’s and M2’s files were unexecuted source in all three archives; their output is now recorded, for three of the five files, in [section 2.6](#). Each `#print axioms` result includes the named declaration’s transitive dependencies; it does not audit every other declaration in the imported libraries.
5. No Mizar run was performed; the Mizar development was not inspected article by article and its axiom closure was not reconstructed.
6. None of the five supplied manuscripts was machine-verified.
7. The archives are articles and implementation designs, not certified Lean packages.
8. A successful future execution of the small illustrative examples would not constitute verification of the full surreal theory or of any manuscript.
9. Source inspection and compilation are different verification levels.
10. Only the  $\text{\LaTeX}$  builds were executed by the audits themselves. That is the only step common to all three. The merged report adds three Lean compilations and their axiom probes, and nothing else.

11. M2’s README refers to a `SHA256SUMS` file which its archive did ship and which this repository does not carry, checksum manifests being outside the collection’s scope.
12. M2’s clone/build/run commands are a reproduction recipe, not commands that were executed. Do not run `lake update` and then describe the result as a check of the pinned revision.

## E.2 On the pinned source inspection

13. The inspection covered named files at one revision via the public GitHub interface, reading source only: no clone, no build, no comprehensive placeholder search, no axiom audit.
14. Module names alone do not prove their strongest imaginable theorems. This study did not verify an installed real-closedness theorem or a complete surcomplex-analysis library by reading module names.
15. `#synth Field Surreal` is not proof of real closedness, normal-form equivalence, or any analytic theorem.
16. A constructor for a Hahn target type, or a comment describing an intended future isomorphism, is not that isomorphism. Normal-form existence, surjectivity of evaluation and compatibility with multiplication remain proof obligations.
17. No complete ordered-field normal-form equivalence, no real-closedness instance for the concrete game-based carrier, no surcomplex algebraic-closure specialization, and no Gonshor exponential was verified in the inspected modules. *This is a scope-limited observation, not a claim of their absence from any other file, branch, repository, project or later revision.*
18. `IsRealClosed` is an interface, not an instance for any particular type.
19. `ZFSet.Definable` and `Classical.allZFSetDefinable` are representative-level liftability conditions, not first-order definability by a formula.
20. The phrase “complete ordered field” in source prose is not a Dedekind-completeness theorem.
21. The `MvPowerSeries.heval` TODO concerns one inspected module and its documentation; it is not a proof that no other project contains a multivariable evaluation theorem.
22. Mathlib and Lean generated or official “latest” documentation pages are *rolling* sources consulted on the inspection date. They are not asserted to be generated from the pinned dependency commit, and they are not substitutes for the repository’s dependency lock.
23. Old Lean 3 surreal documentation is not a reliable inventory of the current separate Lean 4 project, and finding a theorem in that project does not make it available by importing upstream Mathlib alone.
24. Exact typeclass names should be taken from the pinned dependency source rather than invented to match a mathematical phrase.
25. Theorems do not transfer between Mizar and Lean merely because their informal names agree; the two differ in foundational and representation choices.

## E.3 On consistency and “paradox-free”

26. All consistency assurances are relative to the chosen foundation and its metatheory.

27. No absolute consistency proof of ZFC, GBC, KM, any universe principle, or Lean’s foundations is claimed.
28. “Paradox-free” means the explicit exclusion of the specific invalid constructions analysed — the self-cut, the proper-class-member quotient, the unrestricted support sum, incompatible equality principles — and nothing more.
29. The second incompleteness theorem forbids advertising a construction in ZFC or Lean as an absolute proof that its foundation cannot derive a contradiction.
30. An internal model gives relative consistency only, and only after the appropriate syntactic soundness theorem has been formalized.
31. A specific minimal consistency strength cannot be inferred from the use of universe levels in source code. A set-theoretic model with inaccessibles is a sufficient semantic account of Lean universes, but not a theorem that the kernel assumes the corresponding large-cardinal sentences.
32. GBC’s conservativity over ZFC for first-order set sentences must not be strengthened to the claim that every model of ZFC expands to GBC with the same sets, nor that every model of ZFC has a definable global well-order.
33. Feferman reflection conservativity relies on any one proof using finitely many instances. It is not the assertion that ZFC proves the existence of a single transitive set model of all of ZFC in one internal satisfaction statement.
34. No completed translation into NFU or any universal-set theory is claimed.
35. A Grothendieck universe assumption has a cost: an inaccessible is not a theorem of ZFC, and arbitrarily many universes is stronger still. The inaccessible bound is a convenient sufficient condition, not a claim of minimality.
36. A completed Mizar proof establishes a result in Tarski–Grothendieck; extracting a proof in a weaker theory requires dependency analysis.

#### E.4 On the supplied manuscripts

37. All five manuscripts are user-supplied research expositions, not independently certified libraries or refereed publications. Their original files are not duplicated in any archive.
38. Their analytic results are *targets and supplied arguments*, not imported formal certificates.
39. This report is a foundational audit, not a line-by-line independent verification of any manuscript’s analytic proofs.
40. Other archives referenced in their bibliographies were not examined.
41. The fourth, global-divisors manuscript mentioned inside P was never supplied, and no assessment of it is offered.
42. The Analysis manuscript A is the only source read by all three audits. Every claim about the analytic-geometry, trigonometry, polynomial-algebra or Hartogs manuscripts rests on a single audit.
43. Size-correctness of a ring does not certify its Noetherianity, and formalization of an abstract field does not certify a proposed analytic residue formula.

## E.5 On the mathematics proved here

44. The size obstructions, localization arguments, finite-algebra descent, recursion-locality proposition, support-local recursion, topology counterexamples and the two  $t^{\pm n^2}$  examples are elementary foundational or algebraic deductions included to make the architecture precise. *No novelty claim, no priority claim, and no resolution of any published open conjecture is attached to any of them.*
45. [Theorem 11.4](#) specifically carries no priority claim and no machine-verification claim. It is a sharpness result about the scope of the all-scale rigidity theorem of the companion report at [docs/surcomplex/analysis/](#), not a refutation of it ([section 11.5](#)).
46. No sharp reverse-mathematical calibration of normal forms, real closedness, Hahn algebraic closedness or analytic duality is claimed, and no least axiom system for any source manuscript is identified. Determining such a minimum would be a separate investigation.
47. Particular proofs using arbitrary representatives, well-orderings of a coefficient field, or extraction of decreasing sequences may require choice principles not analysed here.
48. [Theorem 9.1](#) does not say that one set field contains all of **SC**, nor that a set of workspaces covers the class.
49. Neither [theorem 9.4](#) nor its source asserts global tensor-product equality for the whole infinite-dimensional analytic ring.
50. [Theorem 6.4](#) is not a blanket certification of every informal class recursion.
51. [Theorem 7.4](#) does not produce closure under all set-indexed Hahn sums; those have unbounded arity.
52. [Theorem 12.1](#) is not a claim that the whole full-class fine topology is discrete; the carrier itself has no isolated points.
53. [Theorem 17.1](#) concludes only that incompatible equality mechanisms cannot be superimposed — *not* that univalent mathematics is inconsistent.
54. The proposed working surreal interface is an interface, not a claimed minimal or independent axiomatization; establishing categoricity would require a stated notion of morphism and a proved isomorphism to a canonical construction.
55. The four-part abstract package is a structural specification, not a proposed minimal finite axiom system.
56. Exact ordinal closure bounds for exponential subfields are theorems from the literature, not consequences of notation, and must be ported with their hypotheses and the van den Dries–Ehrlich erratum.
57. The constructive and HoTT routes are not claimed to already supply the full normal-form, algebraic-closure and analytic package; bare CZF is not claimed to prove every desired large inductive construction, and constructive surreal variants are not claimed equivalent to the classical sign-sequence class without additional logical hypotheses.
58. The cited HoTT, Shulman and Joseph sources do not by themselves constitute a Lean 4 implementation of the complete surcomplex analytic theory.
59. A categorical small-maps framework does not remove the diagonal obstruction.

60. The 2026 Hamkins/Chen/Yang program is work in progress with details still being checked; it is not a completed published axiomatization, not a verified consistency proof, not a proved dependency of this report, and not a statement about the pure real-closed-ordered-field language.

## E.6 On design proposals

61. All proposed module names, layers, architectures, completion criteria and implementation sequences are design suggestions, not claims that files with those paths exist.
62. A dependency gate not yet proved must remain a stated hypothesis in downstream work, never be replaced by a new axiom called a theorem.
63. Declaring one axiom named after each desired theorem makes later deductions conditional; it does not turn a manuscript into a verified result.
64. Passing a finite example does not prove a general Hahn-support lemma; rejecting a self-cut does not prove real closedness.
65. Symbolic checks of specific identities reported in a source manuscript are sanity checks, not substitutes for support theorems, transfinite recursions or all-scale assertions.
66. The regression tests and forbidden-shortcut lists are specification checks, not randomized evidence for any general theorem.
67. Absence of the token `sorry` in one short file is insufficient if a dependency carries a placeholder or a custom postulate.
68. A theorem can be kernel-checked and still mathematically misidentified if its statement uses the wrong summability notion, omits a smallness hypothesis, or proves the result only for a restricted surrogate field.
69. An axiom listing is not a description of the kernel's rules, and an empty list does not establish kernel consistency.
70. M1's class-relation sheaf encoding is a clarification of a workable reading of A, not a proof of its analytic gluing theorem.
71. The illustrative Lean files contain no intended claim to implement surreal arithmetic, Hahn evaluation, Weierstrass preparation or residue theory, and the complexification sketch claims no `Field` instance and no real-closedness theorem.
72. The raw-game constructor shown is an illustration of the size discipline, not the current repository's exact implementation.

## E.7 On what a full formalization would still require

73. Several-variable Weierstrass division in the chosen germ model.
74. Support preservation for coefficientwise extension.
75. Finite-dimensional residue duality.
76. Comparison with ordinary finite analytic deformations.
77. Noetherian and local-completion arguments.

- 78. The precise finite base-change theorem (9.1).
- 79. Coherence and domain conditions for gluing.
- 80. Contour functionals, local evaluation, and global rigidity.
- 81. The ordinary Hartogs and Baire lemmas, whose availability in the chosen dependency revision must be independently checked.
- 82. The final analytic theorems in the proposed architecture remain conditional until those ordinary complex-analytic dependencies have been checked.

## E.8 Recovered on audit

An audit of this merge against its three sources found the following nine limitations stated in a source and missing from the merged text. They are restored here rather than silently dropped, since each narrows a claim the report otherwise appears to make.

- 83. Strong Hahn summability is not ordinary convergence. A formal series with arbitrary ordinary coefficients can be evaluated at an infinitesimal even when its ordinary analytic radius is zero, and doing so does *not* make it an ordinary holomorphic germ (M2).
- 84. The simplest-number theorem, linking Conway option pairs to the unique sign sequence, is a theorem requiring proof. It is not a definition licensing circular recursion (M1).
- 85. A proof that selects the first nonzero coefficient of an arbitrary real-coefficient sequence may conceal excluded middle or a decidability assumption. The supplied support and normal-form proofs use exactly that step (M1).
- 86. No proof-assistant setting turns a full proper class into an ordinary small type merely by renaming it. This is not special to Lean’s `Small` (M1).
- 87. How high an operation’s birthdays can grow and how large the collection of descriptions below a bound becomes are two different questions. A bound solving one need not solve the other (M1).
- 88. That  $\mathbf{No} \cong \mathbb{R}((\omega^{\mathbf{No}}))$  does not mean a fixed-point equation for an unknown field automatically has a unique solution (M2).
- 89. An abstract proof using a basis of a class-sized vector space would need separate justification and must not be substituted for a concrete argument without notice (M2).
- 90. The gluing theorem is not based on an arbitrary infinite union of unrelated well-ordered supports; the connected-overlap hypothesis is doing work (M3).
- 91. No bibliographic entry here attributes authorship to an uploaded manuscript merely from its filename (M3, followed in practice throughout [section A](#)).

## E.9 On this merged document itself

- 83. This report merges three independently written audits.
- 84. Their agreement on the pinned Lean revision reflects *three readings of one source*, not independent execution.

85. Of the claims in the original merged report, the illustrative checks covered only [theorem 3.2](#), the unrestricted-cut obstruction, in the two Lean encodings of [section 2.6](#), each depending on no axioms. The later implementation mappings in [section 2.1](#) have a different and explicitly recorded scope.
86. Where the three audits differ in scope, sources or emphasis, the differences are printed rather than averaged.
87. The historical synthesis does not promote agreement among source audits into stronger verification evidence. Later corrections and expanded arguments are identified in the report's README; the source originals remain available unchanged in Git history.
88. The merged report's greater length, its five source manuscripts, its proved propositions and its bibliography are not execution. The union of three read-only audits is a larger blueprint, not a certificate.

## References

- [1] *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory*. Supplied manuscript, September 21, 2026. Filed as `surcomplex_analysis(3).tex` by M1 and M3 and as `surcomplex_analysis(2).tex` by M2; one file under two labels, see [section A](#). Especially the class-theoretic conventions, fine-topological degeneracy, small-scale evaluation and rescaling, support-preserving gluing with its componentwise convention, affine charts, contour theory, and all-scale polynomial and rational rigidity. Read by all three audits. Not independently proof-assistant verified.
- [2] *Infinitesimal Analytic Geometry over the Surcomplex Numbers: Radius-Free Hahn Germs, a Nullstellensatz, and Conservation of Singular Zeros*. Supplied manuscript, September 21, 2026. File `surcomplex_analytic_geometry.tex`. Hahn-workspace, radius-free-germ, finite-deformation and workspace-invariance sections. **Read by M1 only**; no hash recorded. Not independently proof-assistant verified.
- [3] *Trigonometry on the Surcomplex Plane: Canonical Angles, Infinite Triangles, Infinitesimal Contact, and Hahn-Analytic Oscillation*. Supplied manuscript, September 21, 2026. File `surcomplex_trigonometry.tex`, SHA-256 `e61577ff...7639ad4600`. Size and support conventions, the angle group, and global phases. **Read by M2 only**. Not independently proof-assistant verified.
- [4] *Surcomplex Polynomial Algebra: Order Geometry, Newton Profiles, and Scale-Resolved Stability*. Supplied manuscript, September 21, 2026. File `surcomplex_polynomial_algebra(2).tex`, 112,766 bytes, SHA-256 `9d2ad437...3dfbfe516005`. Set-sized workspaces; support and summability; fixed-degree real-closed-field transfer; support-local Hensel recursion; set-sized rings and polynomial systems; dependency and effectiveness boundary. **Read by M3 only**. Not an independently verified publication.
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- [6] *Surcomplex Analysis: Germs, Coherent Hahn Sections, and Infinitesimal Zero Geometry — a theory of holomorphic functions on  $\mathbf{No}[i]$* . Companion report in this repository at `docs/surcomplex/analysis/`. Its all-scale polynomial rigidity theorem is stated for *class* functions on the whole of  $\mathbf{No}[i]$  and quantifies over a proper class of scales; see [section 11.5](#). Cited by title and repository path; none of its internal labels is referenced here.

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- [8] John H. Conway. *On Numbers and Games*. Second edition, A K Peters, 2001; first edition, Academic Press, 1976. Foundational constructions and arithmetic of surreal numbers.
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- [11] Bjorn Poonen. “Maximally complete fields.” *L’Enseignement Mathématique* (2) **39** (1993), 87–106. [Author-hosted text](#). Corollary 4 supplies the divisible-group, algebraically closed-coefficient Hahn-field input used in [equation \(8.8\)](#) and [theorem 9.1](#).
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